

A Research Project Report for the Degree of Bachelor of Science in Oceanography

Evaluation of Ecosystem Health of the Saint Martin's Island Marine Protected Area of Bangladesh

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**A project report submitted in partial fulfillment of the requirements for the
Degree of Bachelor of Science in Oceanography**

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3. Dedication

For their love, prayers, support, collaboration, sacrifice, and patience during the course, I dedicate this thesis or research paper to my parents.

4. Acknowledgement

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List of Abbreviation

MPA – Marine Protected Area

SMI – Saint Martin’s Island

SMPA – Saint Martin’s Island MPA

ECA – Ecologically Critical Area

SST – Sea Surface Temperature

SSS - Sea Surface Salinity

NOAA - National Oceanic and Atmospheric Administration

IUCN – International Union for Conservation of Nature

SoNG – Swatch of No Ground

CCME – Canadian Council of Ministers of the Environment

AAS - Atomic Absorption Spectrometry

FTIR - Fourier Transform Infrared Spectroscopy

NDVI - Normalized Difference Vegetation Index

WQI – Water Quality Index

HMPI - Heavy Metal Pollution Index

NPI - Nemerow's Pollution Index

PER - Potential Ecological Risk Index

Igeo - Geo Accumulation Index

TRI - Toxicity Risk Indicator

ERI – Ecological Risk Index

Abstract

The Marine Protected Area (MPA) of Saint Martin's Island in Bangladesh is a crucial habitat that sustains a wide variety of marine and land species. This research aims to assess the health of the ecosystem within the MPA by examining the ecosystem services it offers and the impact of the ecosystem's current condition on those services. The study also identifies and assesses the effects of human activities and pollution on the ecosystem health of the MPA. Utilizing field surveys, data analysis, and stakeholder interviews, the research investigates the primary factors influencing the health of the MPA, such as overfishing, habitat destruction, and pollution. The results indicate considerable deterioration in specific regions due to human-induced pressures, posing risks to the ecosystem's long-term viability. Following this evaluation, recommendations are put forth to enhance the health and management of the MPA. Suggested actions include enforcing stricter regulations on fishing methods, improving waste management systems, and encouraging sustainable tourism practices. By contrasting the present condition of the MPA with that of other Ecosystem Conservation Areas (ECAs), this research provides important insights for the conservation and rehabilitation of marine protected areas in Bangladesh and other locations. The findings emphasize the necessity of integrated management strategies that take into account both ecological and socio-economic factors to ensure the resilience and well-being of marine ecosystems.

1. Introduction

Ecosystem health in Maine Protected Areas is crucial for maintaining the well-being of coastal ecosystems and the benefits they provide to society. According to the National Oceanic and Atmospheric Administration (NOAA) MPA is an area of marine environment designated by federal, state, territorial, tribal, or local legislation or regulations to provide long-term protection for some or all of its resources. While MPAs are designed to conserve biodiversity and ecosystem services, studies show that only a third of the research quantifies societal benefits, often focusing on cultural and provisioning services while neglecting to regulate services and health metrics (Arkema et al., 2024). Ecosystem health in MPAs is essential for sustaining marine biodiversity, coastal protection, and carbon sequestration, which are vital for human well-being. To maximize the effectiveness of MPAs, it is recommended to integrate interdisciplinary data for pluralistic valuation methods and ensure social equity by involving local stakeholders in decision-making processes (Villasante et al., 2023). Long-term socioeconomic monitoring of MPAs is also crucial to understanding the broader context in which MPAs operate and to improve outcomes for both ecosystems and local communities (Cook et al., 2024). Studies on Gede Island's reefs (Nisak et al., 2024) emphasize the role of environmental health in supporting ecotourism and fisheries. Research on heavy metal pollution (Bhuyan et al., 2023) and the impact on health literacy in protected areas (Kadir et al., 2024) highlights the importance of monitoring for ecological and human health. Addressing pollution, like cigarette waste, is crucial for ecosystem sustainability (Howlader et al., 2024).

1.1 Background of the Study

Saint Martin's Island (SMI), located in the Bay of Bengal, is the only coral-formed island in Bangladesh and is surrounded by rocky reefs and boulder fields, supporting diverse coral and seaweed (Tomascik et al., 2022; Hossen et al., 2022). The island is a popular tourist destination, particularly between August and April, but faces significant anthropogenic pressure due to inadequate management and high visitor numbers (Kashem et al., 2022).

Biodiversity on the island is threatened by natural disasters, highlighting the need for ecosystem-based disaster risk reduction (Kayum et al., 2022). Local studies reveal that physicochemical properties of the water impact seaweed diversity, with 51 seaweed species recorded (Hossen et al., 2022). SMI exhibits a diverse yet precariously balanced biodiversity, characterized by the presence of multiple endangered and vulnerable species. Among reptiles, the island is home to the critically endangered Hawksbill Turtle, alongside vulnerable species such as the Leatherback Sea Turtle, Loggerhead, Pacific Ridley, and Indian Flap-Shelled Turtle listed by International Union for Conservation of Nature (IUCN) (IUCN Reptile Range Maps). These species highlight the island's significance as a critical habitat for marine reptiles. In terms of fish species, the ecosystem includes the critically endangered Common Shovelnose Ray and Ganges Shark. Additionally, endangered species like the Slender Hammerhead and Sandbar Shark are part of the marine fauna, demonstrating the ecological importance of the waters surrounding the island. Vulnerable fish species such as the Grey Bamboo Shark, Sting Ray, Lowcrown Seahorse, and Big Eye Tuna further emphasize the island's biodiversity richness (Nereus CORU biodiversity). The presence of multiple critically endangered, endangered, and vulnerable species underscores the ecological fragility of SMI and highlights the urgency of implementing effective biodiversity conservation measures.

The island's resources are also impacted by local development and tourism, while energy needs are currently met through diesel generators and solar power; however, wave energy extraction has potential for sustainable energy (Rani et al., 2020b; Habib et al., 2024). In 1999, SMI was designated as an ECA under the Bangladesh Environment Conservation Act aimed to protect the island's rich biodiversity. Coastal ecosystems like SMI are critical for livelihoods but require sustainable management due to threats from climate change and human interventions. Additionally, agricultural challenges include soil issues with high Na^+ and Mg^{2+} levels, coarse sandy texture, and leaching losses that hinder rice cultivation (Bhuyan et al., 2023). The SMPA in Bangladesh, established on January 11, 2021, spans approximately 1,743 km^2 cover 2.5% of marine area of Country, aiming to protect biodiversity and coral habitats from overfishing and habitat degradation (Sarker et al., 2023). The major concern of this Island is the nursing ground of Turtle, Humpback Whale population, and Tourism Pressure.

1.2 Significance as an MPA

Saint Martin is recognized as a significant MPA due to its rich biodiversity and critical ecological functions. Research indicates that the MPA plays a vital role in conserving marine species and habitats, particularly in the face of increasing anthropogenic pressures such as overfishing and coastal development (Howlader et al., 2024). The establishment of the MPA has led to improved fish populations and enhanced ecosystem resilience, which are essential for sustaining local fisheries and protecting marine life (Piechocka-Kłos, 2022). Furthermore, studies highlight the importance of community involvement in MPA management, which fosters sustainable practices and increases public awareness of marine conservation issues. However, challenges remain, including enforcement of regulations and the need for ongoing research to adapt management strategies effectively (Rani et al., 2020). Overall, the significance of SMI as an MPA lies in its potential to balance ecological health with socio-economic benefits for local communities.

The ecological impact of the SMPA is significant, as it plays a crucial role in biodiversity conservation and habitat protection. Research indicates that the MPA has led to an increase in fish populations and improved health of coral reefs, which are vital for maintaining marine ecosystems and supporting local fisheries. Additionally, the MPA has been instrumental in mitigating the effects of overfishing and habitat degradation, thereby enhancing the resilience of marine species to environmental changes (Howlader et al., 2024). However, challenges remain, including the need for effective management strategies to address illegal fishing and pollution, which can undermine the benefits of the MPA (Sarker et al., 2021). Overall, while the SMPA has positively influenced local marine biodiversity, ongoing efforts are essential to sustain these ecological gains and ensure the long-term health of the marine environment.

The pie chart in Fig 1 highlights the significant impact of tourism and related activities on land-use change in SMI. The increasing demand for tourism-related infrastructure and services is driving the conversion of natural habitats into built-up areas. Population growth and natural hazards also contribute to land-use change, but to a lesser extent. To mitigate the negative impacts of land-use change on the island's environment, it is crucial to implement sustainable tourism practices, promote responsible development, and strengthen disaster preparedness and response

measures. Additionally, conservation efforts should focus on protecting critical habitats, such as mangroves and coral reefs, and promoting sustainable land-use practices.

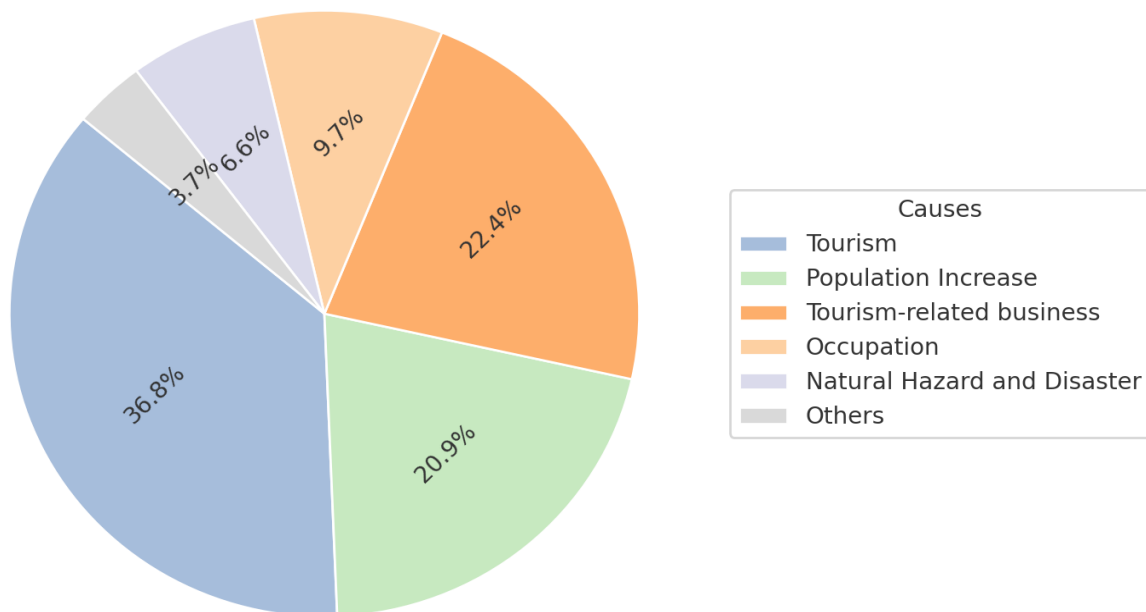


Fig. 1 Major causes of land use changes in SMI (Source: Ahmed et al., 2023)

The pie chart illustrates the major causes of land-use changes in SMI. Tourism is the most significant driver of land-use change, accounting for 36.8% of the total impact. The growing tourism industry has led to increased demand for infrastructure development, such as hotels, resorts, and transportation facilities, resulting in the conversion of natural habitats into built-up areas. Tourism-Related Business contributes 22.4% to land-use change. It includes businesses that cater to tourists, such as restaurants, shops, and transportation services. The expansion of these businesses often requires the conversion of natural areas into commercial zones. With a contribution of 20.9%, population growth is another major factor driving land-use change. As the population increases, there is a growing demand for housing, infrastructure, and agricultural land, leading to the conversion of natural habitats. While less significant compared to the other factors, natural hazards and disasters, such as cyclones and storm surges, can also contribute to land-use change. These events can cause erosion, flooding, and loss of vegetation, leading to changes in land cover. Others account for 3.7% of land-use change and may include factors such as illegal activities, pollution, and climate change.

1.3 Objectives

- To assess the importance of MPA designation assessing the current state of the ecosystem.
- To identify and evaluate the impact of human activities and pollution on the ecosystem health of the MPA.
- To recommend improving the ecosystem health and management of the SMPA comparing with status of ECA.

2. Literature Review

2.1 Successful MPAs in the World

Successful MPAs exhibit several key features that enhance their effectiveness in conserving marine biodiversity and supporting sustainable fisheries. Firstly, effective governance and stakeholder engagement are crucial, as they foster collaboration among local communities, scientists, and policymakers, ensuring that management strategies are inclusive and adaptive to changing conditions (Villasante et al., 2023b). Secondly, the size and location of MPAs significantly influence their success; larger areas with well-defined boundaries tend to provide better protection for marine ecosystems, while strategically placed MPAs can enhance connectivity between habitats (El-Din, 2023; Gonçalves, 2023). Additionally, the enforcement of regulations within MPAs is vital to prevent illegal activities and overfishing, which can undermine conservation efforts (Huang et al., 2024; Villasante et al., 2023b). Finally, ongoing monitoring and research are essential for assessing the effectiveness of MPAs and adapting management practices based on scientific evidence (Gonçalves, 2023). Collectively, these features contribute to the resilience and sustainability of marine environments. According to a 2014 Nature study, sustainable MPAs have five critical characteristics: no-take, well-enforced, more than ten years old, greater than 100 square kilometers, and isolated by deep water or sand. The more of these features present in an MPA, the more effective it will be. By these metrics, the Papahānaumokuākea Marine National Monument in Hawai‘i is a prime example of a successful MPA, checking the boxes on all five features (Brian Owens, 2023). Well-managed MPAs can have big impacts on biodiversity. Mexico established the Cabo Pulmo National Park at the tip of the Baja California peninsula in 1995. By 2009, fish biomass had increased by over 460 percent, and the biomass of key predators such as sharks increased by 11 times. Much of that success can be attributed to the support and leadership of local people (Brian Owens, 2023).

MPAs' efficacy depends on robust rules, enforcement, community engagement, enough funding, and specific management practices that prioritize biodiversity and ecosystem resilience in particular ecological and social contexts.

2.2 Other MPAs in Bangladesh

Naf River MPA

The Naf River estuary is critical to the socioeconomic conditions of coastal communities, providing a primary source of income through small-scale fishing (Chowdhury et al., 2011). The government of Bangladesh established the Naf MPA on May 21, 2024, which covers 734.17 km² under the Marine Fisheries Act of 2020. The waterbody has a depth of around 46 meters. The estuary supports a varied microbenthic community as well as valuable fisheries resources that are critical to the lives of local small-scale fishers. Fishermen confront numerous obstacles, including significant reliance on fishing resources, diminishing yearly harvests, price volatility, limited processing facilities, a lack of financing, and insufficient training, all of which contribute to low economic returns (Chowdhury et al., 2011; Chowdhury et al., 2022). Establishing an MPA could assist to protect these ecological resources while also encouraging sustainable fishing techniques. MPAs can provide an organized strategy to preserving marine biodiversity while also benefiting fisheries, potentially increasing future catches by 20% due to spillover effects. Implementing MPAs in the Naf River could integrate conservation goals with local populations' economic requirements, promoting sustainable fishing methods (Grafton et al., 2012; Cabral et al., 2020).

Nijhum Dwip

The geological aspect of Nijhum Dwip, Bangladesh, involves a comprehensive assessment of the land cover and coastline changes using geospatial analysis. The island's geological features are likely influenced by factors such as sedimentary deposition, erosion, and coastal dynamics, which can impact its land cover and coastline morphology over time. Conducting oceanographic surveys and water sample analysis in Nijhum Dwip can provide valuable insights into the island's hydrological characteristics, water quality, and marine ecosystem dynamics. By studying the geological, geomorphological, and hydrogeological conditions of Nijhum Dwip, researchers can better understand the island's geological history, formation processes, and environmental vulnerabilities (Sobnam and Mamun, 2021).

Nijhum Dwip, situated in Noakhali District, Bangladesh, boasts an abundant and diverse biodiversity, featuring a wide range of plant and animal life within its unique ecosystem. The island is home to approximately 152 plant species from 56 families, including notable varieties like *Sonneratia apetala* (Keora), *Avicennia alba* (Baine), *Bruguiera gymnorhiza* (Kakra), and *Tamarix gallica*. Its animal inhabitants encompass spotted deer, small-clawed otters, golden jackals, water buffalo, monitor lizards, fishing cats, turtles, and dolphins. Particularly famous for hosting the world's largest Indian skimmer bird population, Nijhum Dwip accommodates over 100 species of migratory and resident birds, making it a crucial avian habitat. The surrounding waters attract globally threatened bird species like the Spoon-billed sandpiper, Asian Dowitcher, Nordmann's greenshank, Spotted Redshank, Goliath Heron, and Indian Skimmers. Moreover, the island's tributaries are home to various dolphin species, including the Ganges dolphin, Indo-Pacific humpback dolphin, Indo-Pacific finless porpoise, and Irrawaddy dolphin (Hasan, M. R., 2021). With a mix of marine, brackish, and freshwater ecosystems, Nijhum Dwip's environment supports productive fisheries resources. Its marine biodiversity includes diverse fish species, reptiles such as *Xenochrophis piscator* and *Naja naja*, and mammals like *Canis aureus*, *Herpestes auropunctatus*, *Aonyx cinerea*, *Viverra zibetha*, and *Bandicota indica*.

Swatch of No Ground (SoNG)

The SoNG is a significant geological feature located in the Bay of Bengal near the Ganges–Brahmaputra River mouth. It is a shelf canyon that deeply incises the Bengal shelf, characterized by active gravity-driven currents with erosional characteristics in the upper canyon and non-depositional features. The head of the SoNG submarine canyon provides physical conditions that support extraordinary biological productivity, including globally significant marine mammal diversity, such as Indo-Pacific bottlenose dolphins and Bryde's whales. The sediment composition of the SoNG consists mainly of greenish-gray terrigenous clayey silt and silty sand, with sand-silt and silt-clay layers dominating the area. This geological feature influences ocean dynamics and sediment transport in the region, playing a crucial role in shaping the marine ecosystem and supporting diverse marine life in the Bay of Bengal.

The SoNG, located in the Bay of Bengal, Bangladesh, boasts remarkable biodiversity, particularly in its marine ecosystem. The area supports extraordinary biological productivity, including a

diverse range of marine species such as Indo-Pacific bottlenose dolphins, Bryde's whales, and various other marine mammals like spotted dolphin, spinner dolphin, and false killer whale. The head of the SoNG submarine canyon provides physical conditions that support one of the world's largest aggregations of Indo-Pacific bottlenose dolphins and Bryde's whales, making it a crucial habitat for these species. SoNG has been designated as Bangladesh's first MPA (MPA) due to its high biological value, covering an area of 163,600 hectares, highlighting its importance for marine biodiversity conservation. The area qualifies as an Important Marine Mammal Area (IMMA) under various criteria, including Criterion B2 for aggregations and Criterion D1 for distinctiveness, emphasizing its global significance for marine mammal diversity and conservation efforts. Studies indicate that the cetaceans (Bryde's whales and Indo-Pacific bottlenose dolphins) in the SoNG exhibit distinct genetic characteristics, with the bottlenose dolphin population showing relatively low haplotype diversity but high nucleotide diversity, suggesting unique genetic profiles within the area.

2.3 Ecosystem Services Provided by an Effective MPA

MPAs play a crucial role in providing a variety of ecosystem services that benefit both marine environments and human communities. Effective management of these areas enhances biodiversity, supports recreational activities, and contributes to overall ocean health. The following sections outline the key ecosystem services provided by well-managed MPAs. MPAs are essential for protecting marine biodiversity, which includes various species and habitats. For instance, studies show that MPAs can lead to significant increases in species abundance and diversity, particularly in areas where fishing is restricted (Davies et al., 2022). The protection of sensitive habitats, such as coral reefs, is vital for maintaining ecological balance and resilience against disturbances (Niz et al., 2023). MPAs offer cultural ecosystem services, including recreational opportunities like diving and fishing, which are vital for local economies (Niz et al., 2023; Villasante et al., 2023).

These areas can inspire cultural practices and community engagement, enhancing the social fabric of coastal regions (Villasante et al., 2023). MPAs contribute to climate regulation through carbon sequestration and habitat preservation, which are critical in mitigating climate change impacts (Yu et al., 2024). They also provide essential resources such as food and medicinal compounds,

supporting local and global health (Villasante et al., 2023). While the benefits of MPAs are significant, challenges remain in balancing conservation efforts with local economic needs. Effective management strategies must consider socio-economic factors to ensure equitable distribution of benefits and minimize conflicts among stakeholders (Villasante et al., 2023; Floris et al., 2020).

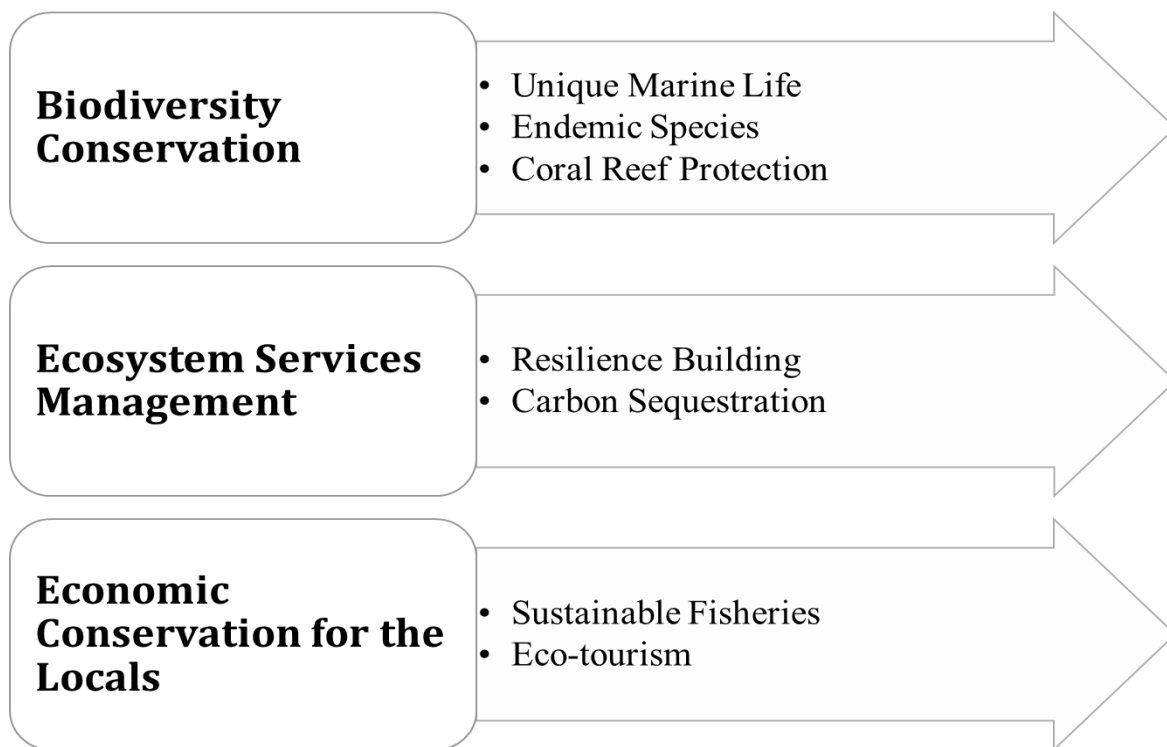


Fig. 2 Services provided by MPA

2.4 Review of previous studies on ESH assessment

The assessment of ecosystem health on SMI, Bangladesh, highlights significant socio-ecological challenges, particularly those stemming from tourism and pollution. Recent research underscores various aspects of these challenges, such as water quality, vegetation health, and the impact of pollution. The socio-ecological environment on the island is reportedly deteriorating due to increasing tourism and population pressure. A study identified alarming levels of electrical conductivity and Total Dissolved Solids in coastal waters, which are linked to pollution from nearby communities (Rahman et al., 2024). Vegetation and forest health on the island have largely been examined through remote sensing, focusing more on forest cover than on health assessments. There is a growing need for indicator-based research to guide the sustainable management of the

island's unique ecosystems (Sumiya et al., 2024). Pollution is another major concern, with a study revealing that 4,481 cigarette butts were collected, highlighting significant littering in ecologically sensitive areas, which poses risks to marine life and the overall health of the ecosystem (Howlader et al., 2024). Despite these challenges, the studies also suggest that there is potential for ecotourism development, which, if managed properly, could encourage sustainable practices on the island (Habib et al., 2023).

Belgrano et al. (2021b) present a comprehensive framework for evaluating MPAs that integrates ecological, social, and economic dimensions. This transdisciplinary approach emphasizes the necessity of assessing MPAs not only for their ecological outcomes but also for their contributions to human well-being and ecosystem services. The authors advocate for the use of the Delphi Method, a structured technique for gathering expert opinions, to facilitate a systematic evaluation of MPAs. This method allows for the incorporation of diverse perspectives from experts across relevant fields, thereby enriching the evaluation process (Gallacher et al., 2016; Hopkins et al., 2018).

Mapping MPAs is essential for understanding their spatial extent and ecological significance, helping to identify areas of high biodiversity and ecosystem service provision while assessing trade-offs in their management. Stakeholder engagement, including local communities and policymakers, is emphasized to balance ecological integrity with human well-being. Recent research highlights the need for multifaceted approaches, integrating ecological indicators like species diversity and habitat quality with social dimensions such as community involvement, economic benefits, and equitable resource distribution. Effective management also depends on governance frameworks and enforcement effectiveness. Advanced methodologies like remote sensing, biodiversity monitoring, and participatory approaches are crucial for holistic evaluations of MPAs. These integrated assessments provide a deeper understanding of MPAs' effectiveness in biodiversity conservation and ecosystem service delivery, facilitating adaptive management strategies that align conservation goals with local needs. Belgrano et al.'s framework offers a multidisciplinary perspective, advancing marine conservation and the sustainable management of marine resources.

- **Ecological Assessments:**
 - **Fish Assemblage Studies:** Research comparing fish populations inside and outside MPAs has shown significant differences in abundance and biomass, particularly in surf zones, indicating MPAs can enhance local fish populations (Marraffini et al. 2024).
 - **Functional Diversity Indices:** Evaluations using functional diversity metrics reveal that while MPAs can protect certain species, they may not fully represent the functional roles of ecosystems, suggesting a need for broader conservation strategies (Hernandez-Andreu et al. 2023).
- **Management Effectiveness:**
 - **Indicator-Based Frameworks:** Tailored frameworks for assessing management effectiveness have been developed, utilizing stakeholder engagement and ecological data to identify deficiencies in MPAs, such as inadequate monitoring and enforcement (Huang et al., 2024).
 - **Community Involvement:** Engaging local communities in the evaluation process has proven essential for identifying management issues and fostering collaborative solutions (Huang et al., 2024).
- **Ecosystem Services:**
 - **Quantifying Benefits:** A focus on ecosystem services, including cultural and provisioning benefits, is crucial for informing MPA design and management, yet many studies overlook regulating services (Arkema et al. 2024).

While these methods provide valuable insights into MPA effectiveness, challenges remain in ensuring that evaluations are comprehensive and inclusive of all ecological and social dimensions. Future research should aim to integrate these diverse approaches to enhance the overall effectiveness of MPAs.

3. Overview of the Methodologies

3.1 Study Area

SMI is in the northeast of the Bay of Bengal, about 9 km south of the Cox's Bazar-Teknaf Peninsula tip and forming the southernmost part of Bangladesh. It is about 8 km west of the northwest coast of Myanmar at the mouth of the river Naf. The island lies between 92°18' and 92°21'E longitudes and 20°34' and 20°39'N latitudes. The local people call it Narikel Jinjira. It is almost flat and is 3.6m above the mean sea level. The 9.66 km wide channel between the mainland and the island is much shallower than the open sea southwest of the island. There are reefs from 10-15 km to the west-northwest. Launches and country boats travel between the island to Teknaf often. The population is approximately 3,700, with the majority being fishermen from 535 households. The island has one hundred eighty-two species of fauna. There are four amphibian species, 28 reptiles, 130 birds, and 20 mammals. The island makes up the Saint Martin's Union Parishad. It has 9 villages which are UP complex of Saint Martin Island, Paschim Para (Western Neighbourhood), Deil Para, Uttar Para (Northern Neighbourhood), Majher Para (Middle Neighbourhood), Purba Para (Eastern Neighbourhood), Konar Para (Edge Neighbourhood), Nazrul Para (Neighbourhood of Nazrul), Golachipa (literally "narrow neck"), Dakkhin Para (Southern Neighbourhood). Tourism exists on the northern part of the island for four months (November to February) (*SMI - Banglapedia*, n.d.). Jinjira is the area with the highest population density. The predominant rock types found on this island are pebbles, boulders, coral rock, limestone concretions, sandstone, shale, and conglomerate (Rahman et al., 2024). Sweet water is scarce on the island.

The SMPA indicated in figure2 Bangladesh, established on January 11, 2021, spans approximately 1,743 km² cover 2.5% of marine area of Country, aiming to protect biodiversity and coral habitats from overfishing and habitat degradation (Sarker et al., 2023). The MPA surrounding SMI is bounded by coordinates from 20°34'N to 20°39'N in latitude and 92°18'E to 92°21'E in longitude (Chowdhury, 2022). The in-situ data is driven from the surrounding of the Island and satellite data cover the whole MPA. Samples were collected 30 km offshore at Dolphin Shoal, a vital habitat for marine life, including Indo-Pacific humpback dolphins and other dolphin species often seen in

groups. The area features vibrant coral reefs that support diverse fish, crustaceans, and other marine species, making it a key site for research, conservation, snorkeling, and diving. However, due to shallow depths and coral reefs, larger boats and ships cannot anchor here.

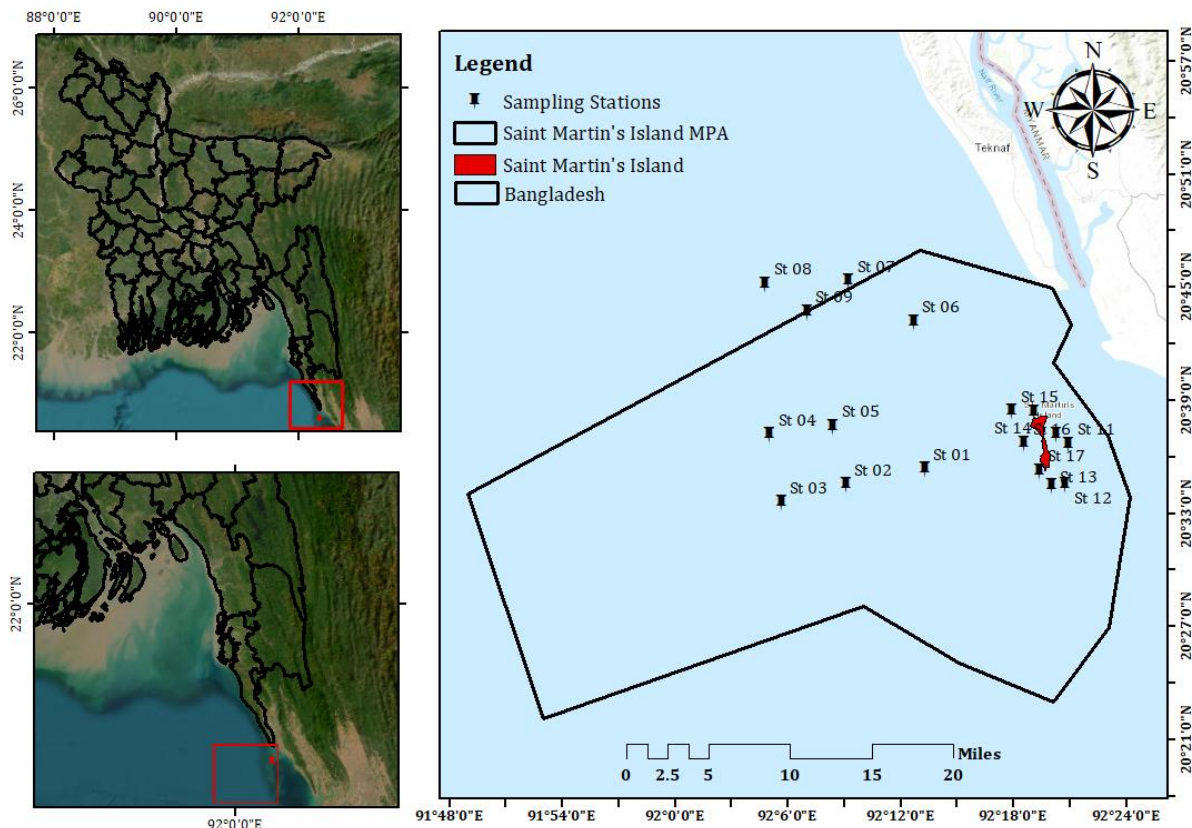


Fig. 3 Study area map (Station 01-18: Nutrients, Physiochemical Parameters; Station 01-10: Heavy Metals, Station 11-18: Phytoplankton and Microplastics)

3.2 Data Collection

In-Situ Data Collection

In this paper, data have been collected from both in-situ and secondary sources. Physiochemical parameters, nutrients, heavy metals, phytoplankton, and microplastic concentrations have been analyzed with in-situ samples, which have subsequently been processed in a laboratory environment. Physiochemical parameters, nutrient concentrations, phytoplankton abundance, and microplastic abundance have been extracted from coastal water samples. Heavy metal concentrations have been analyzed solely from offshore samples. While the majority of samples

have been water-based, seven sediment samples have also been collected specifically for heavy metal analysis. Coastal samples have been collected in January 2023, and offshore samples in December 2021.

Secondary Data Collection

Satellite data have been collected to analyze long-term trends in parameters such as vegetation, rainfall, sea surface temperature, sea surface salinity, chlorophyll concentration, and overall geological and physical changes in the study area. The data have been obtained from Copernicus, along with images from Landsat and Sentinel. These data have been used for comparison with in-situ data, for understanding long-term changes, and for establishing a reference for the study area.

3.3 Data Analysis

Instruments and Software Use for the Analysis

The in-situ samples are collected using different instruments which are listed below:

1. Multiparameter
2. GPS

Samples are being preserved using the Acid solution to analyze later in the laboratory environment. The Laboratory Instrument for later analysis of the preserved samples are;

1. Spectrophotometer for Nutrient Analysis
2. Atomic Absorption Spectrometry (AAS) for Heavy Metal Analysis
3. Compound Microscope to identify Microplastics and Phytoplankton
4. Hotplate for digestion
5. Fourier Transform Infrared Spectroscopy (FTIR) to identify the type of Microplastic

The extracted data is again used to calculate different indices to determine the condition of the study area on a certain scale. In order to calculate the indices, the software is used below;

1. ArcGIS for Spatial Distribution and Map visualization

2. Python for plotting and data visualization
3. Google Earth Engine
4. Microsoft Office (MS Docs and MS Excel)

Vegetation

We download satellite images from Landsat and Sentinel Missions. The images are taken in the post-monsoon (October-November) and winter (December-January) seasons. The image is also quite discernible because the sky is found comparatively clear. The band data is calculated in ArcGIS to get the Normalized Difference Vegetation Index (NDVI).

Equation 1 Normalized Difference Vegetation Index

$$NDVI = \frac{NIR - RED}{NIR + RED}$$

Where NIR is Near Infrared Light and RED is Visible Red Light

The process of NDVI calculation is given below in figure03;

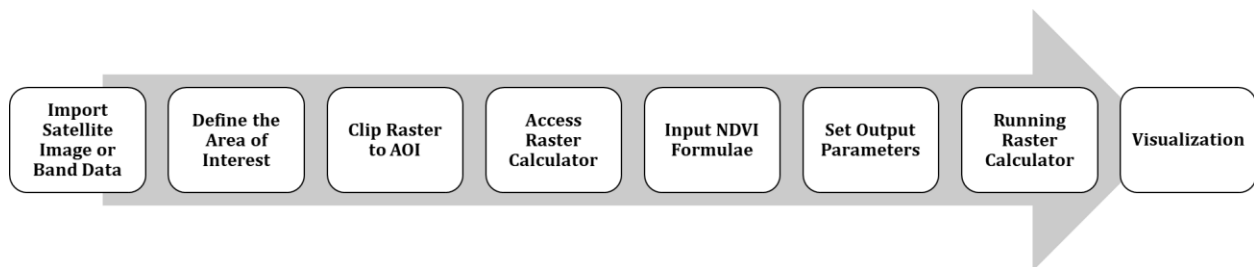


Fig. 4 Steps of NDVI calculation

The NDVI raster is also used to create a plot for the NDVI time series. It is done using Python time series analysis code. It calculates the raster concentration and changes over the time interval.

Temperature and Salinity

The temperature and salinity data are collected from Copernicus Marine Environment Monitoring Service (CMEMS) reanalysis daily data. The system is primarily founded on the real-time global forecasting CMEMS framework. The modeling component utilizes the NEMO platform, influenced at the surface by ECMWF ERA-Interim and subsequently by ERA5 reanalyzes for the

more recent years. A reduced-order Kalman filter is employed for the assimilation of observations. This includes along-track altimeter data (Sea Level Anomaly), Satellite Sea Surface Temperature, Sea Ice Concentration, and in-situ Temperature and Salinity vertical Profiles all being assimilated together. Additionally, a 3D-VAR procedure is implemented to adjust for slowly changing large-scale biases in temperature and salinity. This product offers daily and monthly mean data files for temperature, salinity, currents, sea level, mixed layer depth, and ice parameters, spanning from the surface to the depths. The global ocean output files are presented on a standard regular grid at a resolution of $1/12^\circ$ (approximately 8 km) and across 50 standard levels. The collected data is analyzed in Google Colab Python Platform to calculate spatial and temporal anomaly and time series plots.

Water Quality Status

To determine the water quality status and calculate the Water Quality Index of the study area I use three nutrient values and physiochemical parameters of a total of 17 stations from which 9 stations cover 30 km offshore from the Island. The Physiochemical Parameters (pH, Temperature, Salinity, DO, TDS) were extracted using a multiparameter and listed in the Excel file. The nutrient values are analyzed in the laboratory using a Hach Spectrophotometer. The Samples were preserved with Nitric Acid. The method used for the nutrients individually is given in the table_01:

Table 1 Nutrient analysis method by Hach DR3900

Nutrient Name	Method Name	Reagent Name
Nitrate	Cd Reduction Method	Nitra Ver 5 Nitrate Reagent Powder Pillow
Ammonia	Nessler Method	1. Mineral Stabilizer 2. Polyvinyl Alcohol Dispersing Agent 3. Nessler Reagent
Phosphate	Ascorbic Acid Method	Phos Ver 3 Phosphate Reagent
Silicate	Silicomolybdate Method	1. Citric Acid 2. Molybdate Reagent 3. Acid Reagent

The Water Quality Index is calculated using the Canadian Council of Ministers of the Environment (CCME) Water quality method. The CCME Water Quality Index (WQI) simplifies complex water quality data into a numerical value, making it easier to communicate and comprehend. The CCME Water Quality Index has three key components which are;

- **Scope (F1):** This determines the percentage of water quality metrics that exceeded set guidelines at least once throughout the evaluation period. It represents the level of failure to comply with water quality requirements.
- **Frequency (F2):** This shows how frequently the water quality objectives are not reached, based on the total amount of breaches throughout the period under scrutiny.
- **Amplitude (F3):** This shows the magnitude with which the water quality guidelines are exceeded, indicating the complexity of the exceedances.

Here the formulae are;

Equation 2 Water Quality Index

$$WQI = 100 - \sqrt{((F1)^2 + (F2)^2 + (F3)^2)/1.732}$$

This formula fosters the numbers to ensure they are inside the required range, merging the three components into a single index value.

Table 2 Categories of the stations' water quality according to CCME WQI (Source: CCME Water Quality Index User's Manual 2017 Update)

CCME Water Quality	Category
95-100	Excellent
80-94	Good
65-79	Fair
45-64	Marginal
0-44	Poor

Heavy Metal Analysis

The water and sediment sample are collected from 30 km offshore from the Island. Later on, the samples are analyzed in a laboratory environment with some distinct steps. Here's a summary of the individual steps of heavy metal analysis as outlined in your document:

1. **Sample Collection:** Water and sediment samples were collected using a Van Veen grab from predetermined locations. In-situ measurements of physicochemical parameters were taken immediately, with water samples stored in polyester bottles and sediments in zipper bags for later analysis.
2. **Sample Preparation:** Water samples were analyzed directly, while sediment samples required further drying, grinding, and sieving to obtain fine particles. Both sample types were preserved appropriately before lab processing.
3. **Preparation of Water Samples:** Samples were initially stored at 4°C and then brought to room temperature before analysis. They were filtered using a suction pump and Whatman filter paper, acidified with nitric acid, and partially digested on a hot plate to prepare them for metal analysis.
4. **Preparation of Sediment Samples:** Sediments were dried, ground, and homogenized before storage. The fine particles obtained were used for further analysis.
5. **Sample Digestion:** Sediment samples were digested with concentrated nitric acid on a hot plate until fumes ceased, allowing for metal leaching. The solution was filtered and diluted with deionized water for analysis.
6. **Atomic Absorption Spectrophotometer (AAS):** Heavy metals in samples were quantified using AAS. Flame AAS was used for most metals (e.g., Pb, Cd, Cr), while hydride generation and cold vapor methods were applied for As and Hg, respectively.

The extract concentration of the heavy metals from water and sediment samples is visualized with spatial distribution. Some indices are also calculated which help us to determine their impact on the environment. The indices are calculated in Excel and plotted with Python. The Indices used in this paper are discussed below;

Heavy Metal Pollution Index (HMPI)

The Heavy Metal Pollution Index (HMPI) is a quantitative measure used to assess the level of heavy metal contamination in water bodies. The HMPI is calculated based on the concentrations of specific heavy metals and their respective standards or guidelines. The HMPI is generally calculated using the following formula:

Equation 3 Heavy Metal Pollution Index

$$HMPI = \frac{\sum_{i=1}^n (C_i/S_i)}{n}$$

Where; n = Total number of heavy metals analyzed; C_i = Concentration of the heavy metal i in the water sample; S_i = Standard or guideline value for the heavy metal i.

Table 3 Different levels of pollution of the stations according to HMPI (Source: Edet and Offiong, 2002)

Heavy Metal Pollution Index	Description
HMPI < 15	Low Concentration
$15 \leq HMPI \leq 30$	Medium Concentration
HMPI > 30	High Concentration

Nemerow's Pollution Index (NPI)

The Nemerow index is a weighted multi-factor environmental quality metric that takes into consideration extraordinary or exceptional maximum values. It aggregates the effects of many pollutants into a single index, making it easier to assess overall water quality. The NPI is calculated using the following formula:

Equation 4 Nemerow's Pollution Index

$$P_N = \sqrt{(P_{avg})^2 + (P_{max})^2}$$

Where, P_N = Nemerow's Pollution Index; P_{avg} = Average value of the single-factor pollution indices; P_{max} = Maximum value of the single-factor pollution index among the pollutants being measured.

Table 4 Different levels of contamination of the stations according to NPI (Source: Dey et al., 2021; Su et al., 2022)

Nemerow's Pollution Index (NPI)	Description
$NPI \leq 1$	Clean
$1 \leq NPI \leq 2.5$	Low Contamination
$2.5 \leq NPI \leq 7$	Moderate Contamination
$NPI \geq 7$	High Contamination

Ecological Risk Index (ERI)

The Ecological Risk Index (ERI) is a methodology for evaluating the possible ecological dangers associated with pollutants, particularly heavy metals, in environmental media such as sediments, soils, and water bodies. Formulae for calculating ERI is;

Equation 5 Ecological Risk Index

$$ERI = \sum_{i=1}^n T_i * C_i$$

Table 5 Levels of risk according to ERI (Source: Liu et al., 2021)

Ecological Risk Index (ERI)	Description
$ERI < 40$	Low Risk
$40 \leq ERI < 80$	Moderate Risk
$80 \leq ERI < 160$	Considerable Risk
$160 \leq ERI < 320$	High Risk
$ERI \geq 320$	Very High Risk

Potential Ecological Risk (PER) Index

Håkanson established the Potential Ecological Risk Index (PERI), which is frequently used in ecological risk assessments. This index assesses the ecological hazards associated with heavy metal pollution in sediments and soils by taking into account both heavy metal concentrations and toxic response variables. The formula for calculating the PERI is given as:

Equation 6 Potential Ecological Risk Index

$$PER = \sum_{i=1}^n E_r * C_n$$

Where, E_r = Potential Ecological Risk factor for each heavy metal; C_n = Concentration of the heavy metal in the sample.

Table 6 Interpretation of PER values (Source: Guo et al., 2010)

Potential Ecological Risk (PER) Index	Description
$PER < 65$	Low Risk
$65 \leq PER < 130$	Moderate Risk
$130 \leq PER < 260$	Considerable Risk
$PER \geq 260$	High Risk

Geo Accumulation Index (I_{geo})

Muller (1969) developed the geo-accumulation index (I_{geo}) to assess the level of heavy metal and metalloid elements in the sediment by comparing the current concentration with the pre-industrial level. The formula for calculating the Geo Accumulation Index is:

Equation 7 Geo accumulation Index

$$I_{geo} = \log_2\left(\frac{C_n}{1.5 \cdot B_n}\right)$$

Where, I_{geo} = Geo Accumulation Index; C_n = Mean concentration of the heavy metal in the samples analyzed (mg/kg); B_n = Geochemical background value of the heavy metal (mg/kg).

Table 7 Classification of Geo Accumulation Index (Source: Barbieri, 2016)

Geo Accumulation Index (I_{geo})	Grade	Description
$I_{geo} \leq 0$	Grade 0	Uncontaminated
$0 < I_{geo} \leq 1$	Grade 1	Uncontaminated to moderately contaminated
$1 < I_{geo} \leq 2$	Grade 2	Moderately contaminated
$2 < I_{geo} \leq 3$	Grade 3	Moderately to heavily contaminated
$3 < I_{geo} \leq 4$	Grade 4	Heavily contaminated
$4 < I_{geo} \leq 5$	Grade 5	Heavily to extremely contaminated
$I_{geo} > 5$	Grade 6	Extremely contaminated

Toxicity Risk Indicator (TRI)

According to Zhang et al., 2016, the toxicity Risk Indicator (TRI) is calculated using the Threshold Effect Level (TEL) and probable Effect Level (PEL) to assess the toxicity risk level of elements in sediment. TRI Value is measured using the formulae below;

$$TRI_i = \sqrt{\frac{(C_i/TEL)^2 + (C_i/PEL)^2}{2}}$$

Equation 8 Toxicity Risk Index

$$TRI = \sum_{i=1}^n (TRI_i)$$

Where, C_i = Concentration of toxic metal i;

Table 8 Level of toxicity according to TRI value (Source: Guo et al., 2010)

Toxicity Risk Indicator (TRI)	Description
$TRI \leq 5$	No Toxic Risk
$5 < TRI \leq 10$	Low Toxic Risk
$10 < TRI \leq 15$	Moderate Toxic Risk
$15 < TRI \leq 20$	Considerable Toxic Risk
$TRI > 20$	Very High Toxic Risk

Phytoplankton Analysis

Phytoplankton samples were collected at eight stations on St. Martin's Island. Using a 55 µm mesh net, 351.19 L of seawater was filtered at each station, and the concentrate was preserved in a vial with Lugol's solution for lab analysis. In the lab, phytoplankton density was assessed using a Sedgewick Rafter Counting Chamber at 400x magnification. The abundance of phytoplankton was calculated as cells per liter using the formula;

Equation 9 Phytoplankton Abundance

$$\text{Phytoplankton Abundance; } N = n \times \frac{v}{V}$$

where n = Average cell number in 1 ml of sample; v = Volume of Concentrate and V = Volume of Seawater filtered.

The Population density of phytoplankton used in different indices analysis further to understand the condition of the ecosystem for biodiversity. The indices are discussed below;

- **Shannon-Wiener Diversity Index (H):** Calculated to assess species diversity, where a higher value indicates greater diversity within the phytoplankton community. The general formulae; $H = -\sum_{i=1}^n p_i \cdot \ln(P_i)$; Here, P_i =the proportion of individuals belonging to species i in the total community. Higher Values indicate greater diversity, suggesting a

balanced distribution of species. Lower Values reflect lower diversity, potentially dominated by one or a few species. A value of $H = 0$ signifies no diversity, meaning only one species is present.

- **Margalef's Species Richness Index (R):** Margalef's Species Richness Index, introduced by Ramón Margalef in 1958, quantifies biodiversity by measuring the number of species in an area while adjusting for sample size. The general formulae; $R = \frac{S-1}{\ln(N)}$; Here, S= Total number of species observed in the sample, N = Total number of species observed in the sample. Higher Values indicate greater species richness, suggesting a more diverse ecosystem. Lower Values reflect lower species richness, indicating less biodiversity.
- **Evenness Index (E_H):** Calculated to determine the distribution uniformity of species, with values closer to 1 indicating even distribution. General formulae; $E_H = \frac{H}{\ln(S)}$; Here, H = Diversity Index, S = Total number of species observed in the sample. This index ranges from 0 to 1, where 1 indicates complete evenness among species.
- **Simpson Dominance Index (D):** The Simpson Dominance Index, developed by Edward Simpson in 1949, quantifies species diversity in an ecological community by calculating the chance that two randomly selected individuals belong to the same species, showing species dominance levels. General formulae; $D = \sum_{i=1}^S [(n_i/N)]^2$; Here, n_i = the number of individuals of species i; N = the total number of individuals of all species; S = the total number of species. The index ranges from 0 to 1. 0 indicates infinite diversity (i.e., all individuals belong to different species). 1 indicates no diversity (i.e., all individuals belong to a single species).

This methodology provides a comprehensive analysis of phytoplankton composition and biodiversity, reflecting the health and stability of the aquatic ecosystem. To assess long term comparison with the in-situ data we also collect the chlorophyll concentration data from 2001 to 2020. That data was analyzed to show the long-term anomaly.

Microplastic Analysis

Microplastic samples are collected from 10 stations on the coast. The collected samples were carried in 250 ml of sample bottles to prepare it for laboratory analysis. Microplastic detection

involves several steps which are suggested by the National Oceanic and Atmospheric Administration (NOAA).



Fig. 5 Steps of microplastic detection

Wet Sieving

The wet sieving technique involves pouring the sample through a set of stacked stainless steel mesh sieves with mesh sizes of 5.6 mm (No. 3.5) and 0.3 mm (No. 50). A squirt bottle filled with distilled water is used to rinse the sample thoroughly, transferring all residual solids to the sieves and removing any salts from the field sample. This rinsing process is repeated as necessary to ensure all material is captured. The sieves are then thoroughly rinsed with distilled water to clean, drain, and sort the material. Depending on the study objectives, material retained on the 5-mm sieve may be discarded or archived. Begin by weighing a clean, dry 500-mL beaker to the nearest 0.1 mg. Transfer the solids collected on the 0.3-mm sieve into the tared beaker using a spatula, rinsing minimally with distilled water to ensure all solids are transferred. Place the beaker in a 90°C drying oven for 24 hours or until completely dry.

Wet Peroxide (WPO)

Wet Peroxide (WPO) in water samples has significance for detecting microplastics because of its capacity to decompose and eliminate organic materials. This stage of the process implies adding 20 mL of 0.05 M Fe (II) solution and 20 mL of 30% hydrogen peroxide to the beaker with the 0.3 mm solids fraction. Let stand for 5 minutes, then add a stir bar, cover with a watch glass, and heat to 75°C on a hotplate. Remove as soon as bubbling occurs, and place in a fume hood to allow boiling to subside. If needed, add distilled water to control overflow. Reheat to 75°C for 30 minutes. If organic material remains, add 20 mL hydrogen peroxide, repeating until clear. Finally, add ~6 g NaCl per 20 mL to increase solution density, heating to 75°C until dissolved.

Density Separation

This procedure was used to remove low-density plastic particles from high-density solutions. Transfer the WPO solution to the density separator, rinsing the beaker with distilled water to include any remaining solids. Cover the separator loosely with aluminum foil and let solids settle overnight. Check for microplastics, removing any with forceps, then drain and discard settled solids. Collect any floating solids using a clean 0.3-mm sieve, and rinse the separator with distilled water to ensure all solids transfer to the sieve. Allow the sieve to air dry under loose foil cover for 24 hours.

Microscopic Exam

The microscopic investigation was performed using a Euromax digital microscope at 4X and 10X magnification, which allowed us to detect the color and size of microplastics using the associated program Image Focus Plus V2.

Abundance Calculation Formula;

Equation 10 Microplastic Abundance

$$Abundance = \frac{Total\ Microplastics\ Found\ in\ 100\ ml\ of\ Water\ (N) * 1000}{100}$$

4. Results

4.1 Vegetation

The Normalized Difference Vegetation Index (NDVI) is illustrated in Fig. 6 for SMI, revealing a gradual decline in vegetation health from 1990 to 2015.

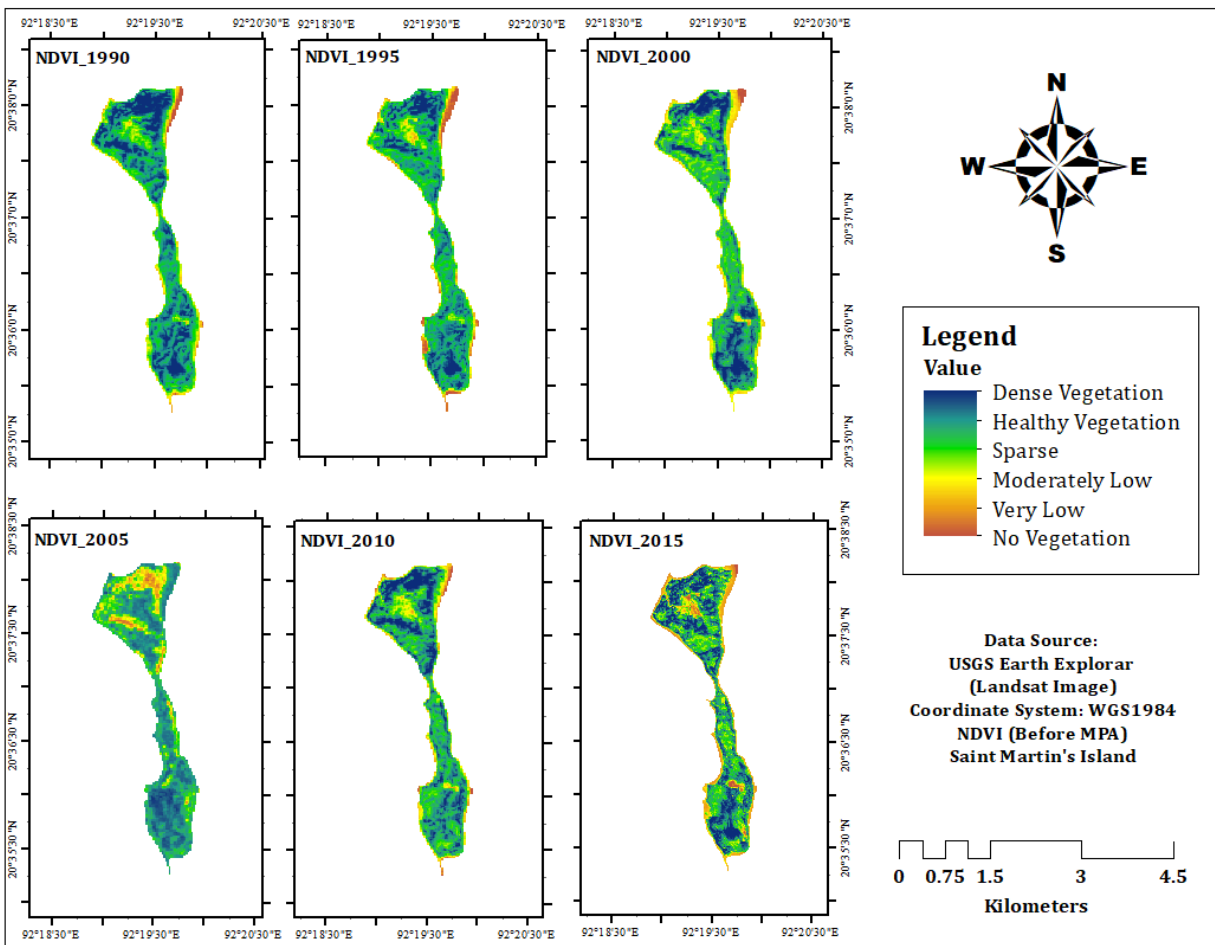


Fig. 6 Changes of Vegetation concentration in SMI over the time from 1990 to 2015

In 1990, a significant portion of the island was covered in dense vegetation, as indicated by the dark green areas. However, by 1995, the extent of dense vegetation had noticeably reduced, with a shift towards areas displaying healthy but less dense vegetation, represented by light green. The trend of diminishing dense vegetation continued into 2000, with further expansion of moderately healthy vegetation. By 2005, the island's vegetation health had deteriorated further, with large

areas transitioning to moderate and low vegetation cover, reflected by yellow and orange colors. This decline persisted in 2010, with vegetation health continuing to diminish. By 2015, the NDVI analysis showed a stark decrease in vegetation health, with extensive areas exhibiting very low vegetation cover, marked in red, and some regions showing no vegetation at all, indicated by brown.

The NDVI data for 2020 and 2024, presented in Fig. 7, provide a detailed overview of the island's vegetation health just prior to the announcement of the MPA. The 2020 analysis reveals a landscape characterized by a mix of vegetation conditions. There are areas with moderate vegetation cover, indicated by yellow shades, and regions with very low vegetation, marked in red. Despite the presence of these degraded areas, a significant portion of the island still displays healthy vegetation, represented by light green areas.

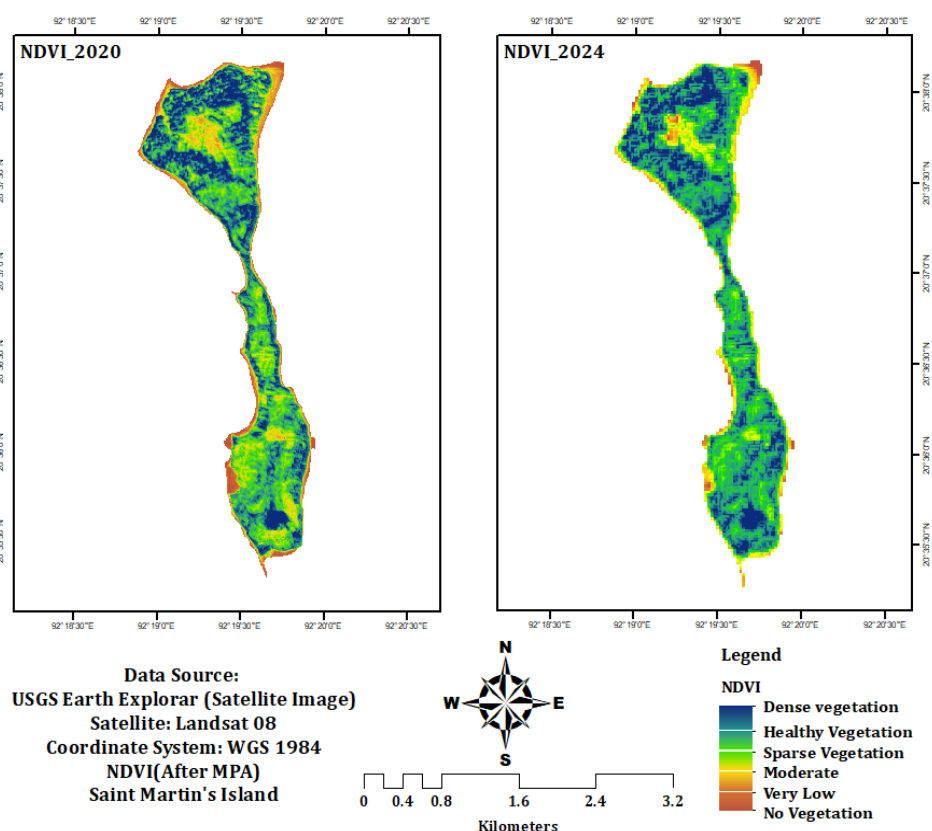


Fig. 7 Recent condition of vegetation after establishment of MPA

This suggests that, although the island has experienced decline in previous decades, there is clear evidence of recovery in vegetation health by 2020. By 2024, the vegetation condition has markedly

improved. The red zones, which represent areas of very low vegetation cover, have substantially diminished, while regions with dense, robust vegetation (dark green) have expanded significantly. This shift indicates a positive and encouraging trend in the recovery of the island's ecosystems. The data for 2020 and 2024, when compared to the downward trend observed in the NDVI from 1990 to 2015, show a remarkable turnaround in vegetation health, pointing to a potential reversal of the previous degradation.

This improvement suggests that various factors may be contributing to the recovery, such as effective conservation measures, natural processes of regeneration, or favorable changes in environmental conditions. However, it is important to note that the NDVI, while a useful tool for assessing vegetation health, is not a comprehensive indicator on its own. Other key factors, including soil fertility, water availability, local climate variations, and human interventions, also play crucial roles in shaping the overall state of the island's ecosystem. Therefore, while the NDVI data indicate a positive trend, a more holistic approach is required to fully understand the dynamics at play.

To gain a comprehensive understanding of the long-term impact of the MPA and other conservation measures, it is imperative to continue monitoring NDVI and integrating other environmental and socio-economic data. Regular assessments and detailed analyses will help identify the drivers behind these changes and ensure the formulation of effective, science-based strategies for sustaining and enhancing the island's ecological health and biodiversity.

4.2 Precipitation

The precipitation plot in Fig. 8 consists of four time series subplots, each representing average precipitation (in mm) across different seasons from the year 2000 to 2020. During winter, represented by the blue line, precipitation levels remain relatively low throughout the period. Peaks are infrequent, with a few spikes observed in the mid-2000s and minor increases around 2015, but overall, winter precipitation remains consistently low, rarely exceeding 20 mm. In contrast, the premonsoon season, shown by the green line, exhibits a more variable pattern. Early years up to 2006 display frequent fluctuations, with notable peaks.

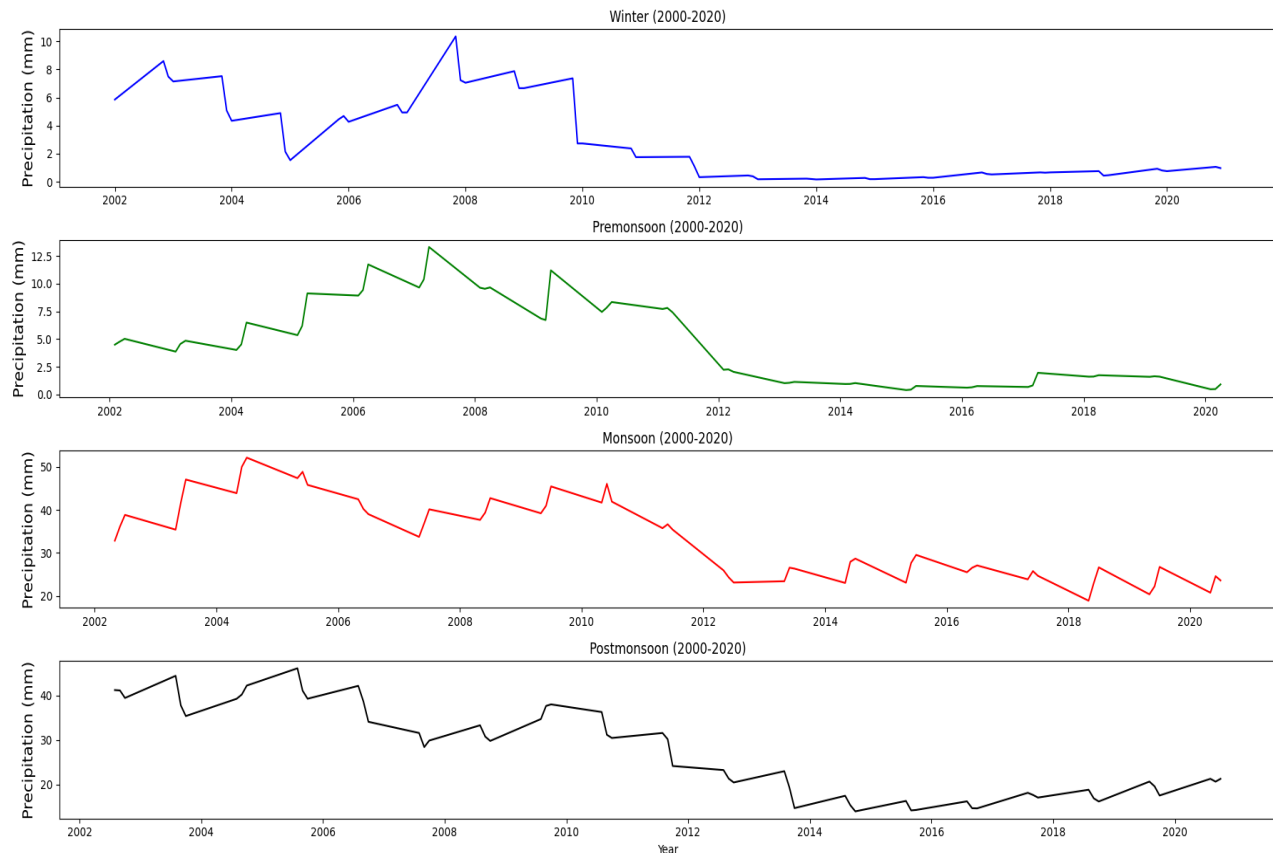


Fig. 8 Seasonal precipitation rate change

However, after 2010, the premonsoon rainfall decreases significantly, stabilizing at lower levels with minor fluctuations. This suggests a shift in premonsoon rainfall dynamics, with fewer intense rain events in the later years of the study period. The monsoon season, represented by the red line, stands out for its consistently high precipitation levels. Monsoon rainfall dominates the overall pattern, frequently ranging from 40 mm to over 60 mm. There is a noticeable cyclical pattern of peaks and drops, indicating a regularity in the monsoon's intensity. Despite these periodic fluctuations, monsoon precipitation remains high compared to other seasons, underscoring its critical role in the annual rainfall cycle. Lastly, the post monsoon season, depicted by the black line, shows a gradual upward trend. Starting at lower levels around 20 mm in the early 2000s, post monsoon precipitation gradually increases, approaching 60 mm by 2020. Unlike the sharp fluctuations seen in other seasons, post monsoon rainfall shows a steadier and more progressive rise, hinting at a potential shift towards wetter post monsoon conditions over time.

4.3 Temperature

The region surrounding SMI in Bangladesh is undergoing significant climatic shifts, with rising temperatures being one of the most prominent indicators of these changes. As climate change accelerates, the local marine environment is increasingly vulnerable to the impacts of warming temperatures. The analysis of spatial temperature anomalies from 2001 to 2020, across different seasons, sheds light on the extent and variability of these warming trends. These changes not only reflect the broader global climate shifts but also highlight the specific vulnerabilities of the Marine Protected Area (MPA) within the region. With its sensitive ecosystem, the MPA faces unique challenges that could affect its biodiversity, ecological balance, and overall resilience.

Fig. 9 represents the spatial distribution of temperature anomalies during the winter season, from November to January, for the period between 2001 and 2020 in the area surrounding SMI. The analysis places particular emphasis on the MPA within this region. Over the two decades, a distinct warming trend is evident, as most years display positive temperature anomalies. This suggests that winter temperatures during this time frame have generally been higher than the baseline reference period. Such a trend aligns with the global impacts of climate change, highlighting the ongoing rise in temperatures within this environmentally sensitive and ecologically significant area. The spatial patterns of temperature anomalies show considerable variation across the region. Within the MPA, some areas consistently record higher positive anomalies compared to the surrounding regions. This suggests that the MPA may be experiencing more intense warming, potentially due to localized climatic or environmental factors. These changes could have far-reaching effects on the marine ecosystem, potentially altering species habitats, ecological dynamics, and overall environmental stability.

The data also reveal noticeable interannual variability in temperature anomalies. In some years, the warming trend is widespread across the entire region, while in others, temperature increases are concentrated in specific hotspots. This variability reflects the complex interaction between global climate systems and regional environmental processes, which together influence temperature distributions. The findings underline the importance of ongoing monitoring and the need for adaptive conservation strategies to address the potential ecological impacts of these warming trends within the MPA and its surrounding regions.

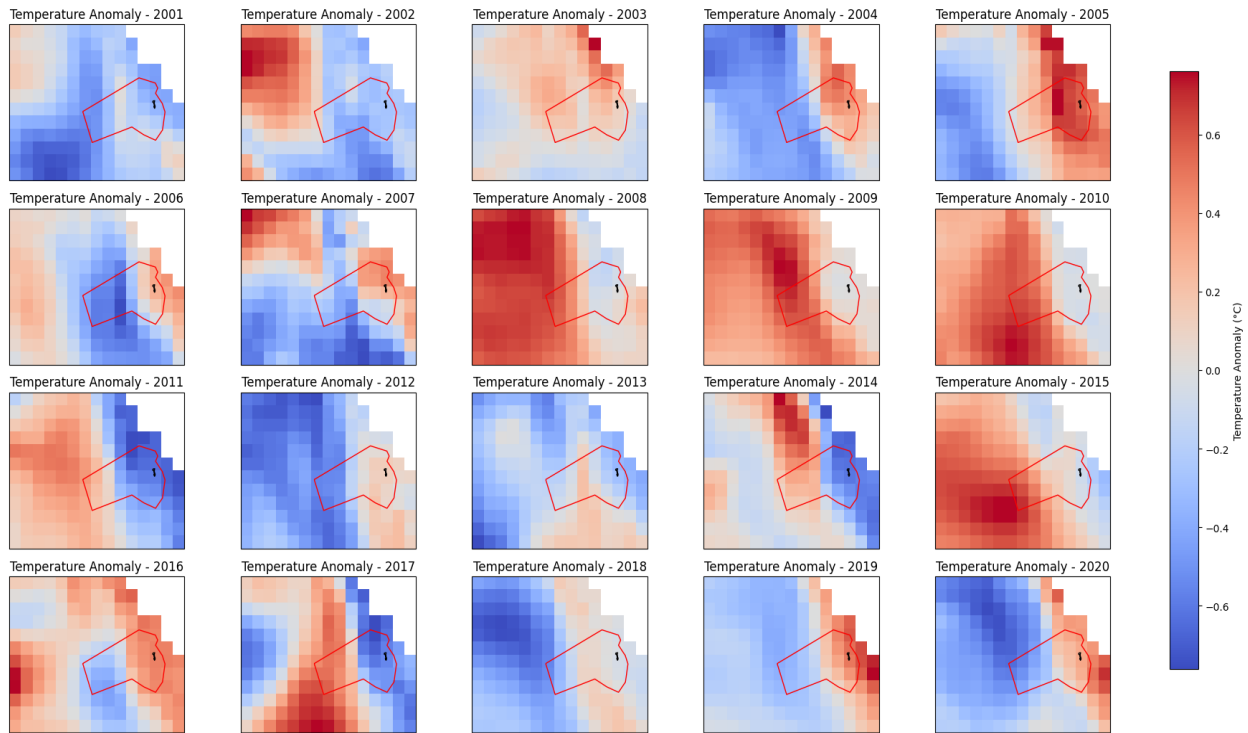


Fig. 9 Spatial anomaly of temperature in winter (Nov-Jan)

The images in Fig. 10 illustrate the spatial temperature anomalies during the premonsoon season (February to April) from 2001 to 2020 in the region of SMI, Bangladesh, with a particular emphasis on the MPA. The analysis reveals a general warming trend, as most years exhibit positive temperature anomalies, indicating an increase in premonsoon temperatures compared to a reference period. This trend highlights a gradual shift towards warmer conditions in the region during this season. The spatial distribution of temperature anomalies shows significant variability across the region, with certain areas within the MPA consistently displaying higher temperature anomalies than the surrounding zones. This suggests that the MPA might be experiencing more pronounced warming trends, raising concerns about potential impacts on the marine ecosystem. Moreover, the data indicates year-to-year variability in the spatial patterns, with some years characterized by widespread warming and others featuring localized hotspots of elevated temperatures. As the precipitation rate shows a decline in premonsoon season, it leads to increase the temperature. The winter becomes shorter in duration and the intensity of sunlight is also a factor in this increasing of temperature.

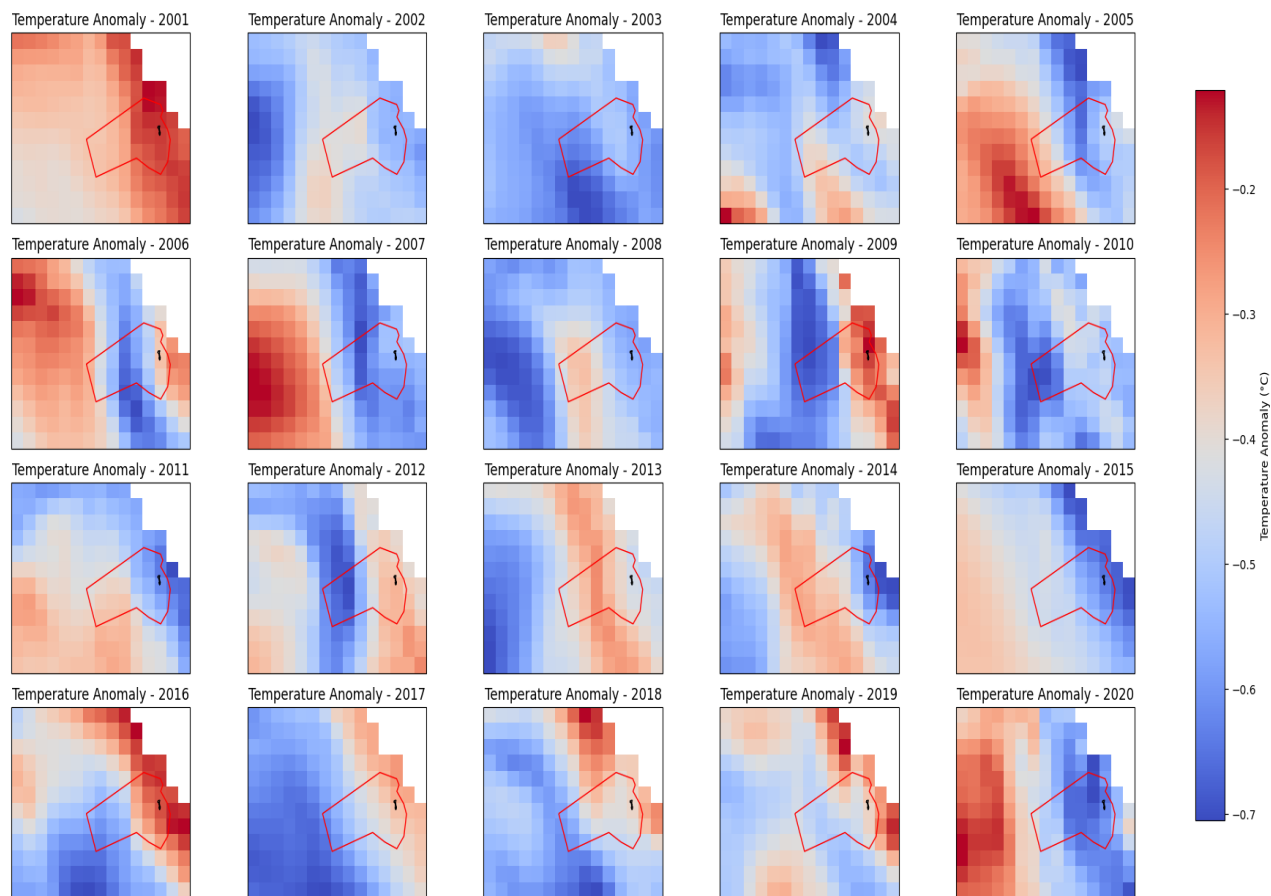


Fig. 10 Spatial anomaly of temperature in premonsoon (Feb - Apr)

The spatial temperature anomaly of the monsoon season (May to July) from 2001 to 2020 in the region surrounding SMI visualized in Fig. 11, Bangladesh, with a particular focus on the MPA (MPA), reveals a discernible warming trend. Over the two-decade period, the majority of years display positive temperature anomalies, signifying an overall increase in monsoon temperatures relative to a reference baseline. This warming trend is not spatially uniform across the region. Within the MPA, specific areas consistently exhibit higher temperature anomalies compared to their surroundings, indicating that the protected area might be subject to more pronounced warming. Such localized temperature increases raise potential concerns for the resilience and sustainability of the marine ecosystem within the MPA boundaries. Additionally, the analysis highlights significant interannual variability in the spatial distribution of temperature anomalies. Certain years are characterized by more extensive warming, while others reveal concentrated hotspots of elevated temperatures. The irregular polygon delineating the MPA frequently aligns with regions of intensified anomalies, underscoring the area's susceptibility to climate-induced

changes. The observed warming trends are likely attributable to a combination of global and localized factors. On a broader scale, the findings align with the established patterns of global climate change. Locally, elevated anomalies within the MPA may be influenced by factors such as alterations in ocean circulation, localized pollution, or anthropogenic activities. The pronounced variability further suggests that the region's climatic patterns are shaped by intricate interactions between natural processes and human-induced influences.

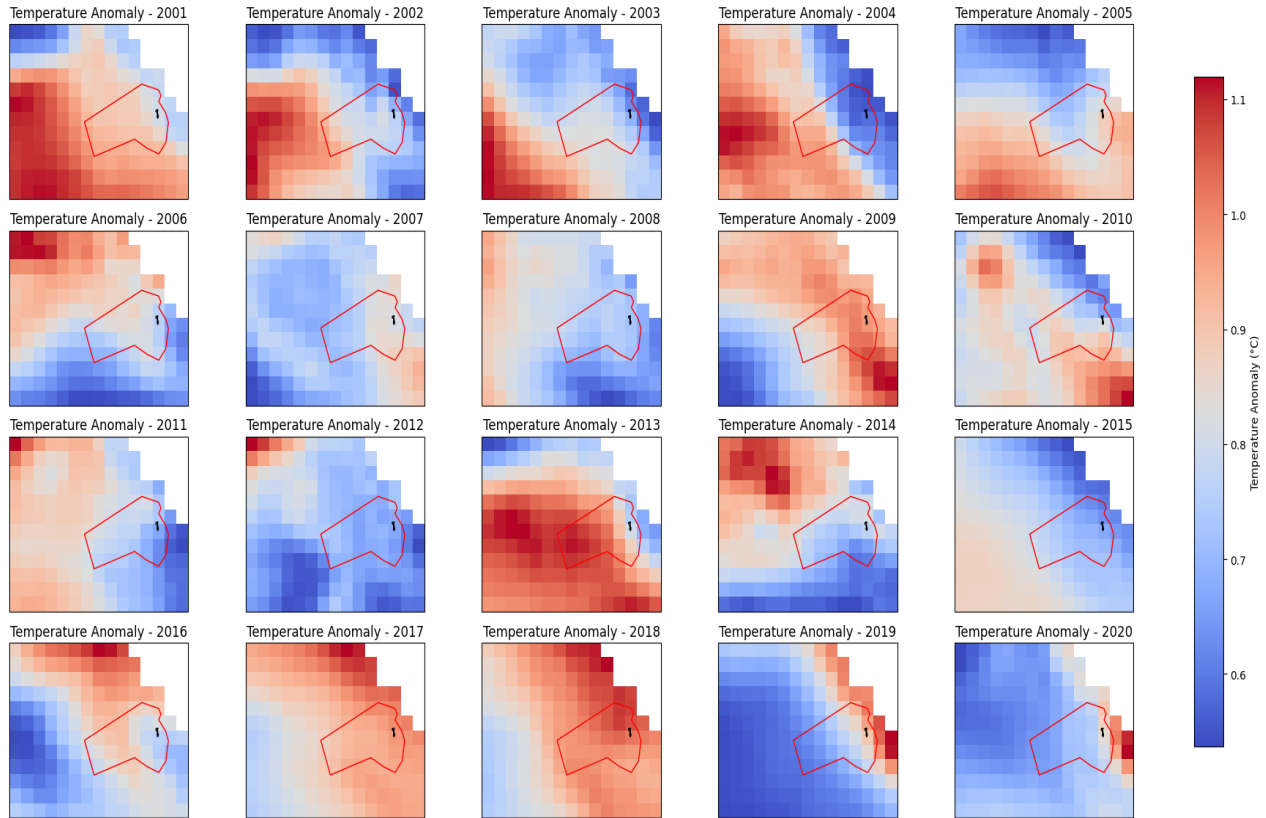


Fig. 11 Spatial anomaly of temperature in monsoon (May - July)

The provided series of maps in fig 12 illustrates the spatial temperature anomalies during the post monsoon season (August to October) from 2001 to 2020 in the region surrounding SMI, Bangladesh, with a particular focus on the MPA. Over the two decades, the data reveal considerable variability in temperature anomalies, with alternating periods of warming and cooling across the region. A notable pattern emerges in the irregular spatial distribution of anomalies, as some years display widespread warming trends, indicated by positive temperature anomalies (red shades), while other years exhibit cooler-than-average conditions (blue shades). Within the MPA, certain regions consistently align with areas of significant temperature deviations, either warmer

or cooler, compared to surrounding waters. This suggests that the MPA might be particularly susceptible to localized climatic and environmental influences.

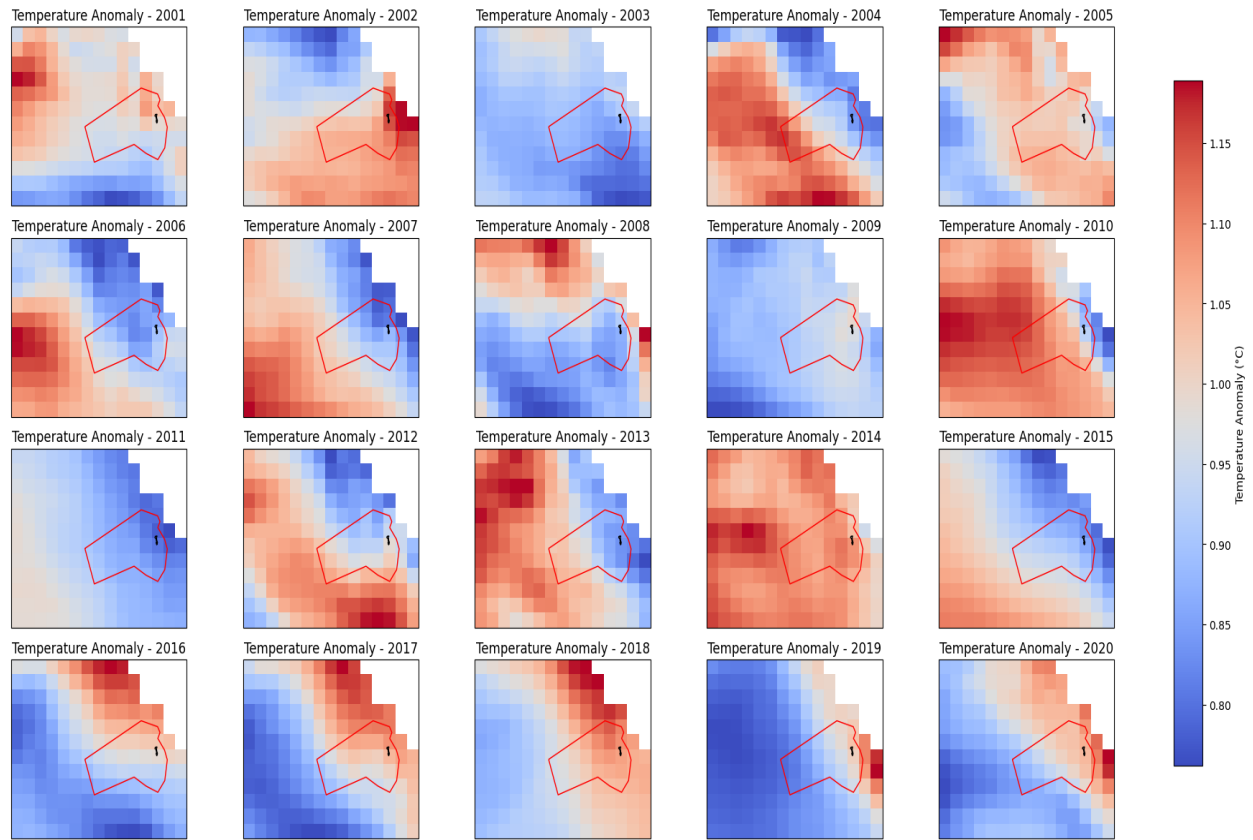


Fig. 12 Spatial anomaly of temperature in post monsoon (Aug - Oct)

The interannual variability in the anomaly patterns highlights the complex interactions of natural and anthropogenic drivers shaping the region's monsoon climate. While some years, such as 2007, 2010, and 2015, are characterized by pronounced warming within and around the MPA, others, such as 2011 and 2020, show a cooling trend. The warming episodes align with broader climate change trends and could be associated with regional factors such as shifts in monsoonal circulation, changes in sea surface temperatures, or localized human impacts. Conversely, the cooler periods may reflect natural variability or transient atmospheric and oceanic phenomena such as cyclonic activity or altered wind patterns.

4.5 Salinity

The region surrounding SMI is experiencing notable climatic shifts, with changing salinity patterns emerging as a key concern. As climate change progresses, the marine environment is becoming increasingly vulnerable to salinity fluctuations, driven by factors such as altered precipitation, ocean currents, and evaporation rates. The salinity in the region is influenced by a complex interplay of natural and anthropogenic factors. Further in-depth research is needed to fully understand the underlying causes of the observed variability and its potential impacts on the marine ecosystem. An analysis of salinity anomalies from 2001 to 2020 reveals seasonal variability and trends, mirroring the impacts of rising temperatures in the area. The MPA within this region is particularly affected, as shifts in salinity, combined with warming temperatures, could disrupt species habitats, ecological dynamics, and the overall stability of this sensitive marine ecosystem.

From 2001 to 2020, the spatial salinity anomalies showed in Fig. 13 around SMI during the winter season (November to January) exhibit notable year-to-year variability but in an irregular drift. Some years like 2001, 2006, 2011, 2012, 2014, 2016, 2018 and 2020 show lower salinity than the average. Whereas 2013 and 2017 shows a very high salinity distribution comparative to other years. The lowest salinity can be found in 2020. 2003, 2004, 2005, 2009, 2010, 2015 shows very high salinity anomalies in the northern and eastern areas, likely driven by reduced freshwater inflow or heightened evaporation rates. Conversely, other years reveal low salinity anomalies in the same regions, potentially caused by increased freshwater input or decreased evaporation. The offshore salinity is higher than the coastal salinity because of less interaction, wave activity, sedimentation, deposition and other natural phenomena.

The winter anomaly might be influenced by seasonal changes in water circulation, temperature, and freshwater input. The anomaly of 2020 could be influenced by natural climate variability, such as El Niño-Southern Oscillation (ENSO) or other oceanic-atmospheric oscillations. Longer-term climate trends, like global warming, could also contribute to the variability of the winter anomaly.

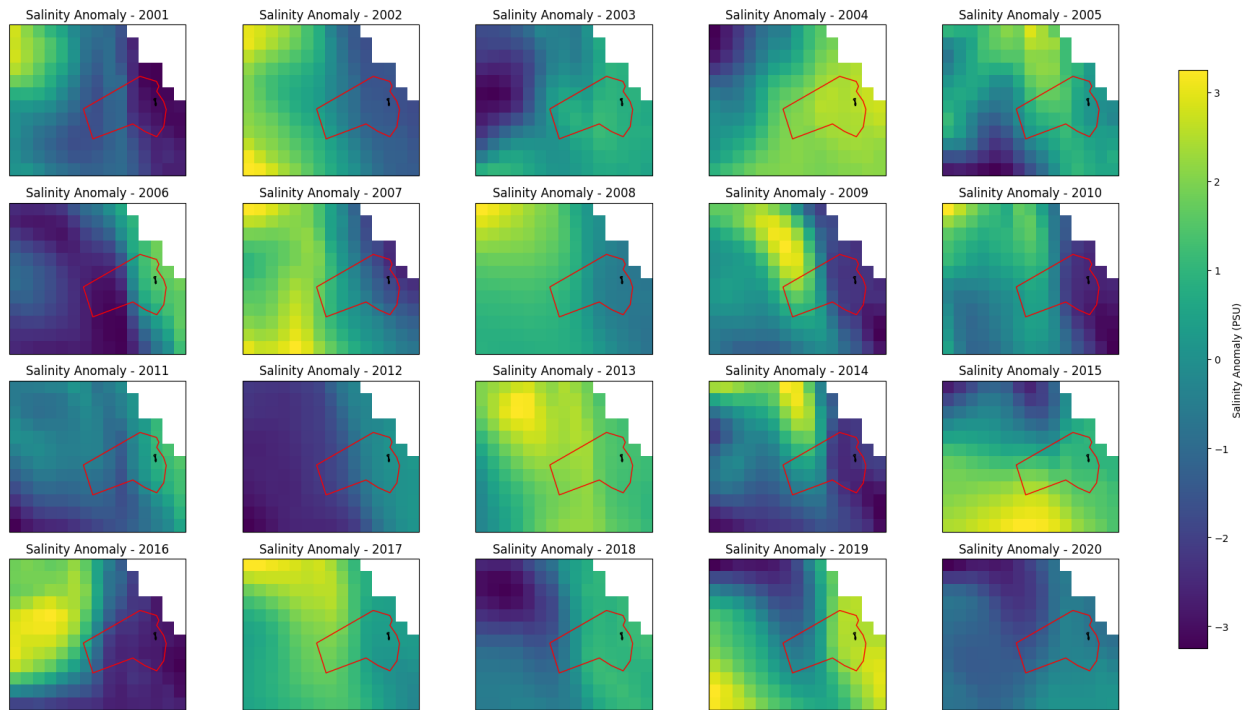


Fig. 13 Spatial anomaly of salinity in winter (Nov-Jan)

Fig. 14 depicts the spatial distribution of salinity anomalies during the pre-monsoon season (February to April) over a 20-year period from 2001 to 2020. The Fig. utilizes a color scale, measured in practical salinity units (PSU), to represent variations in salinity levels across the region. A clear trend is observed, marked by a general decline in salinity anomalies, particularly within the area highlighted in red on the map. This region stands out due to its significant year-to-year fluctuations in salinity. While some years display relatively mild variations, others are characterized by pronounced negative salinity anomalies. These consistent anomalies suggest that the area is undergoing more substantial shifts in salinity levels compared to its surrounding regions. The observed decrease in salinity anomalies may indicate an increased influx of freshwater into the area, potentially driven by factors such as enhanced river discharge, higher precipitation, or localized runoff during the pre-monsoon period.

The concentration of negative anomalies in this specific region underscores its sensitivity to both climatic and environmental changes. Such patterns could reflect broader hydrological shifts or localized impacts, highlighting the importance of continued observation and analysis to understand the underlying drivers and their potential ecological consequences.

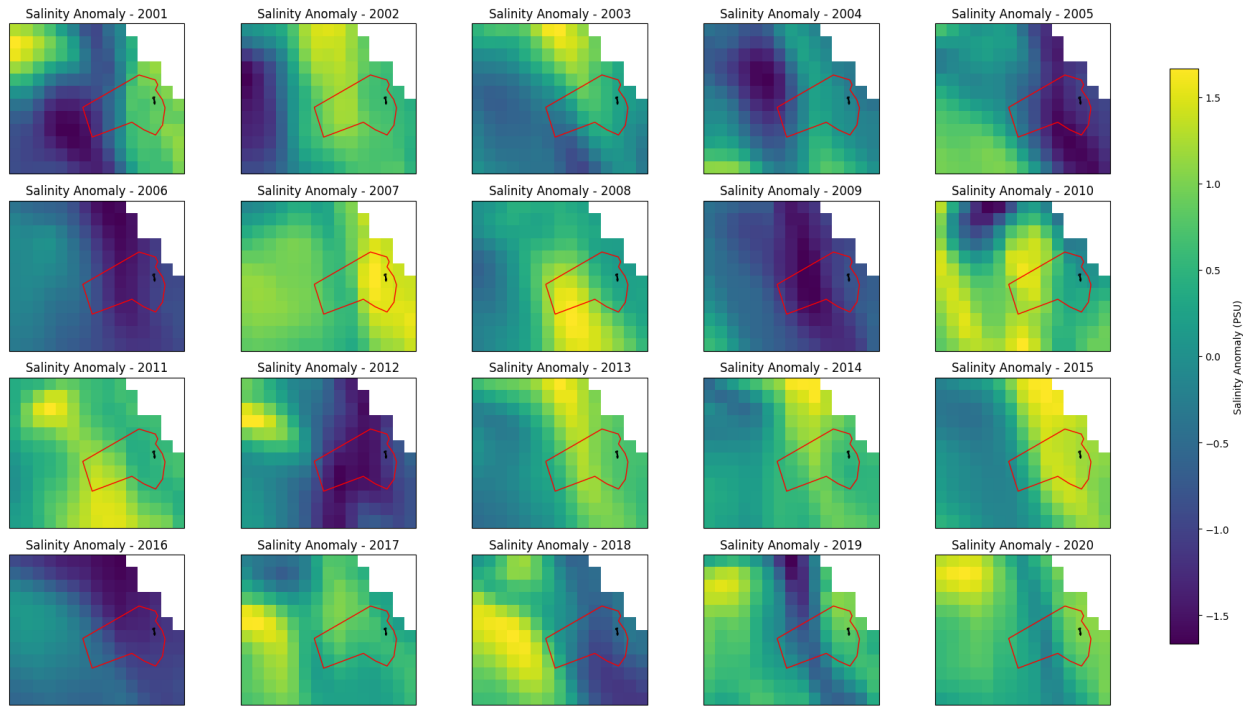


Fig. 14 Spatial anomaly of salinity in premonsoon (Feb-April)

Fig. 15 presents the salinity anomalies in a specific region over a 20-year period, from 2001 to 2020, with each image illustrating the deviation from the average salinity levels for a particular year. The salinity anomaly data reveal a fluctuating pattern over the years, alternating between periods of higher (positive) and lower (negative) anomalies. During the monsoon season (May to July), significant negative anomalies are observed in years like 2001, 2003, 2004, 2007, and 2009. These negative anomalies are indicative of increased freshwater input, which could result from heightened river discharge, intense precipitation, or greater runoff during a particularly strong monsoon. This pattern suggests that these years experienced robust monsoonal activity, which diluted the salinity levels in the region. Notably, the majority of these negative anomalies occurred before 2013, hinting at potential changes in climatic or hydrological patterns in recent years. In contrast, positive salinity anomalies are more prominent during the pre-monsoon and post-monsoon periods. These increases in salinity levels likely result from reduced freshwater input coupled with elevated evaporation rates, creating conditions that favor higher salinity concentrations. This pattern reflects the influence of seasonal climatic factors such as reduced rainfall and increased temperatures during these periods.

Overall, the temporal variability in salinity anomalies highlights the intricate interplay between natural processes such as monsoon dynamics, precipitation, and evaporation, alongside potential anthropogenic influences. This variability underscores the importance of monitoring salinity trends to understand their implications for the region's marine and coastal ecosystems.

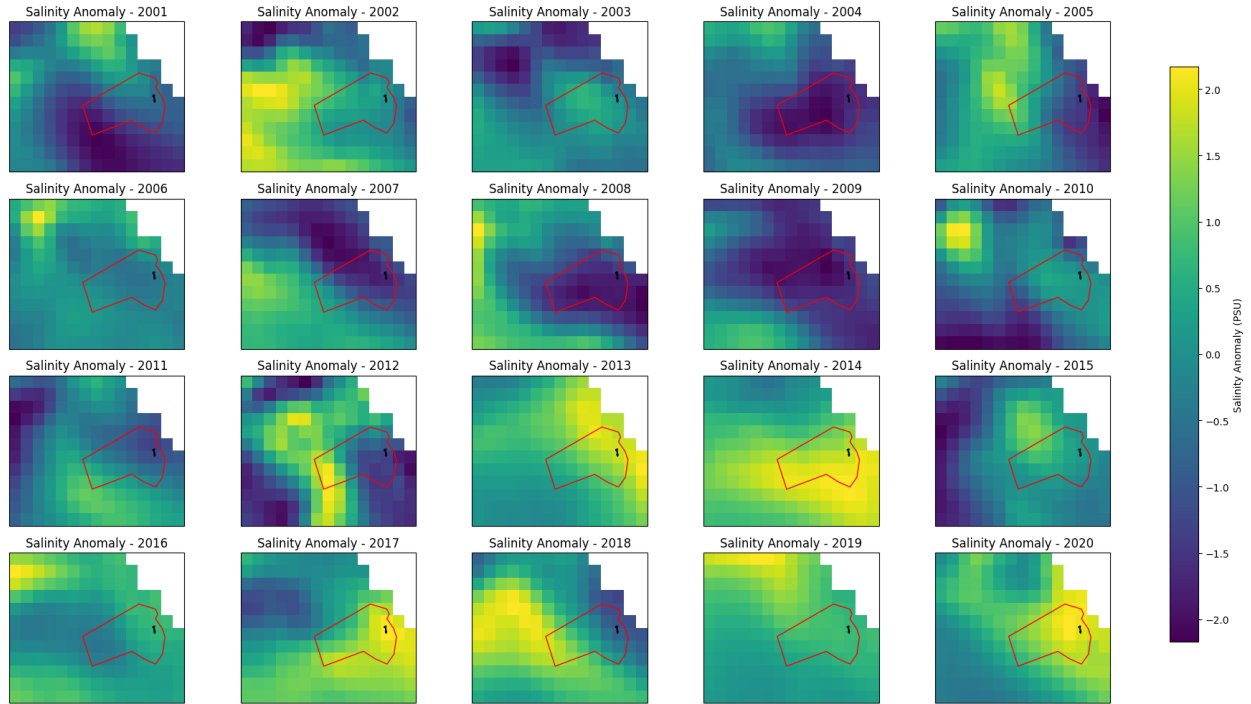


Fig. 15 Spatial anomaly of salinity in monsoon (May-July)

The pre-monsoon season in the region is typically characterized by lower rainfall and higher salinity due to evaporation. The monsoon season brings heavy rainfall, which can dilute the salinity, resulting in a lower salinity anomaly (negative values). Post-monsoon, the salinity can increase again due to reduced rainfall and evaporation, leading to a higher salinity anomaly (positive values).

The Fig. 16 shows the Salinity Anomaly during the post monsoon season (August-October) from 2001 to 2020. The anomaly is concentrated around the area marked by the red contour. The salinity anomaly is generally negative in the marked area from 2001 to 2003. This means that the salinity in this area is lower than the long-term average during these years. Starting from 2004, the anomaly becomes positive, indicating higher salinity than the long-term average. This trend of positive salinity anomaly continues through the following years with some fluctuation. It appears that the salinity anomaly tends to shift towards more positive values during the 2010s. The pre-monsoon

season in the region is typically characterized by lower rainfall and higher salinity due to evaporation. The monsoon season brings heavy rainfall, which can dilute the salinity, resulting in a lower salinity anomaly (negative values). Post-monsoon, the salinity can increase again due to reduced rainfall and evaporation, leading to a higher salinity anomaly (positive values).

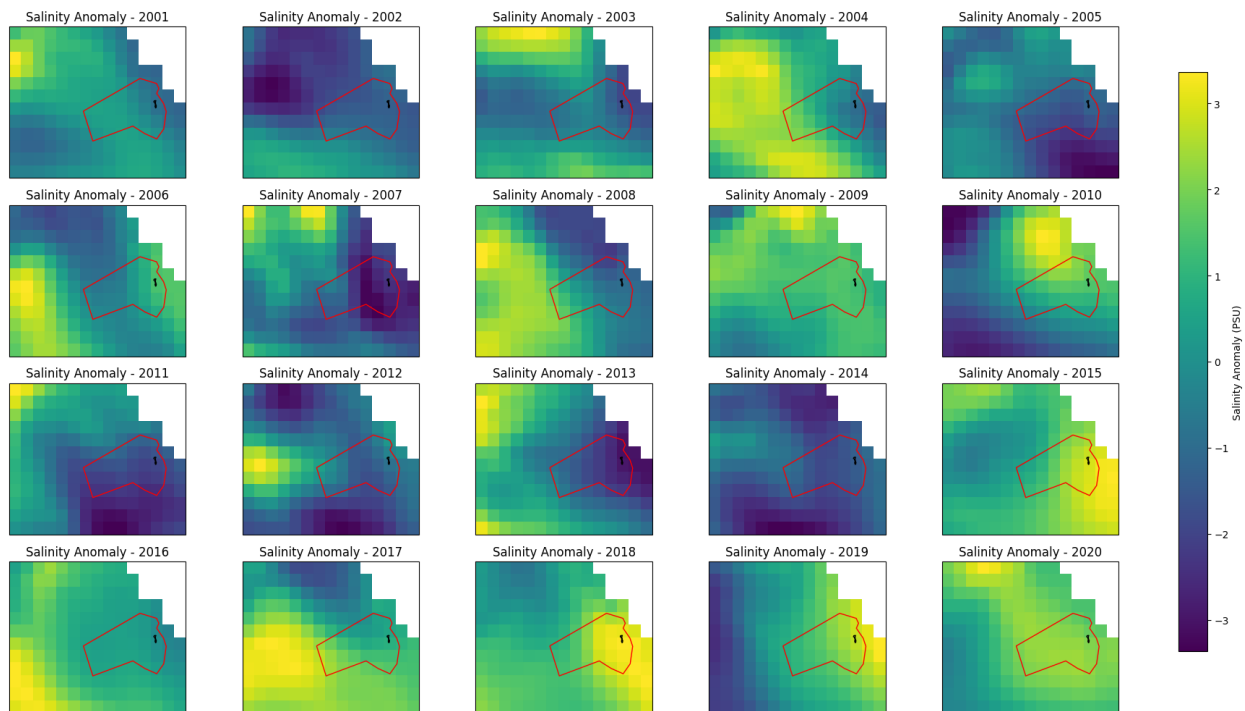


Fig. 16 Spatial anomaly of salinity in post monsoon (August-October)

This variability is influenced by natural climatic factors such as shifts in precipitation, wind patterns, and sea surface temperatures, as well as human activities like land use changes and water resource management. Additionally, oceanographic processes, including upwelling, downwelling, and current dynamics, play a key role in shaping the salinity distribution.

4.6 Water Quality Status

Nutrients Availability

The maps in Fig. 17 show the spatial distribution of four key nutrients: Phosphate, Silicate, Nitrate, and Ammonia. Each nutrient is represented by a color-coded map, with warmer colors indicating higher concentrations and cooler colors indicating lower concentrations.

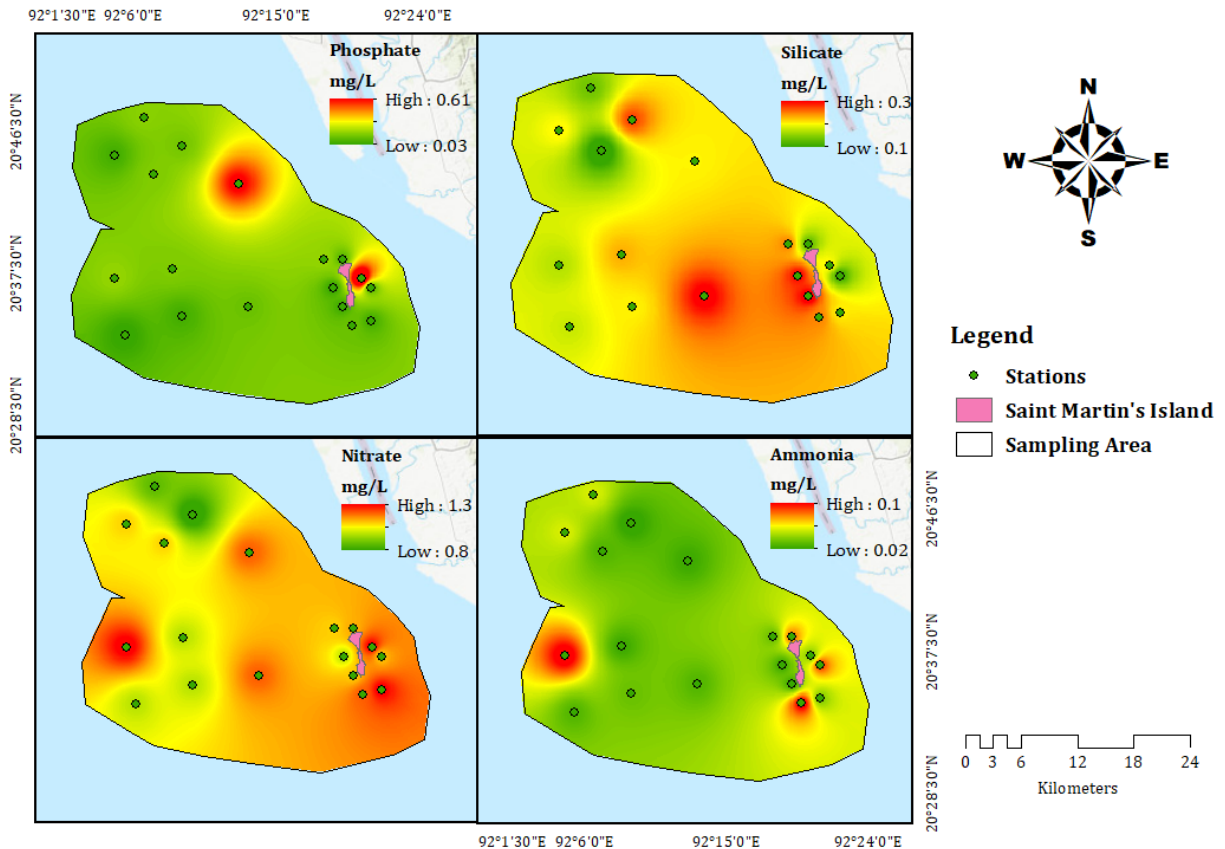


Fig. 17 Nutrient distribution in different stations

Phosphate concentrations appear to be higher in the offshore areas, with a general trend of decreasing concentrations towards the coast. Coastal areas show lower phosphate concentrations, particularly in the northern and eastern parts of the island. Similar to phosphate, silicate concentrations are also higher in the offshore areas, with a gradual decrease towards the coast. Coastal areas exhibit lower silicate concentrations, especially in the northern and eastern regions. Nitrate concentrations are generally higher in the offshore waters. Coastal areas show lower nitrate concentrations, with some localized hotspots of higher concentrations. Coastal areas show lower nitrate concentrations, with some localized hotspots of higher concentrations. Coastal areas exhibit higher ammonia concentrations, particularly in the northern and eastern regions. Overall nutrient concentrations on average from the different years found the sequence $\text{NO}_3 > \text{SiO}_2 > \text{PO}_4 > \text{NH}_3$.

Coastal areas may receive higher nutrient inputs from land-based sources such as rivers, sewage, and agricultural runoff. These inputs can increase the concentration of nutrients like ammonia and nitrate in coastal waters. Offshore areas may benefit from upwelling, a process where cold,

nutrient-rich water from the deep ocean is brought to the surface. This can lead to higher concentrations of nutrients like phosphate and silicate in these areas. Phytoplankton uptake and biological processes can influence the distribution of nutrients. Areas with higher phytoplankton activity may have lower nutrient concentrations due to uptake by organisms. Areas with higher nutrient concentrations, particularly in the offshore waters, may support higher primary productivity by fueling phytoplankton growth. High nutrient concentrations in coastal areas, especially ammonia and nitrate, can lead to eutrophication, which can result in algal blooms and oxygen depletion. Nutrient availability can influence the distribution and abundance of fish populations. Areas with higher nutrient concentrations may support higher fish populations.

Physiochemical Parameters

The provided map in Fig. 18 offers a comprehensive view of the physiochemical parameters of SMI, revealing distinct patterns between coastal and offshore regions. There five parameters which are salinity, temperature, Total Dissolve Solids, Dissolve Oxygen and pH.

Salinity is generally higher in the coastal areas, with values ranging from 29.94 to 32.39 PSU. This suggests a stronger influence of seawater in these regions. Temperatures also tend to be slightly higher, ranging from 23.71 to 27°C. In contrast, salinity is lower offshore, with values ranging from 0.02 to 25 PSU. This indicates a greater influence of freshwater sources like rivers or groundwater. Temperatures are also lower, ranging from 20.67 to 23.71°C. In contrast, salinity is lower offshore, with values ranging from 0.02 to 25 PSU. This indicates a greater influence of freshwater sources like rivers or groundwater. Temperatures are also lower, ranging from 20.67 to 23.71°C. DO levels are generally higher in the coastal areas, with values ranging from 6.10 to 10.03 mg/L. This suggests better oxygenation, possibly due to wave action and mixing with the atmosphere. DO levels are lower offshore, with values ranging from 3.45 to 7.95 mg/L. This could be attributed to lower oxygen exchange with the atmosphere and potentially higher organic matter decomposition. pH values are slightly higher in the coastal areas, ranging from 7.06 to 8.22. This indicates a more alkaline environment. pH values are lower offshore, ranging from 6.95 to 7.46, suggesting a more acidic environment. TDS levels are higher in the coastal areas, ranging from 16.23 to 24.77 ppt. This suggests a higher concentration of dissolved salts and minerals. TDS

levels are lower offshore, ranging from 3.13 to 23.73 ppt, indicating a lower concentration of dissolved substances.

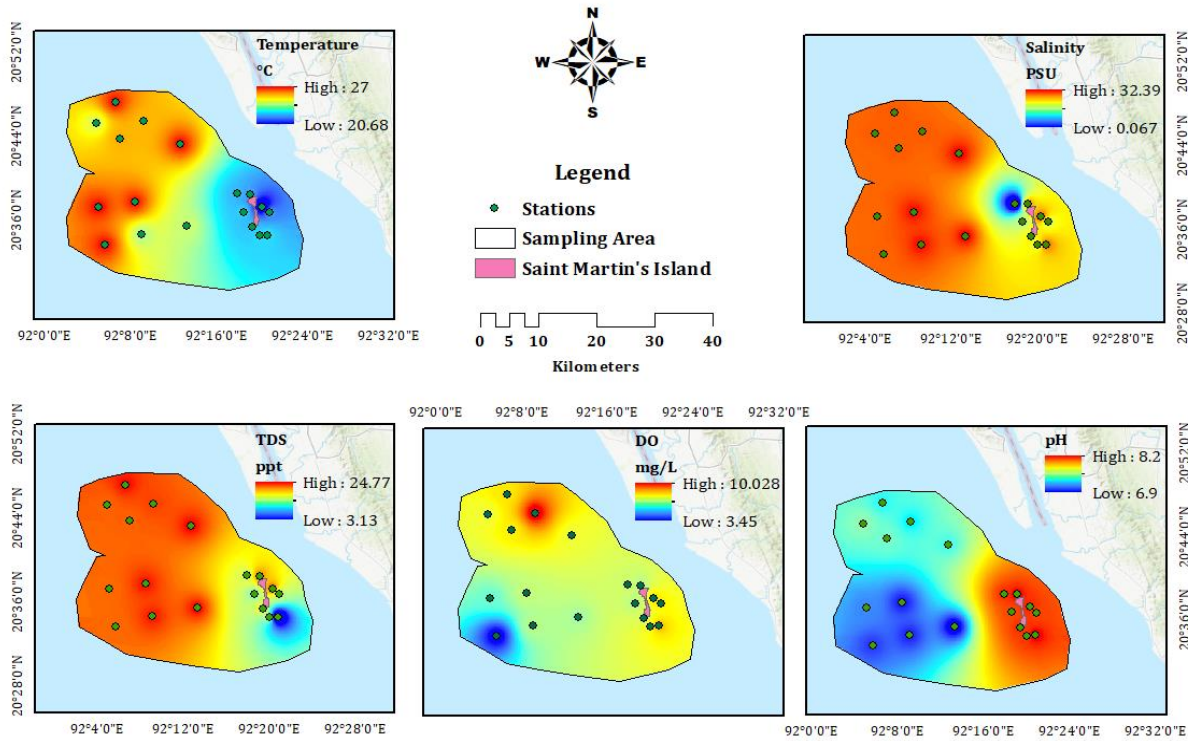


Fig. 18 Spatial distribution of physiochemical parameters

Overall, the physiochemical parameters exhibit distinct spatial variations between the coastal and offshore regions of SMI. These variations can be attributed to factors such as water depth, proximity to land, freshwater inputs, and biological activity. Following table 9 represents the statistical analysis of physiochemical parameters. The scree plot in Fig. 19 shows that a few principal components are sufficient to capture most of the variance in the original data. This suggests that PCA is an effective technique for dimensionality reduction in this dataset.

Table 9 Statistical analysis of physiochemical parameters

Stations	Mean	Max	Min	Median	Mode	SD	SE
DO (mg/L)	7.184	10.03	3.45	7.22	6.63	1.282	0.302
pH	7.672	8.22	6.95	7.54	8.11	0.473	0.1114
Salinity (PSU)	27.331	14.405	0.0228	30.04	30.17	7.632	1.799
Temperature (°C)	24.818	27.00	20.67	24.74	27.00	1.807	0.426
TDS (ppt)	21.08	24.77	3.09	23.16	-	5.29	1.247

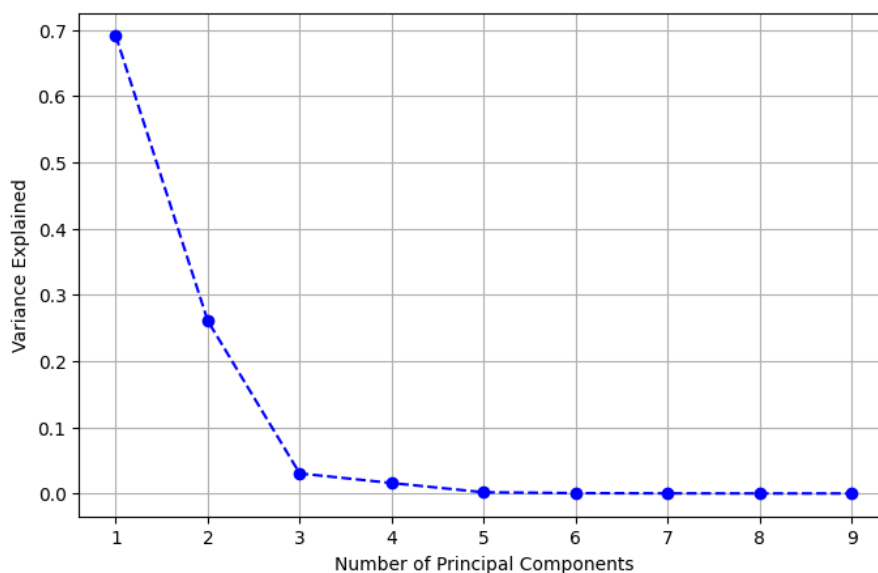


Fig. 19 Explained variance by principal components

The graph in Fig. 19 illustrates the outcomes of a Principal Component Analysis (PCA) applied to physiochemical data, effectively reducing its dimensionality while retaining the most significant information. This analysis condensed multiple original variables into two principal components: PC1 and PC2. PC1, displayed along the horizontal axis, accounts for 34.89% of the variance, while PC2, on the vertical axis, explains 23.23% of the variance. Each point on the plot represents a sampling station, and their positions reflect their relative physiochemical characteristics in the reduced multidimensional space.

The distribution of stations reveals considerable variability in their physiochemical profiles. Stations located close to each other, such as St 01 and St 02, exhibit similar water quality parameters, while those farther apart, like St 03 and St 11, demonstrate distinct differences. Some degree of clustering is visible, indicating shared characteristics among specific groups of stations, but overall, the stations are widely spread, reflecting diverse environmental conditions. The spatial arrangement of stations on the plot may be influenced by environmental factors, such as hydrological conditions, proximity to pollution sources, or local ecological variations. Further analysis could enhance the interpretation of the PCA results. For instance, examining variable loadings on the principal components would identify which physiochemical parameters contribute most significantly to the observed variance. Cluster analysis could further group stations with similar profiles, while spatial mapping could link these groupings to geographic or environmental patterns.

In summary, the PCA plot in Fig 20 offers valuable insights into the relationships between stations based on their physiochemical characteristics, highlighting similarities, differences, and potential spatial trends. This information is crucial for understanding water quality variability and identifying areas that may require targeted conservation or management efforts.

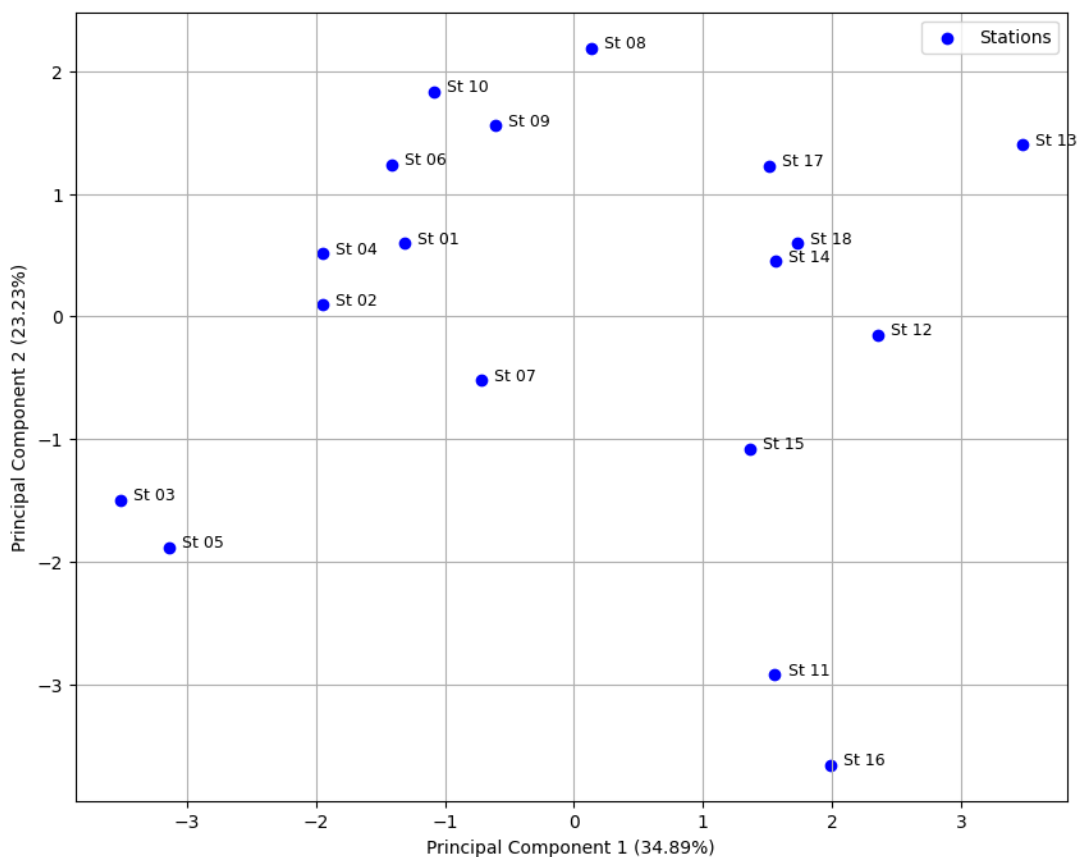


Fig. 20 PCA plot for physiochemical parameters

4.6 Heavy Metal Pollution

Heavy Metals in Water

Heavy metal contamination in marine environments poses significant risks to ecosystems, human health, and biodiversity. The spatial maps in Fig. 10 illustrate the concentrations of six heavy metals—Cr, Fe, Zn, Cu, Mn, Ni, Hg, As, Pb, and Cd—in the water around SMI, Bangladesh. Each map shows the spatial distribution of the metal concentrations in ppb (Parts Per Billion), with contour lines and a color gradient representing levels from low to high. The island is represented by a pink polygon. Green dots indicate sampling stations where water samples were collected.

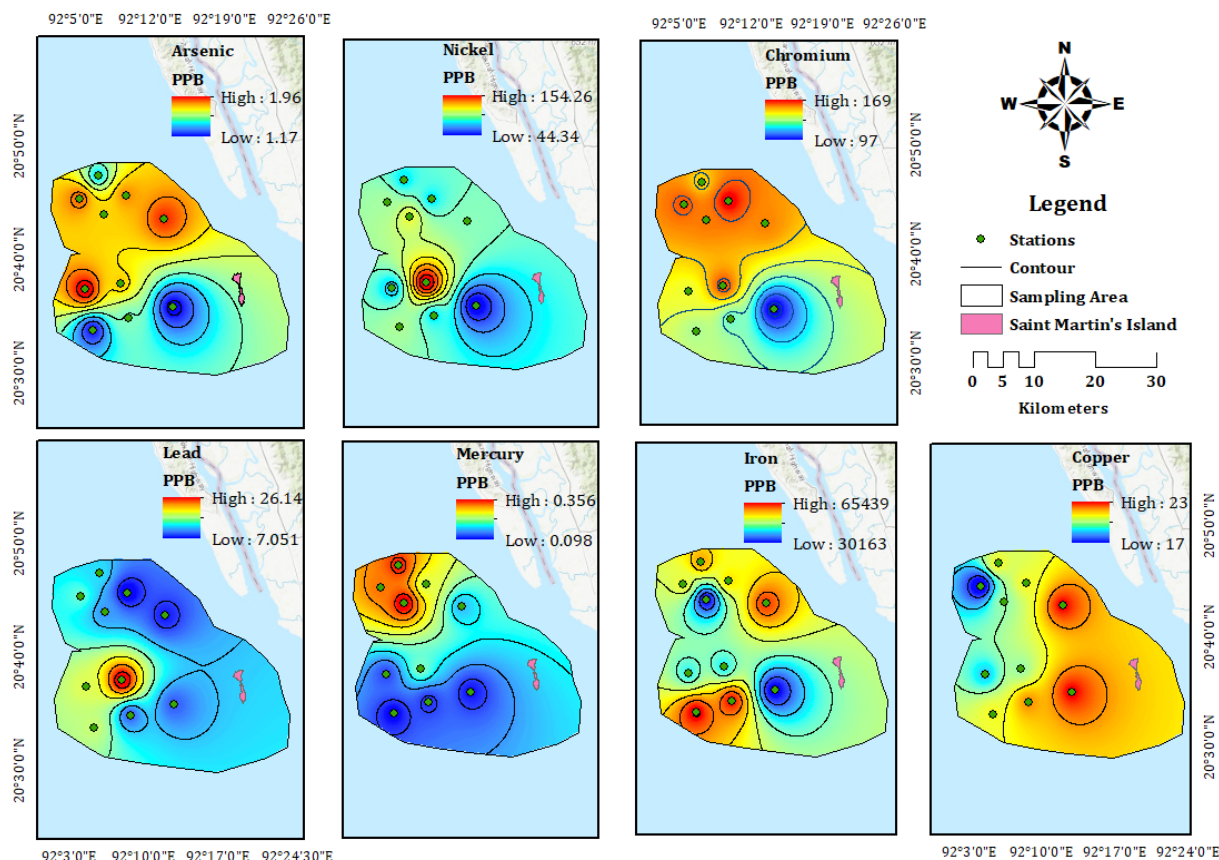


Fig. 21 Spatial distribution of heavy metal concentration in water

As concentrations are higher in the northern and central waters around the island, ranging from 1.17006 to 1.95991 ppb, while lower levels are found in the southeastern areas. Ni shows peak concentrations in the central regions mainly in station 5, with values decreasing outward, ranging from 44.3403 to 154.255 ppb. Cr levels are elevated in the northwestern waters, tapering off toward the east, with concentrations between 97.0062 and 168.998 ppb. Pb is more concentrated in the western areas station 5, with lower levels observed in the east, ranging from 7.05067 to 26.1359 ppb. Hg levels are high near the northern boundary, with a gradual decrease toward the south and ranges between 0.0983704 and 0.356532 ppb. Fe displays the highest concentrations in the western regions, with the lowest in the southeastern areas, ranging from 30163.1 to 65439.3 ppb. Copper concentrations range from 17.002 to 22.9997 ppb, with elevated levels observed in the southern and eastern areas. We also analyzed the Zn, Mn and Cd but these elements don't show a lot variation in spatial distribution. The highest concentration of Zn and Mn are 10 ppb and 130 ppb successively found in station 2. Cd shows 1 ppb constantly all the 10 stations.

The table 10 summarizes key descriptive statistics for the concentrations of various heavy metals in water samples from SMI. Fe has the highest concentrations by far, with significant variability. Ni and Cr also have relatively high concentrations and variability. Cu shows notable variability, likely due to a wide range in values. Zn and Mn have the lowest variability, indicating more uniform concentrations. As, Pb and Hg have moderate concentrations and variability in their distribution across samples. Also, the standard value of the element suggested by Department of Public Health, Bangladesh help us to evaluate the comparative analysis of the concentration of each element in the water. Here appears that the concentration of Cr and Fe exceeds the suggested value.

Table 10 Statistical condition of heavy metal concentration (ppb) in water

Element	Mean	Std	Min	Max	Median	Mode	SD	SE
As	1.622	50	1.17	1.96	1.723	1.17	0.281	0.089
Pb	13.321	50	7.05	26.15	11.955	26.15	5.671	1.793
Hg	0.205	1	0.0983	0.357	0.183	0.09835	0.102	0.0321
Ni	89.211	100	44.33	154.33	84	44.33	28.293	8.947
Cr	145.6	50	97	169	153	97	21.829	6.903
Fe	50118	300-1000	30160	65445	52989	30160	13278.86	4199.15
Mn	20.7	100	17	23	21	23	1.829	0.578
Cu	17.5	1000	5	130	5	5	39.528	12.5
Zn	8.2	5000	8	10	8	8	0.632	0.2

Heavy Metals in Sediment

The spatial map of heavy metal concentration in the sediment represented by Fig. 11 and 12 of SMI of Bangladesh shows that the highest concentrations of Cr, Fe, Zn, Cu, Mn, Ni, Hg, As, Pb, and Cd are found in the central and western parts of the island.

Cr concentrations in the sampling area range from approximately 14.13 to 73.67 ppm, with higher levels observed in the northern and central regions. In marine environments, Cr sources typically include industrial discharges, particularly from metal processing, electroplating, and leather tanning industries. Fe concentrations range from about 30.16 to 65.44 ppm, with elevated levels found mainly along the island's periphery. Natural sources of Fe include the weathering of rocks

and soils, while anthropogenic inputs come from industrial effluents, especially from steel manufacturing and other metal industries.

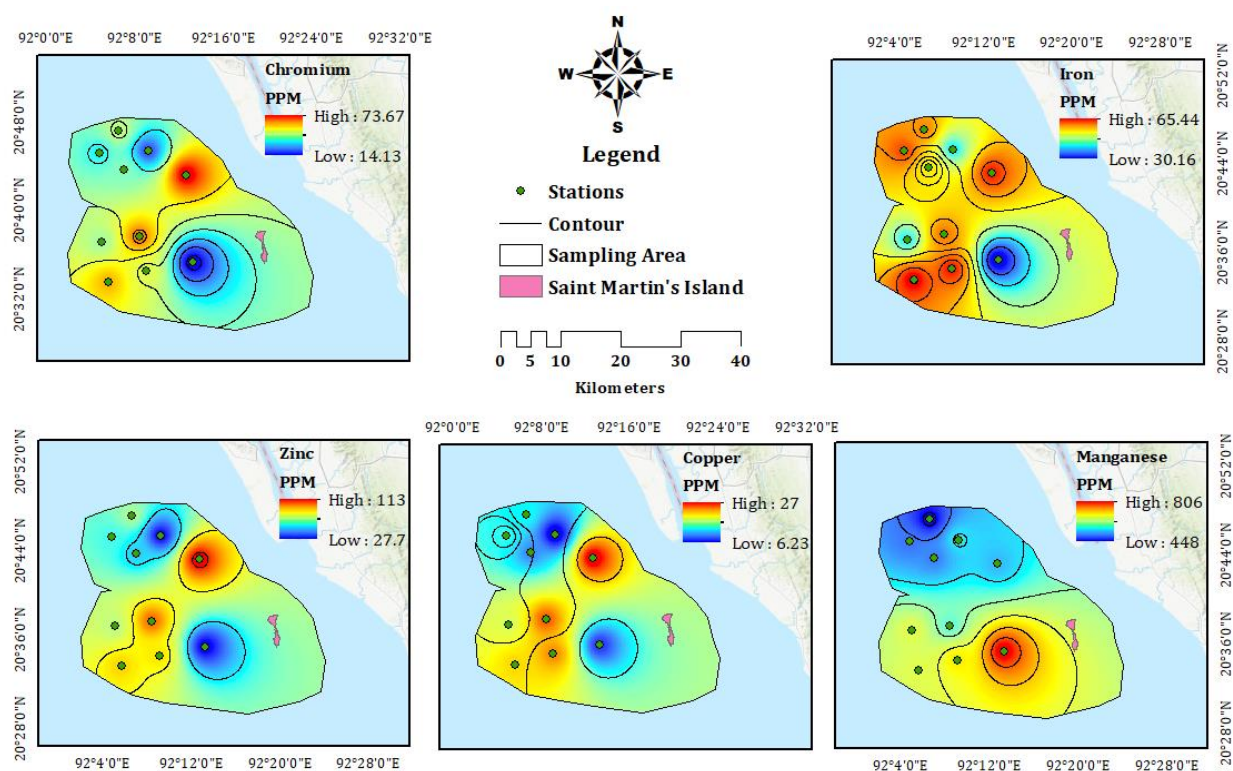


Fig. 22 Spatial distribution of heavy metals (Cr, Fe, Zn, Cu, Mn) concentration in sediment

Zn concentrations vary between 27.7 and 113 ppm, with high levels detected in both the northern and central regions, similar to Cr. Zn contamination in marine sediments often originates from galvanization, urban runoff, sewage sludge, and tire wear. Cu concentrations range from approximately 6.23 to 27 ppm, with the highest levels concentrated in the central areas. Cu in marine environments can stem from antifouling paints on ships, agricultural runoff, mining, and metal plating industries. Mn concentrations span from around 448 to 806 ppm, with elevated concentrations scattered throughout the sampling area. While Mn in marine sediments is mainly natural, coming from the weathering of Mn-bearing rocks, anthropogenic contributions can arise from its use in industrial processes.

Ni concentrations in the sampling area range from approximately 32.255 to 143.39 ppm, with higher levels found in the northern and central regions. Ni in marine environments can originate from natural sources like rock weathering, as well as from anthropogenic sources such as industrial discharges, particularly from metal refineries and fossil fuel combustion. Hg concentrations vary

between around 0.0292071 and 0.0957758 ppm, with localized areas of higher concentration in the northern part. Hg often enters marine ecosystems through mining activities, industrial discharges, and atmospheric deposition, and is notably hazardous due to its tendency to bioaccumulate in the food chain.

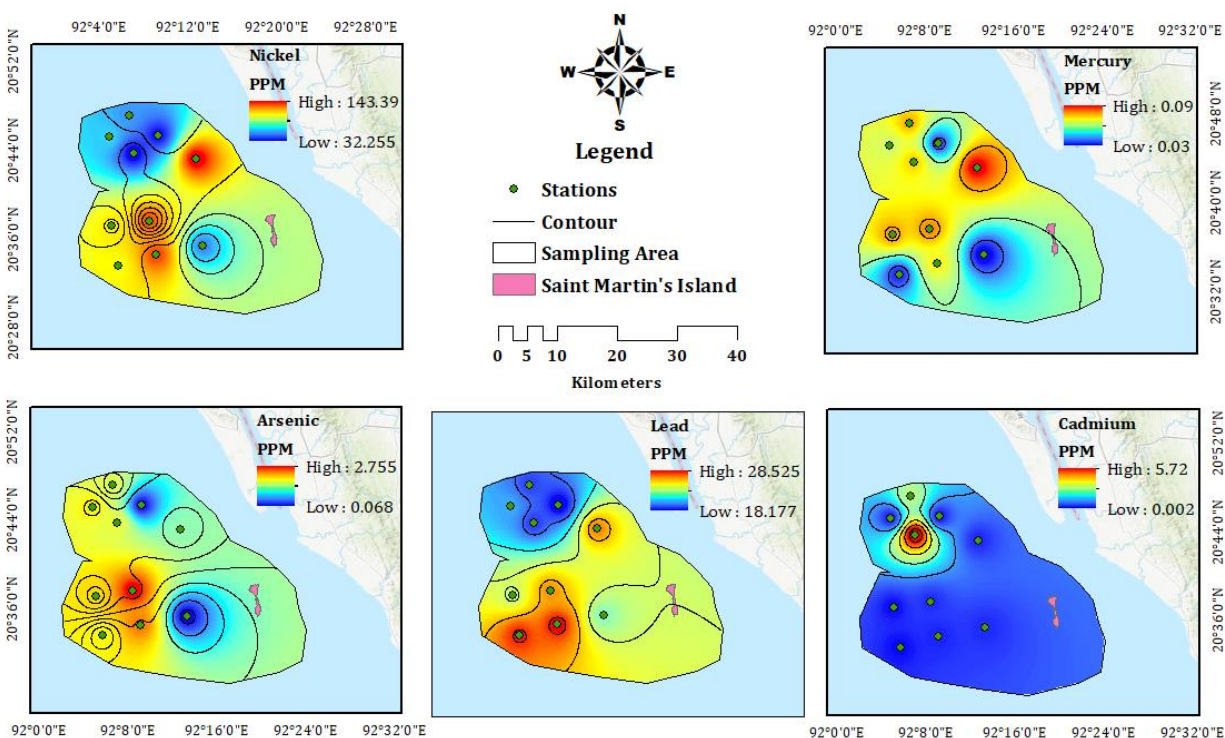


Fig. 23 Spatial distribution of heavy metals (Ni, Hg, As, Pb, Cd) in sediments

As concentrations range from about 0.0678195 to 2.755 ppm, with the highest levels occurring in the central and southern regions. As can enter marine environments through natural mineral deposits or through human activities such as agricultural runoff, industrial waste, and mining. Pb concentrations range from approximately 18.1774 to 28.5249 ppm, with higher levels concentrated in the central areas. Pb contamination is often linked to industrial processes, vehicle emissions, and past use of leaded gasoline, as well as from products like paint, batteries, and other metal industries. Cd concentrations range from about 0.0024907 to 5.72428 ppm, with higher levels found in the northern regions, while other areas exhibit lower concentrations. Cd in marine sediments commonly arises from industrial discharges, phosphate fertilizers, and metal plating industries.

This table 11 shows the statistical condition of heavy metal concentration in sediment. The table contains the average, standard deviation, minimum, maximum, median, mode, standard deviation, and standard error for each metal. The metals included are As, lead, Cd, Hg, Ni, Cr, iron, Mn, copper, and zinc. The standard deviation and standard error are given in milligrams per liter (mg/L). The maximum permissible limit (MPL) (Aremu et al, 2011) by WHO is given to evaluate the comparative analysis of the concentration of heavy metals in sediment.

Table 11 Statistical condition of heavy metal concentration (PPM) in sediment

Element	Average	MPL	Min	Max	Median	SD	SE
As	1.533	Nil	0.0675	2.756	1.7315	0.86	0.272
Pb	23.438	85	18.175	28.525	23.3	3.932	1.243
Cd	0.89	0.05	0.002	5.725	0.137	1.868	0.59
Hg	0.065	Nil	0.0292	0.096	0.0749	0.024	0.0076
Ni	86.247	0.05	32.25	143.4	84.58	41.742	13.2
Cr	44.304	0.01	14.125	73.68	43.795	18.422	5.826
Fe	54.564	0.3	30.16	65.44	59.9	12.11	3.83
Mn	598.85	0.01	448	806	577	105.77	33.449
Cu	16.61	1	6.2	26.86	16.1	7.007	2.216
Zn	68.869	5	27.7	112.94	69.4	28.44	8.994

4.4 Phytoplankton Abundance

The provided Fig. 13 illustrates the spatial distribution and percentage composition of phytoplankton abundance across different sampling stations, highlighting significant variations in their distribution patterns.

The highest abundance of phytoplankton was recorded at Station 2, with a remarkable value of 67,082.57, making it a dominant hotspot within the study area. In contrast, the lowest abundance was observed at Station 4, with a minimal value of 2,444.07, reflecting its limited contribution to the overall phytoplankton population. This distinct difference underscores a spatial gradient in phytoplankton abundance across the study area, where the central region exhibits higher abundance values compared to the peripheral zones. Such a gradient could indicate the influence of localized environmental factors, such as nutrient availability, water currents, or light penetration, which might favor phytoplankton growth in the central area.

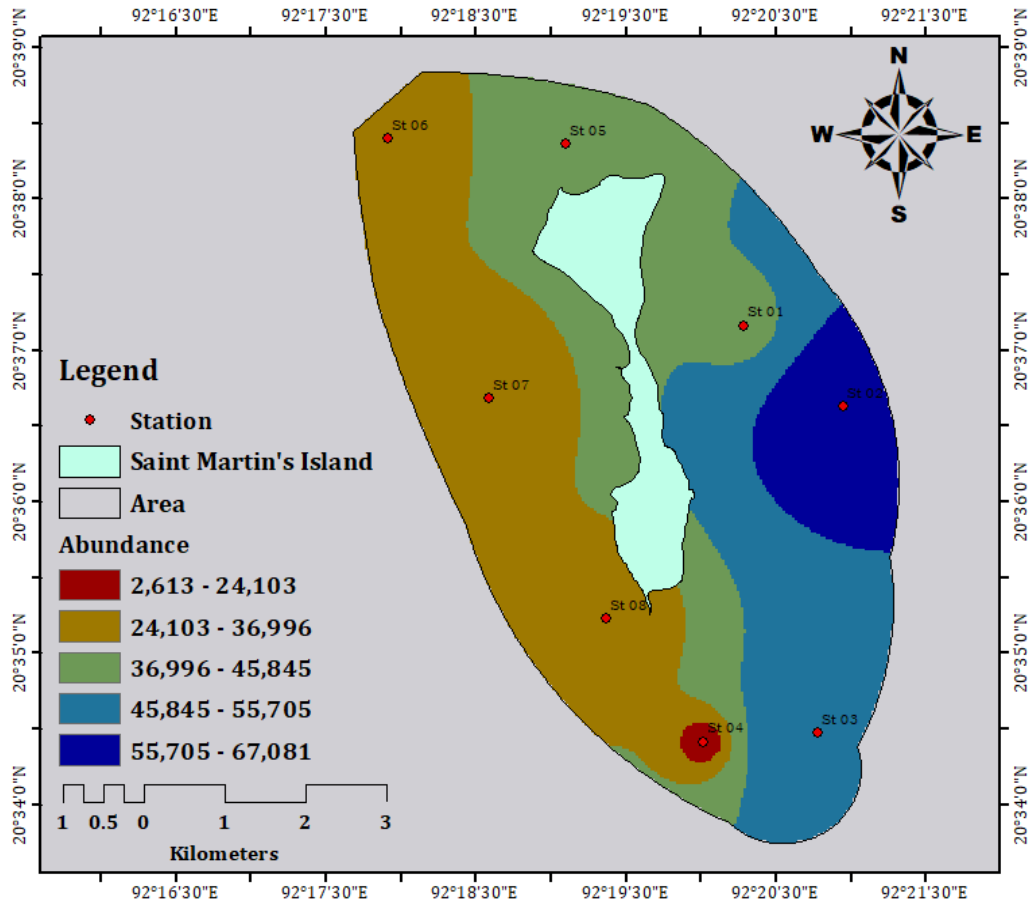


Fig. 24 Spatial distribution of phytoplankton

To further enhance the spatial analysis, Fig. 14 includes a pie chart that illustrates the relative contributions of each station to the overall phytoplankton abundance in the study area. This chart provides a clear visualization of how the phytoplankton population is distributed across the different stations, offering valuable insights into the dominance and ecological significance of each site.

Among the stations, **Station 2** stands out as the most influential contributor, accounting for **23.5%** of the total phytoplankton abundance. This makes Station 2 the most important station in terms of supporting and sustaining the local phytoplankton community, highlighting its ecological significance within the study region. Following Station 2, **Station 3** contributes **18%** of the total phytoplankton population, positioning it as the second-largest contributor. This underscores the station's considerable role in maintaining the local aquatic ecosystem, suggesting that it supports a healthy and vibrant phytoplankton population as well. On the other hand, **Station 4** contributes

a much smaller portion, with only **0.9%** of the total phytoplankton abundance. This makes Station 4 the least significant in terms of phytoplankton population, indicating that its ecological contribution to the phytoplankton community is minimal compared to the other stations. This disparity in the relative contributions of each station suggests varying environmental conditions or factors at play that influence the abundance and distribution of phytoplankton across the study area.

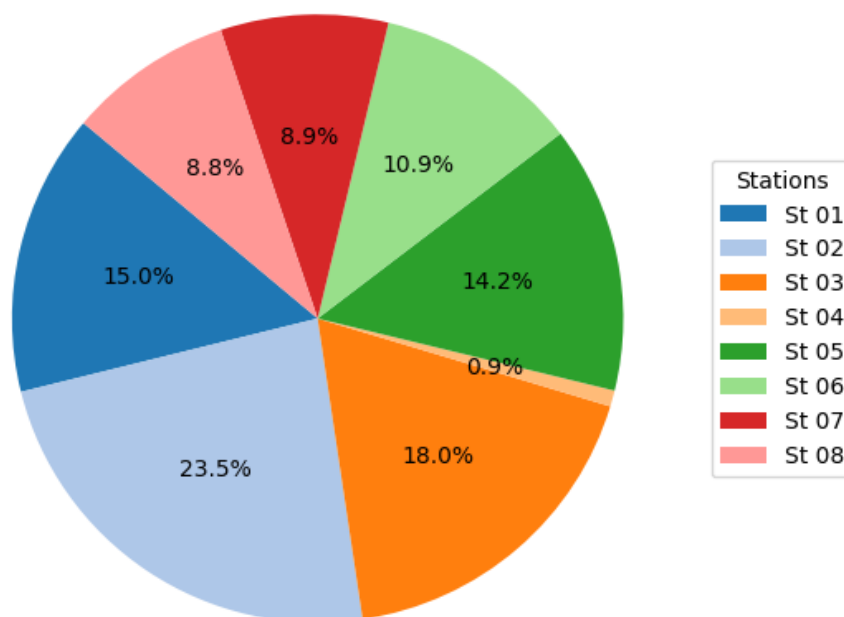


Fig. 25 Percentage of phytoplankton abundance

Overall, the pie chart not only provides a numerical breakdown of phytoplankton distribution but also helps identify key stations that play pivotal roles in the local marine ecosystem, offering deeper insights into the factors that drive phytoplankton abundance in this region.

4.5 Microplastic Abundance

Microplastic pollution poses significant threats to marine ecosystems, impacting marine life, disrupting food webs, and potentially affecting human health. Fiber-based microplastics, in particular, can harm marine organisms through ingestion and entanglement, and may also facilitate the transfer of toxic pollutants. Microplastic has been found of three types which are fiber, foam and fragment in seven colors like black, blue, brown, green, grey, red, and transparent. The overall abundance of Microplastic found in the study area is 2155 MPs/L.

$$Abundance = \frac{215.5 * 1000}{100} = 2155 (\pm 927.437 \text{ SD})$$

The pie chart in figure15 illustrates the distribution of microplastic abundance by color and types in the samples collected from SMI.

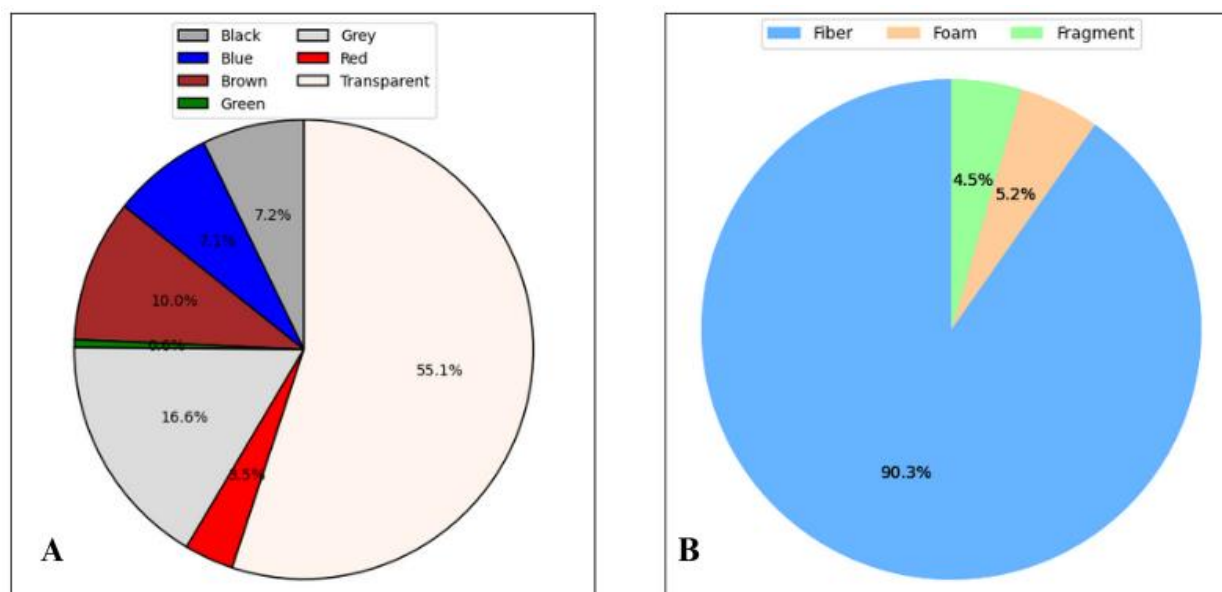


Fig. 26 Microplastic found by A) Color, B) Type

Transparent microplastics account for the largest proportion, constituting 55.1% of the total microplastic abundance. This suggests that transparent plastics, often derived from packaging materials, significantly contribute to the area's microplastic pollution. Other colors, such as black, blue, grey, red, brown, and green, contribute to varying degrees. Black and grey microplastics are relatively abundant, accounting for 10.0% and 7.2%, respectively. This indicates the presence of microplastics from various sources, including industrial activities, urban runoff, and fishing gear. Green, brown, and red microplastics have a relatively low abundance, contributing less than 10% each. This suggests that these colors may be less prevalent in the local sources of microplastic pollution. Fibers constitute the majority of microplastics, accounting for 90.3% of the total abundance. This suggests that fibers, often derived from textiles and fishing gear, significantly contribute to the area's microplastic pollution. Foam and fragments contribute relatively smaller proportions, at 5.2% and 4.5%, respectively. This indicates that while fibers are the dominant type, other forms of microplastics are also present in the environment. The pie chart reveals that microplastic pollution in SMI is dominated by fibers. This finding highlights the importance of

addressing sources of fiber pollution, such as textile waste and fishing activities, to mitigate the impact of microplastics on marine ecosystems.

The pie chart in figure16 illustrates the distribution of microplastic abundance among different stations (St 01 to St 10). Each slice of the pie represents a station, and its size corresponds to the percentage of microplastics found at that station. Station St 05 and St 09 have the highest microplastic abundance, each accounting for 14.5% of the total. Station St 06 has the lowest microplastic abundance, with only 2.5%. The remaining stations have moderate levels of microplastic abundance, ranging from 5% to 13.2%.

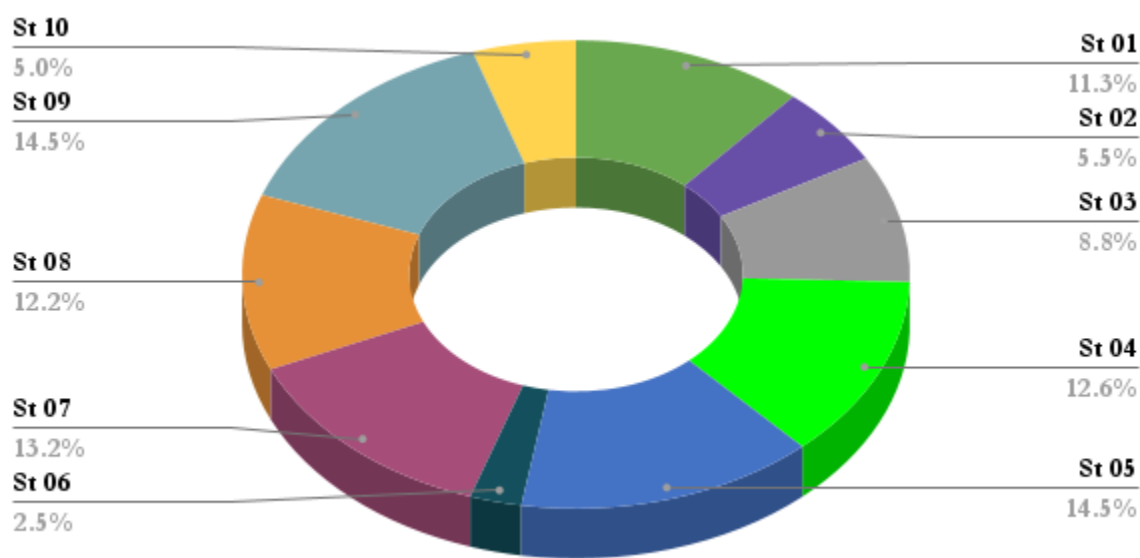


Fig. 27 Percentage of microplastic abundance

The spatial distribution map in Fig. 26 of microplastic abundance shows a clear pattern of variation across the sampled stations. The abundance of microplastics varies significantly between different stations. This suggests that factors such as water currents, proximity to pollution sources, and local environmental conditions play a crucial role in influencing the distribution of microplastics.

Stations 02, 07, and 09 exhibit higher microplastic abundance, indicated by the darker shades of blue. The highest abundance value is 3120 which can be shown in station 9. These areas may be more susceptible to microplastic pollution due to various factors, such as proximity to urban centers, industrial activities, or shipping lanes. Stations 06 and 04 show lower microplastic abundance, represented by the lighter shades of yellow. The lowest number of abundances is 540 which can be found in station 6. These areas might be relatively less impacted by microplastic

pollution, possibly due to their distance from pollution sources or the influence of natural processes like sedimentation. The spatial distribution map highlights the uneven distribution of microplastic pollution in the study area. This information is crucial for understanding the sources and pathways of microplastic pollution, as well as for developing targeted strategies for monitoring, mitigation, and remediation efforts.

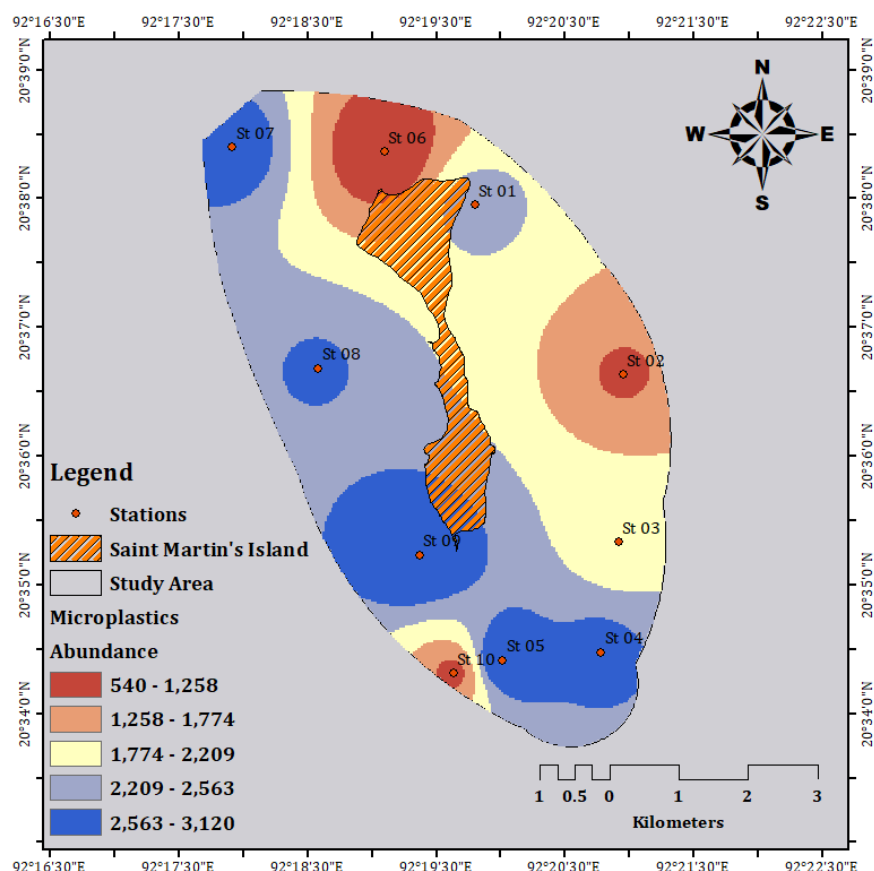


Fig. 28 Spatial distribution of microplastic

The graph provides in Fig. 27 information about the average size of different types of microplastics (fiber, foam, and fragment) across ten stations. The graph highlights that fiber microplastics in the study area tend to be larger compared to foam and fragment microplastics. This information can be useful for understanding the potential sources and environmental impacts of different microplastic types. Larger microplastics may pose greater risks to marine organisms due to their potential for ingestion and entanglement.

Fibers consistently exhibit the largest average size across all stations, with values ranging from approximately 0.2 to 0.8 mm. Foam and fragment microplastics generally have smaller average

sizes compared to fibers. Their sizes fluctuate between 0.1 and 0.3 mm. While fiber size shows a considerable variation across stations, with some stations exhibiting larger average sizes than others, the variation in foam and fragment sizes is less pronounced.

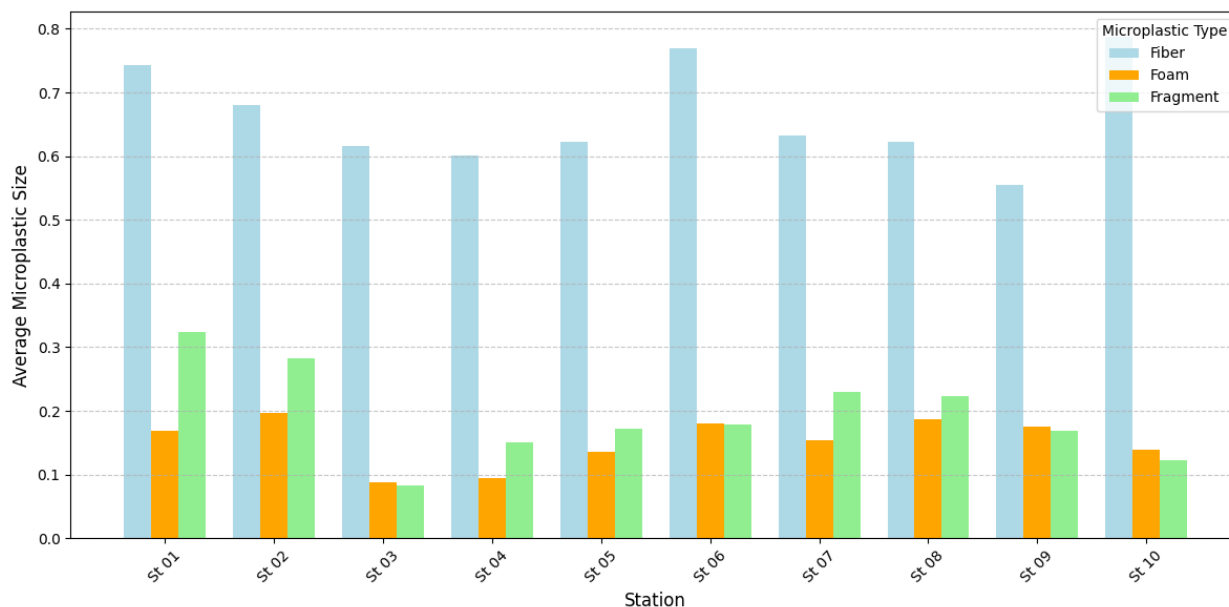


Fig. 29 Average size (mm) of different types of microplastic over the stations

Both terrestrial and marine ecosystems are seriously threatened by microplastic contamination, which also has an effect on biodiversity, human health, and environmental health. These microscopic plastic particles are ubiquitous in soil, water, and the air and are frequently the result of larger plastic waste or industrial operations. Plankton and larger predators are among the marine species that frequently consume microplastics because they mistake them for food. In addition to chemical pollution from harmful compounds linked to plastics, this intake may result in physical injury, such as internal injuries and obstructions. Microplastics build up in the food chain over time, and eating seafood may have an adverse effect on human health. Microplastics also damage habitats, upset biological equilibrium, and thwart conservation initiatives. Stricter laws, international cooperation, and creative waste management techniques are needed to combat this pollution in order to preserve ecosystems and guarantee sustainability.

5. Discussion

5.1 Assessing the Effectiveness of ECA and the Necessity for MPA Designation

Saint Martin Island, designated as an ECA in 1999 and an MPA in 2021, offers a chance to evaluate the impact of these conservation measures on its marine ecosystem. This analysis examines the island's ecological condition across four periods—pre-ECA (1990-2000), post-ECA (2001-2010), pre-MPA (2011-2020), and post-MPA (2021-2024)—to assess whether these designations have enhanced the health and resilience of its marine environment.

Fig. 30 presents the Normalized Difference Vegetation Index (NDVI) time series for SMI from 1990 to 2024, showcasing significant fluctuations in vegetation health over the 34-year period. The observed decline in NDVI is influenced not only by changes in terrestrial vegetation but also by coastal factors such as algae, phytoplankton, coral, and dynamic processes like erosion, accretion, and shoreline shifts. Temporary submersion of certain areas during image capture may also create the appearance of low or no vegetation. The northern part of SMI, particularly Modhopper, is densely developed with hotels, resorts, bazaars, and a higher population density, resulting in significantly reduced vegetation cover compared to the less developed southern region. These trends underscore the complex interplay between natural and human factors influencing vegetation dynamics. The time series reveals distinct phases of declines and recoveries in NDVI values, driven by a combination of natural events, human activities, and conservation efforts.

The sharp decline in NDVI from 1990 to 1993 may be linked to deforestation driven by agricultural expansion, urban development, and increased tourism infrastructure on the island. The establishment of large resorts during this period intensified habitat destruction. Moreover, severe cyclonic storms, such as those common in the Bay of Bengal, could have caused extensive damage to vegetation. A recovery phase from 1993 to 1995 indicates natural regeneration, likely as the environment rebounded from earlier disturbances. Reduced human activity or less extreme weather events might have contributed to this temporary recovery. Another decline starting around 1995 and persisting until 2005 reflects increasing human activities, including expanded tourism

and resort development. During this time, the ecosystem faced sustained pressure from deforestation, land use changes, and unsustainable practices. Cyclones like Sidr (2007) and other severe storms during this period might also have played a role in degrading vegetation. Post-2006, NDVI values fluctuated but exhibited a gradual upward trend, possibly due to increased conservation efforts. The establishment of the MPA in 2022 was significantly important to enhance the vegetation recovery.

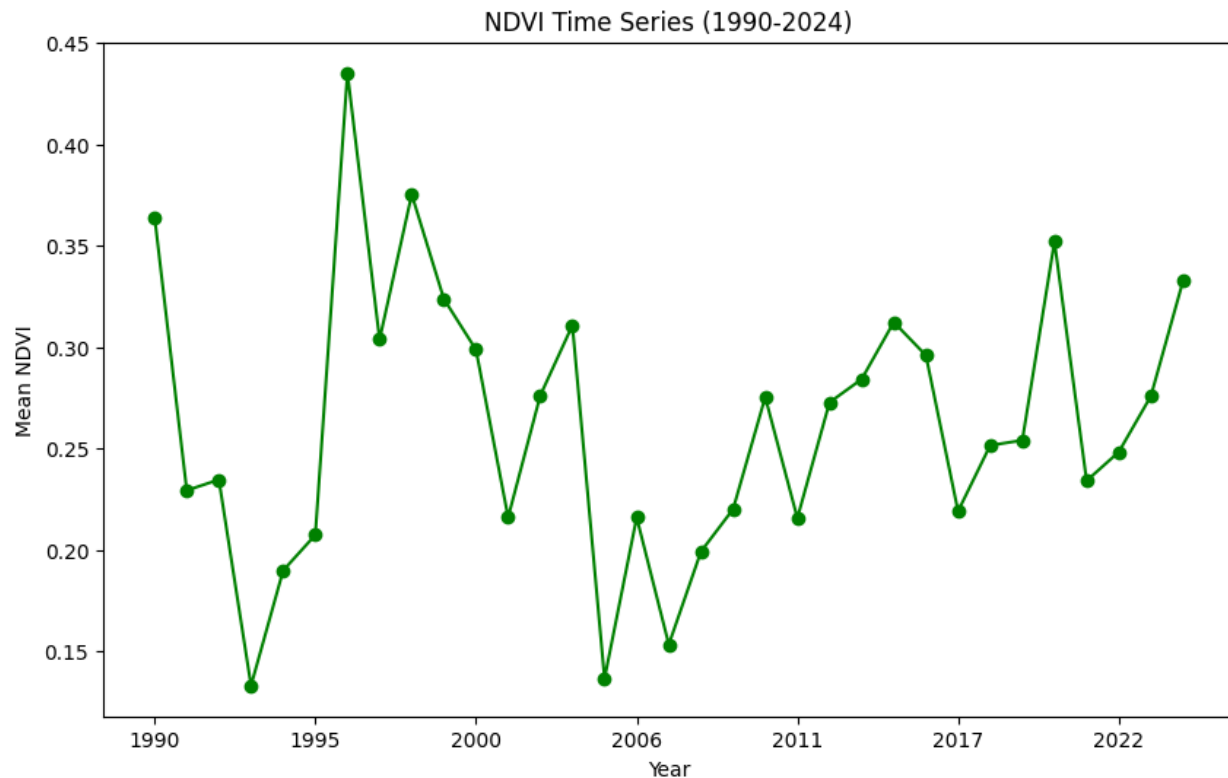


Fig. 30 Time series plot of change in NDVI of SMI from 1990 to 2024

The dynamic interaction between human and natural effects over a number of decades is reflected in the NDVI time series for SMI. The early deterioration of vegetation health highlights the effects of aggressive human activity and potentially unfavorable weather patterns. The following oscillations show times of environmental stress and recovery caused by different levels of natural variability and human intervention. The NDVI's recent rising trend after the MPA was established shows that focused conservation efforts have had positive results, indicating that they are successful in preserving and restoring the health of the vegetation. To maintain these improvements and handle recurring issues brought on by climate change and demands from human development, regular monitoring and adaptive management will be crucial going forward.

The time series plot in Fig. 31 shows the mean temperature over time. The temperature fluctuates in a sinusoidal pattern, with peaks in the summer months and lows in the winter months. The temperature generally ranged 22-30°C. The analysis of the daily temperature time series spanning from 2001 to 2020 highlights a steady and discernible rise in average temperatures over the two-decade period, signaling a gradual but consistent warming trend. A prominent feature of this dataset is the well-defined seasonal cycle, characterized by markedly cooler temperatures in the winter months, particularly in January and February, and significantly warmer peaks during the summer, notably in May and June. This seasonal rhythm persists uniformly across the entire timeframe, demonstrating the predictable nature of seasonal temperature variations.

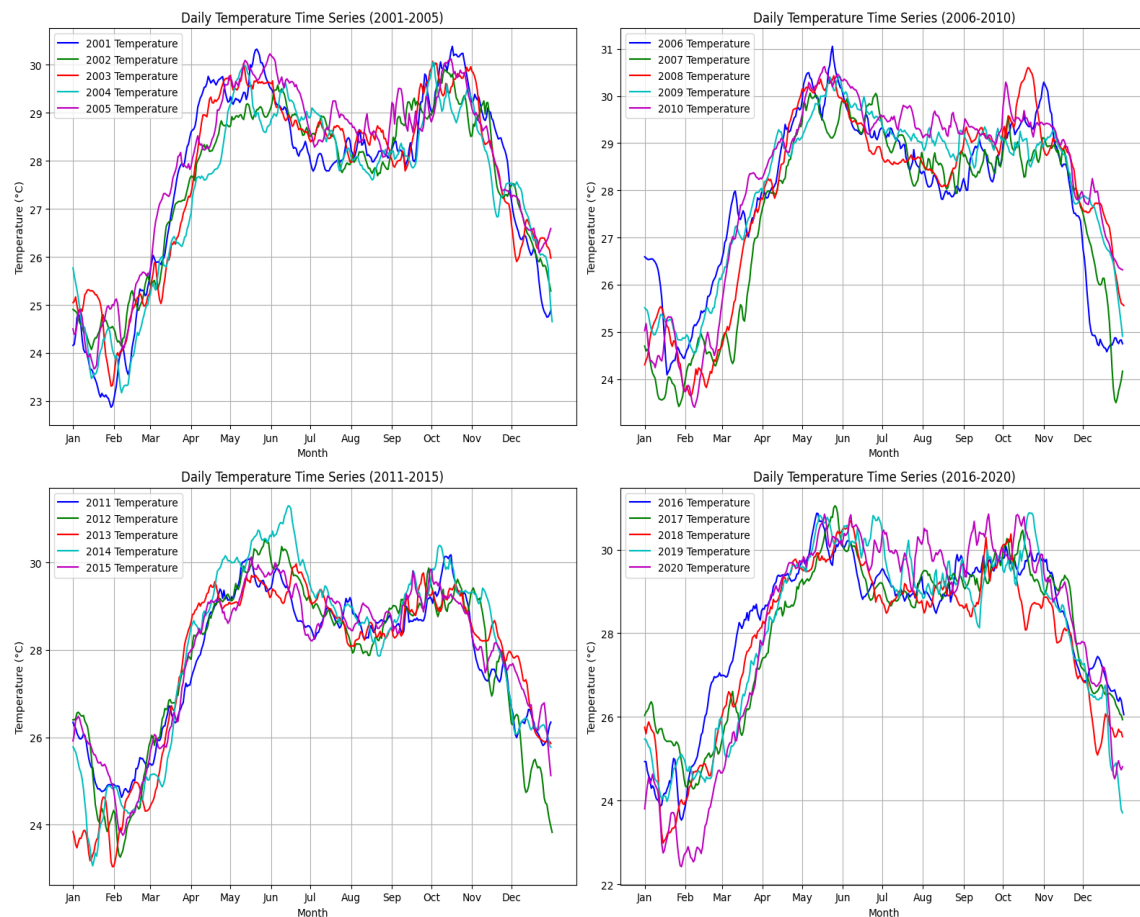


Fig. 31 Time series plot of temperature

However, the dataset also reveals substantial year-to-year fluctuations, particularly during the summer months, where temperature deviations from the norm are more pronounced. These variations, while adding complexity to the overall pattern, do not disrupt the broader warming trend observed over the years. Notably, the upward trajectory of temperatures becomes

increasingly prominent as the years progress, underscoring the impact of long-term climatic shifts. Despite this overarching warming trend, the seasonal cycles remain remarkably consistent, reflecting the enduring natural rhythm of cooler winters and hotter summers. The interplay between this stable seasonal framework and the progressive warming underscores the dual influence of predictable climatic patterns and gradual environmental change, painting a nuanced picture of temperature dynamics over the past two decades.

The plot in Fig. 32 shows the temperature anomalies for four different seasons: Winter, Premonsoon, Monsoon, and Post monsoon. The x-axis represents the year, ranging from 2001 to 2020. The y-axis represents the temperature anomaly in degrees Celsius ($^{\circ}\text{C}$). Each season's temperature anomalies are plotted as a blue line, with a black dashed line representing the trend line for that season.

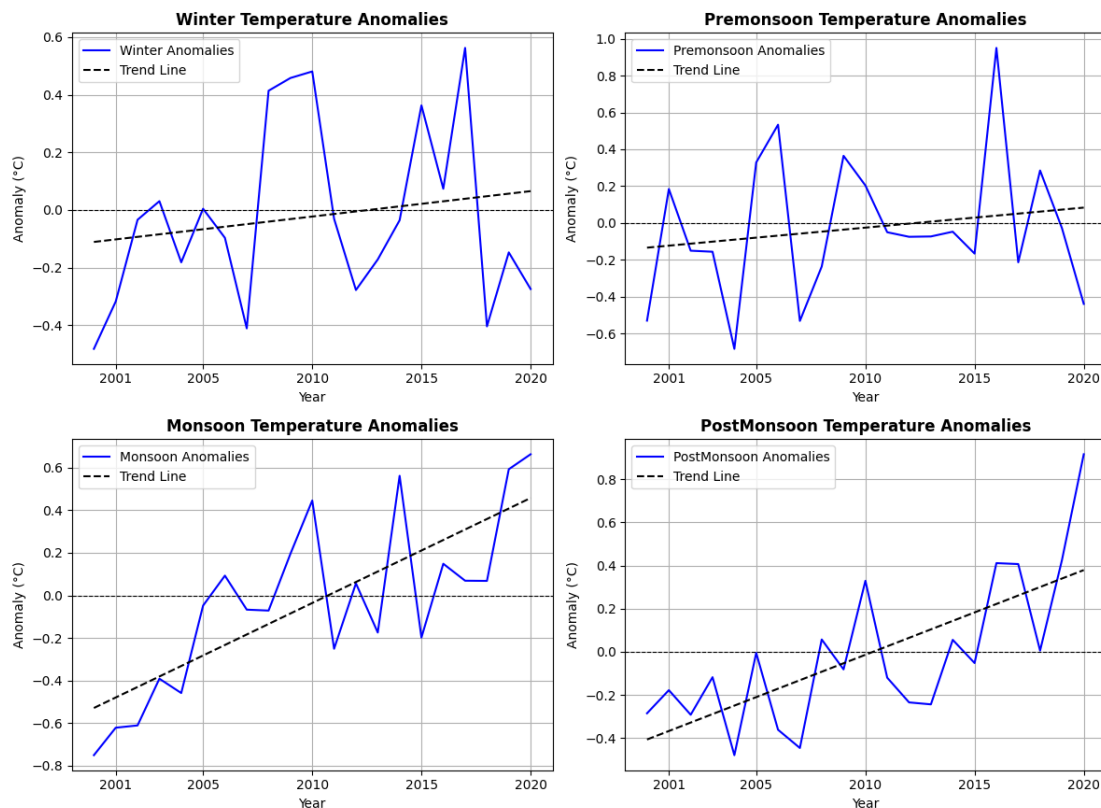


Fig. 32 Seasonal temperature anomaly from 2001 to 2020

The plot in Fig. 33 shows the salinity time series for five-year ranges. The salinity time series shows a consistent seasonal pattern with high salinity in summer and low salinity in winter over the 20-year period, indicating a similar trend over time.

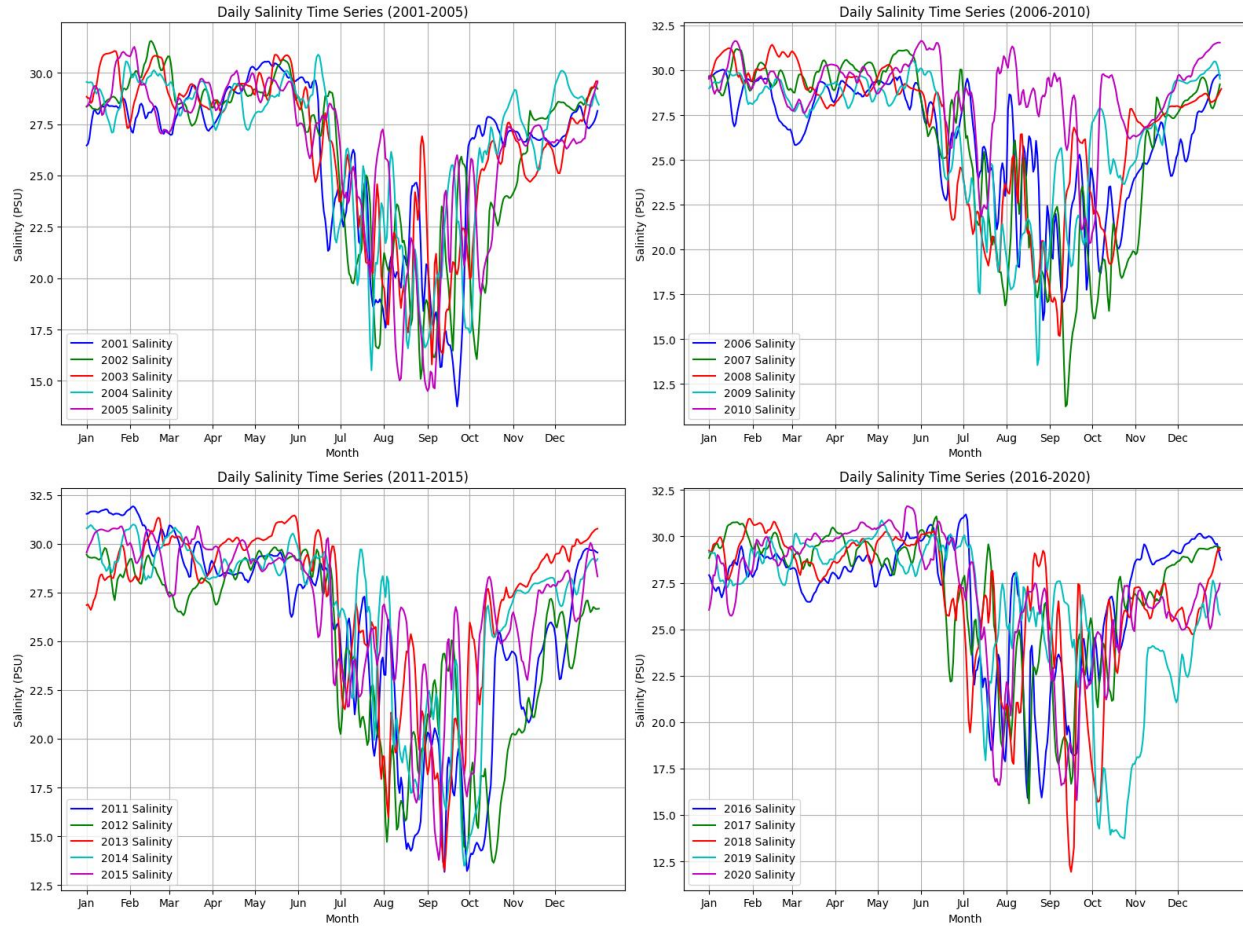


Fig. 33 Time series plot of salinity

The average salinity is slightly increased over the 20 years. In 2001 the highest peak was 31 PSU whereas in 2020 the peak was around 32 PSU. The Lowest peak in 2020 can be found around 16 PSU whether 2001 shows around 15 PSU in post monsoon season. The increase around 1 PSU. The lowest trend in these 20 years found in 2007 which ranged from 11 to 31 PSU and highest trend found in 2010 which was ranged from 20 to 33 PSU. 2012 and 2019 shows some uncommon seasonal shift because of climate variability.

The seasonal anomaly of salinity in Fig. 32 shows a distinct steep increase of salinity accept in winter season. The trend of the salinity in winter shows a decreasing rate. Highest increasing trend found in monsoon season because of the decreasing rate of precipitation. On the other hand, the increase of temperature leads to high evaporation cause the increase of salinity.

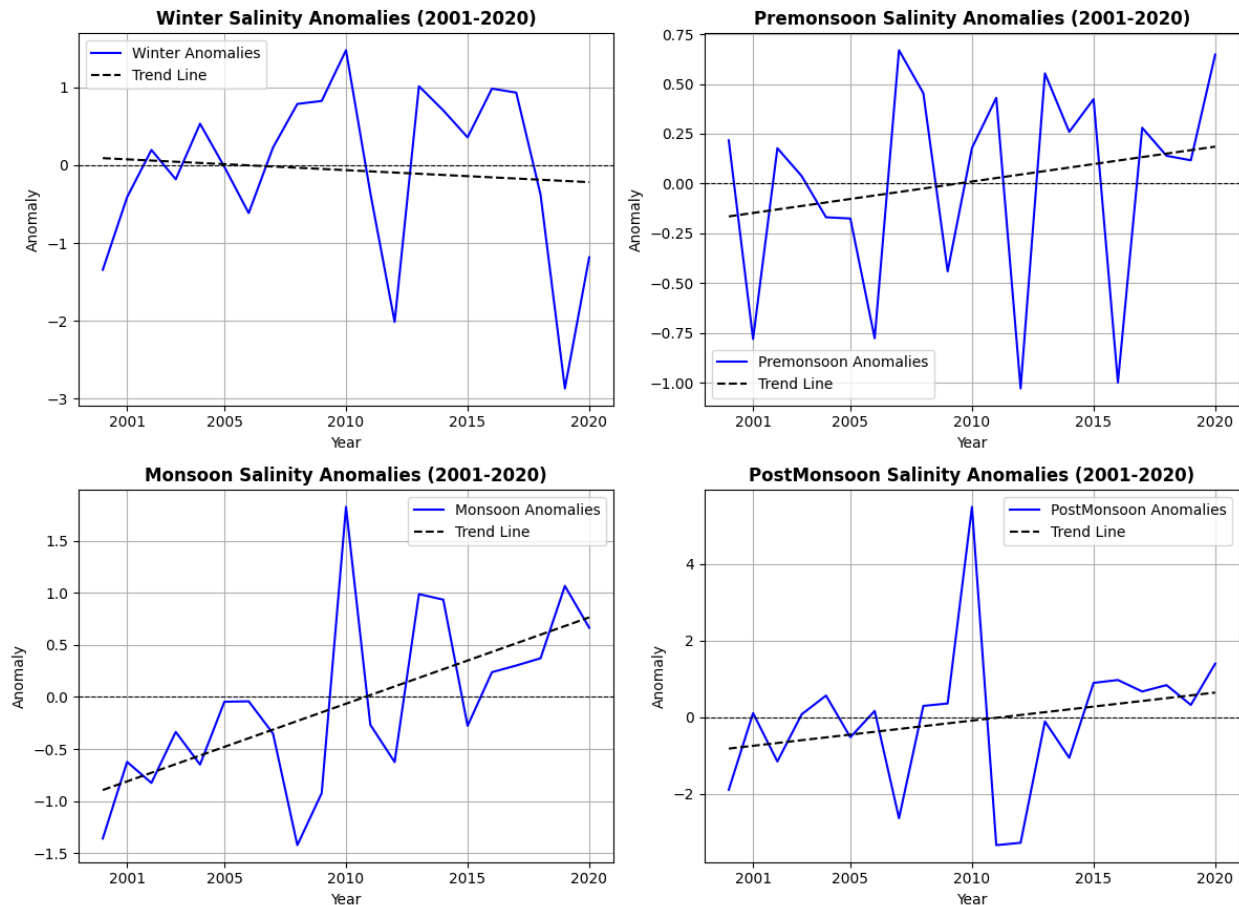


Fig. 34 Seasonal salinity anomaly from 2001 to 2020

Figure 27 illustrates the seasonal chlorophyll anomaly trends for SMPA from 2001 to 2020, offering a detailed view of chlorophyll concentration variations over two decades. These anomalies represent the seasonal deviations from the long-term average chlorophyll concentrations, with the x-axis indicating the years and the y-axis showing the anomaly values in mg/m^3 . Across all four seasons, the plots reveal notable fluctuations in chlorophyll anomalies over the 20-year span. Periods of positive anomalies indicate higher-than-average chlorophyll concentrations, while negative anomalies reflect lower-than-average levels. By analyzing these seasonal anomalies, distinct trends and patterns emerge, shedding light on the temporal dynamics of chlorophyll concentration in the region. This seasonal perspective highlights the complex interactions between environmental factors and chlorophyll production over time.

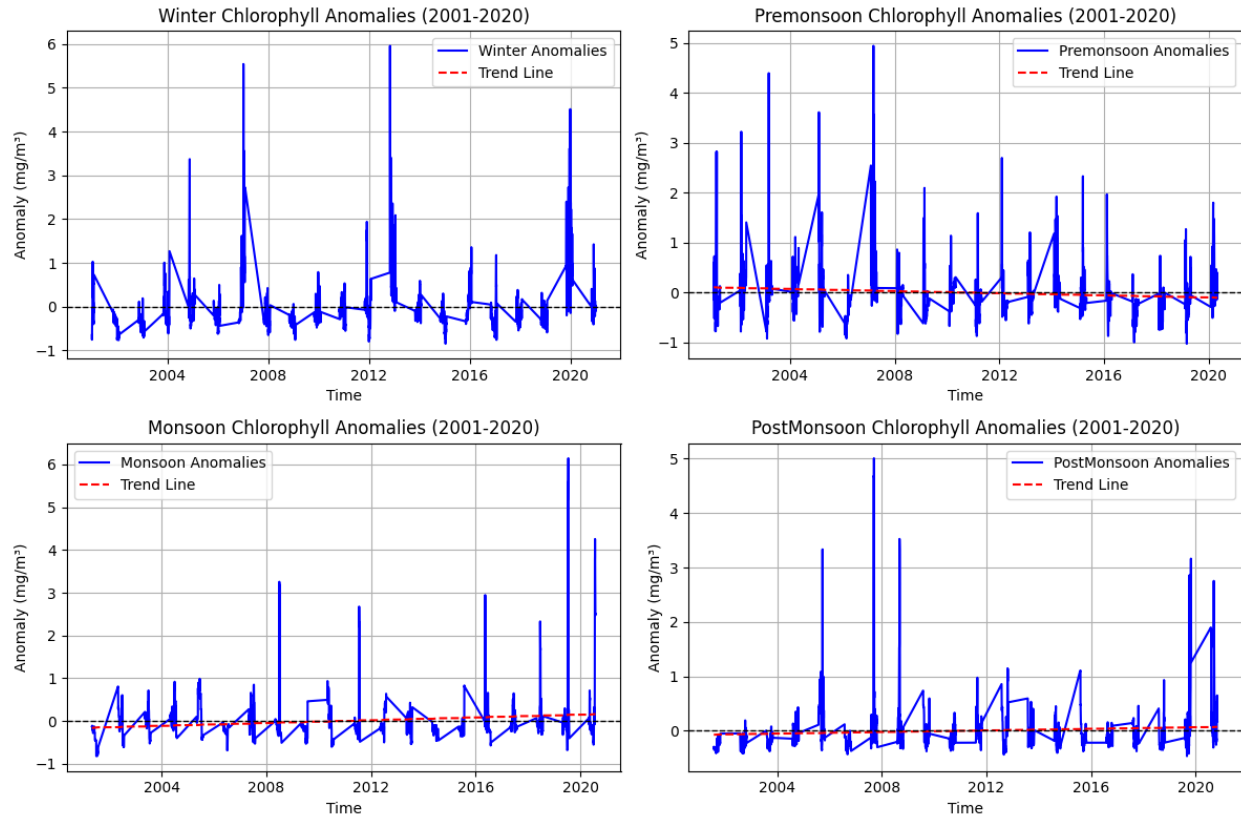


Fig. 35 Time Series Anomaly of Chlorophyll Concentration

During the winter season, the trend line indicates a slight upward trajectory, suggesting a gradual increase in chlorophyll concentration. In contrast, the premonsoon season displays a subtle downward trend, pointing to a marginal decline in chlorophyll levels during this period. For the monsoon season, the trend line remains relatively flat, indicating minimal change in chlorophyll concentration over time. Meanwhile, the post monsoon season shows a modest upward trend, reflecting a general increase in chlorophyll concentration during this phase. The highest peak can be found in 2008. In Winter there are less contribution of river influx and circulation events which leads to a slight decrease of chlorophyl. Whereas the precipitation and river flow lead to increase the chlorophyl in monsoon and post monsoon season.

These variations in chlorophyll concentrations are ecologically significant as they directly correlate with primary productivity in the region. Positive anomalies, characterized by higher chlorophyll levels, indicate increased phytoplankton abundance, which forms the foundation of the marine food web. This abundance enhances primary productivity, supporting larger populations of herbivores such as zooplankton, small fish, and other organisms that depend directly

or indirectly on phytoplankton as a food source. Consequently, higher primary productivity can boost the abundance and diversity of marine species across different trophic levels, potentially enriching the overall ecosystem. Conversely, negative anomalies, or periods of lower chlorophyll concentrations, signify reduced phytoplankton abundance, resulting in diminished primary productivity. This decreases food availability for herbivores and higher trophic levels, potentially leading to declines in the abundance and diversity of marine organisms. Such reductions in primary productivity can disrupt the balance of the marine ecosystem, altering interactions among species and triggering shifts in community composition. Changes in chlorophyll levels, whether due to natural variability or anthropogenic influences, underscore the delicate interdependence within the marine ecosystem, highlighting the importance of monitoring and managing these variations to ensure ecological stability and resilience.

Over the 24-year study period, analysis of Sea Surface Temperature (SST), Sea Surface Salinity (SSS), and Precipitation data provides a comprehensive understanding of the physical conditions influencing SMI, illustrating both the impacts of the ECA (ECA) designation in 1999 and projections for the MPA (MPA) designation in 2021. The long-term trends of SST, SSS, and precipitation reveal environmental shifts with varying degrees of annual and seasonal variability. The ECA designation initially helped maintain stable conditions; however, by itself, it was inadequate in curbing the growing pressures of overfishing, habitat degradation, and climate change. The MPA designation, implemented in 2021, represents a critical opportunity to address these ongoing threats. For the MPA to succeed, it must adopt comprehensive strategies that prioritize stricter fishing regulations, habitat restoration projects, sustainable tourism management, and active monitoring of SST and SSS. Furthermore, adaptive management plans that respond to ongoing climate changes, including measures to mitigate coral bleaching and protect vulnerable species, will be vital.

5.2 Ecosystem Health and Conservation Efforts

Assessing ecosystem health is vital for effective conservation. This study focuses on key indicators like water quality, phytoplankton dynamics, and heavy metal contamination in aquatic ecosystems to understand their stability and the influences on overall environmental conditions. By identifying these factors, we aim to support more targeted conservation strategies.

The bar plot from Fig. 36 illustrates the phytoplankton condition of the Saint Martin Ecosystem. Diversity Index measures the variety of phytoplankton species present. A higher value indicates greater diversity. Simpson Dominance Index measures the dominance of a particular species. A lower value indicates higher diversity and less dominance. The Evenness Index measures the distribution of abundance among different species. A higher value indicates more even distribution. Species Richness Index measures the number of different species present. A higher value indicates greater species richness.

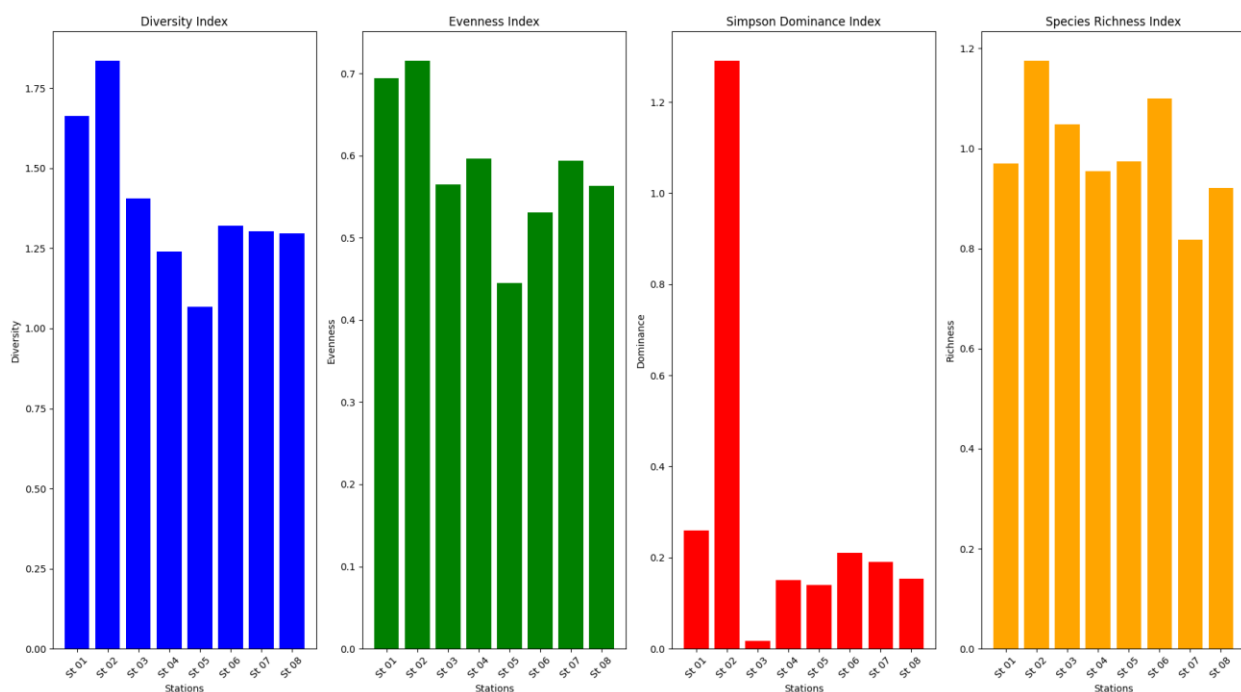


Fig. 36 Condition of Phytoplankton according to different Indices

There is a general trend of decreasing diversity from Station 1 to Station 5, followed by a slight increase and then a stable level. Stations 1 and 2 exhibit relatively high diversity. Stations 3 to 5 show a decline in diversity. Stations 6 to 8 have moderate to high diversity. The Simpson Dominance Index shows a fluctuating pattern, with peaks and troughs at different stations. Station 2 has the highest dominance, indicating a few dominant species. Stations 1, 3, 6, and 7 show relatively low dominance, suggesting a more even distribution of species. The Evenness Index generally shows a decreasing trend from Station 1 to Station 5, followed by a slight increase and then a stable level. Stations 1 and 2 have relatively high evenness, indicating a more balanced distribution of species abundance. Stations 3 to 5 show lower evenness, suggesting dominance by a few species. Stations 6 to 8 have moderate to high evenness. The Species Richness Index shows

a similar pattern to the Diversity Index, with a decrease from Station 1 to Station 5, followed by a slight increase and then a stable level. Stations 1 and 2 have higher species richness. Stations 3 to 5 have lower species richness. Stations 6 to 8 have moderate to high species richness.

Based on these indices, the overall phytoplankton condition appears to be moderate to good, with some stations exhibiting higher diversity and evenness. However, there are also stations with lower diversity and higher dominance, indicating potential ecological imbalances.

The Pearson correlation heatmap in Fig. 37 illustrates the relationships between various physiochemical parameters (pH, Salinity, Temperature, TDS, DO) and nutrients (PO₄, NO₃, NH₃) in the marine ecosystem.

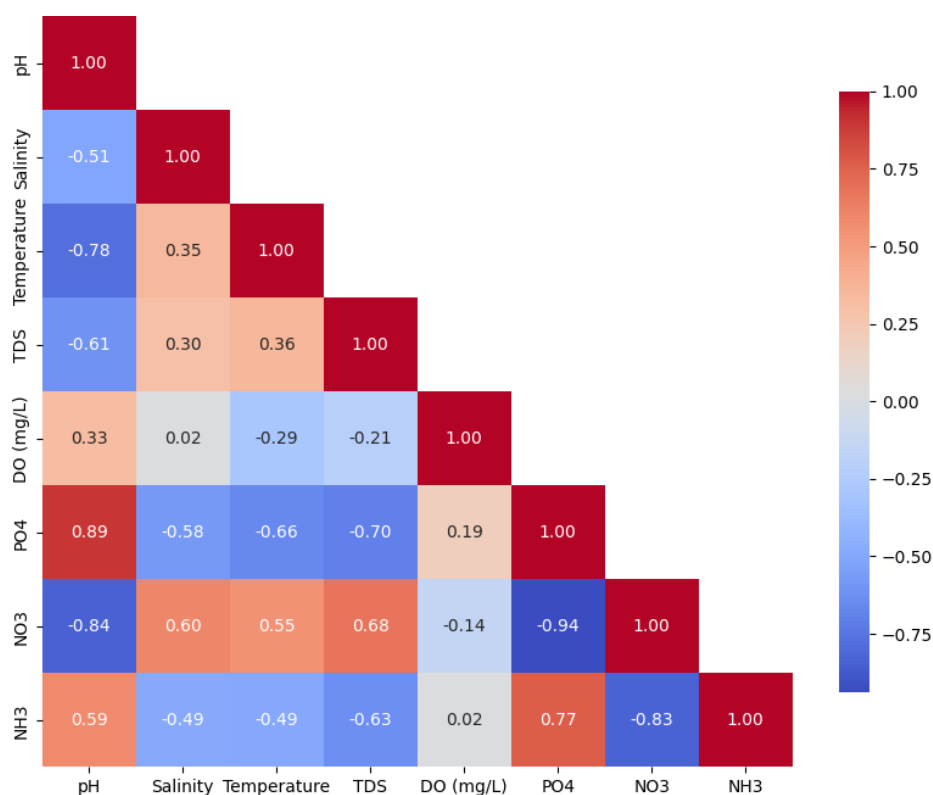


Fig. 37 Pearson Correlation plot between physiochemical parameters and nutrient availability

There is a strong positive correlation (+0.89) between pH and phosphate (PO₄). This suggests that higher pH levels could be associated with higher concentrations of phosphate. A strong negative correlation (-0.84) exists between pH and nitrate (NO₃), indicating that as pH increases, nitrate levels tend to decrease. There's a moderate positive correlation (+0.59) with ammonia (NH₃), suggesting that higher pH may support slightly higher ammonia levels. There is a moderate

positive correlation (+0.55), which implies that higher temperatures might be associated with increased nitrate concentrations. A negative correlation (-0.66) exists between temperature and phosphate, suggesting that as temperature rises, phosphate levels tend to decline. Similar to PO₄, there is a moderate negative correlation (-0.49) with ammonia, implying a decrease in NH₃ levels as temperature increases. A strong positive correlation (+0.68) indicates that as TDS levels rise, nitrate levels also tend to increase, which could suggest higher mineral or nutrient content in areas with more dissolved solids. There is a moderate negative correlation (-0.70) with phosphate, implying that higher TDS levels might be associated with lower phosphate concentrations. A moderate negative correlation (-0.63) with ammonia suggests a decline in NH₃ with higher TDS levels.

Balanced nutrient levels, regulated by factors like pH and temperature, can support healthy algal growth and promote primary productivity in the ecosystem, providing food for various marine organisms. Favorable temperature and pH conditions that align with nutrient availability can support diverse marine life, from primary producers to higher trophic levels. High nutrient levels, particularly from nitrate (NO₃) and phosphate (PO₄), can lead to eutrophication. This causes excessive algal blooms, leading to oxygen depletion, which can be detrimental to aquatic organisms. Strong correlations between parameters like temperature and nutrient levels can indicate that shifts in temperature (e.g., from climate change) may lead to changes in nutrient availability. For example, higher temperatures reducing phosphate levels might impact nutrient-sensitive organisms, affecting ecosystem balance. Increased ammonia (NH₃) levels, especially in higher pH conditions, can be toxic to marine life, leading to stress or mortality in sensitive species.

The water quality in Fig. 38, as represented by the average WQI of 76.18, falls within the "Fair" category. This indicates that the water quality is generally acceptable but may have some limitations or require attention in certain areas. The overall water quality, as represented by the average WQI of 76.18, falls within the "Fair" category. This indicates that the water quality is generally acceptable but may have some limitations or require attention in certain areas. Most stations fall within this category, indicating that the water quality is generally acceptable for various uses, such as irrigation, and recreation but the salinity doesn't allow to drink. However, there may be specific contaminants or parameters that require monitoring and potential treatment. Most stations fall within this category, indicating that the water quality is generally acceptable for

various uses, such as drinking, irrigation, and recreation. However, there may be specific contaminants or parameters that require monitoring and potential treatment.

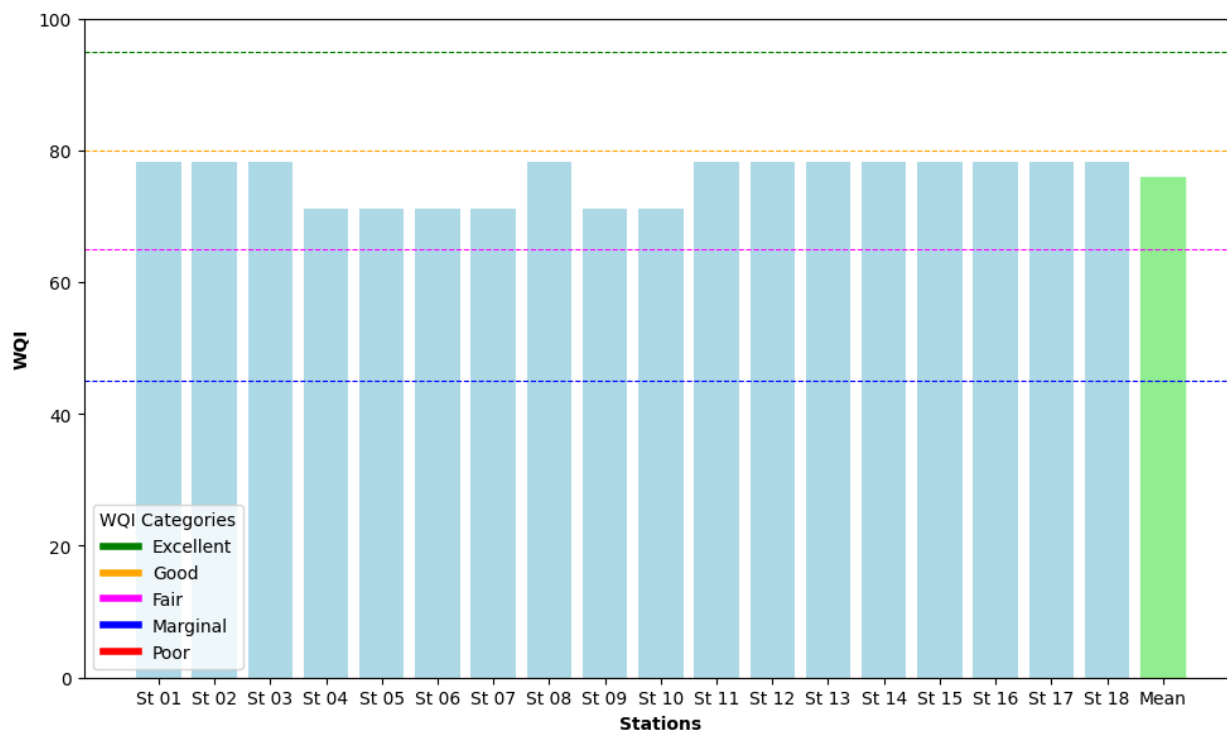


Fig. 38 CCME Water Quality Index

The provided violin plot in Fig. 39 illustrates the distribution of the Nemerow's Pollution Index (NPI) values for SMI. The mean and median values are relatively close, indicating a symmetric distribution. The mean NPI value is approximately 1.69, and the median is around 2.02. The mean and median values are relatively close, indicating a symmetric distribution. The mean NPI value is approximately 1.69, and the median is around 2.02. The violin plot shows a somewhat skewed distribution, with a longer tail towards the higher NPI values. This suggests that there might be a few stations with significantly higher levels of pollution compared to the majority. While not explicitly shown in the plot, the presence of outliers could be indicated by the extended tails of the violin plot. Outliers might represent stations with unique environmental conditions or anthropogenic influences.

The NPI values suggest a low to moderate level of pollution in SMI. The majority of stations fall within the "clean" or "low contamination" categories. However, the presence of higher NPI values indicates areas of concern. The variation in NPI values across different stations suggests spatial

heterogeneity in pollution levels. Factors such as proximity to pollution sources, hydrological conditions, and socioeconomic factors might influence the distribution of pollutants. Elevated levels of pollution can have adverse effects on the marine ecosystem, including toxicity to organisms, bioaccumulation, and disruption of food chains.

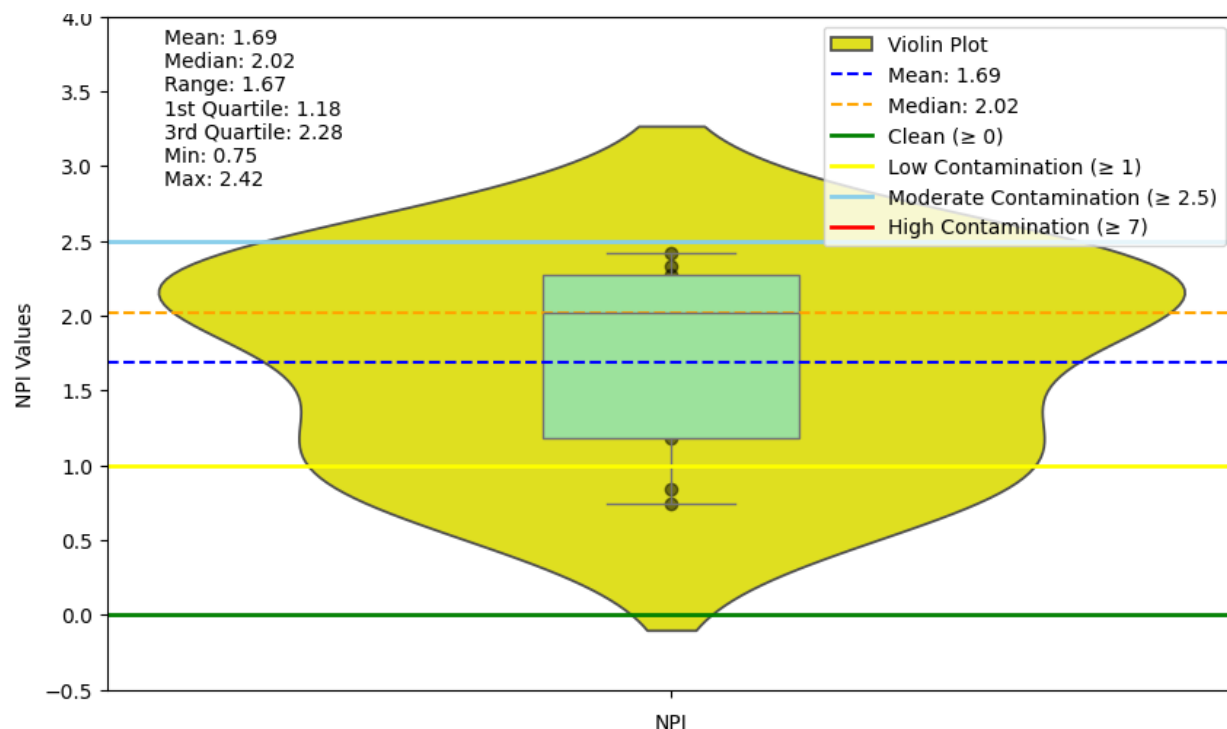


Fig. 39 Nemerow's Pollution Index of SMI

The provided violin plot in Fig. 40 illustrates the distribution of the Heavy Metal Pollution Index (HMPI) values for SMI. The provided violin plot illustrates the distribution of the Heavy Metal Pollution Index (HMPI) values for SMI. The range of HMPI values is 23.05, with a minimum of 12.60 and a maximum of 35.65. The interquartile range (IQR), which represents the middle 50% of the data, is 10.08. The violin plot shows a somewhat skewed distribution, with a longer tail towards the higher HMPI values. This suggests that there might be a few stations with significantly higher levels of heavy metal pollution compared to the majority. While not explicitly shown in the plot, the presence of outliers could be indicated by the extended tails of the violin plot. Outliers might represent stations with unique environmental conditions or anthropogenic influences.

The HMPI values suggest a moderate level of heavy metal pollution in SMI. While the majority of stations fall within the "low concentration" or "medium concentration" categories, the presence

of higher values indicates areas of concern. The variation in HMPI values across different stations suggests spatial heterogeneity in heavy metal pollution. Factors such as proximity to pollution sources, hydrological conditions, and sediment characteristics might influence the distribution of heavy metals. Elevated levels of heavy metals can have adverse effects on the marine ecosystem, including toxicity to organisms, bioaccumulation, and disruption of food chains.

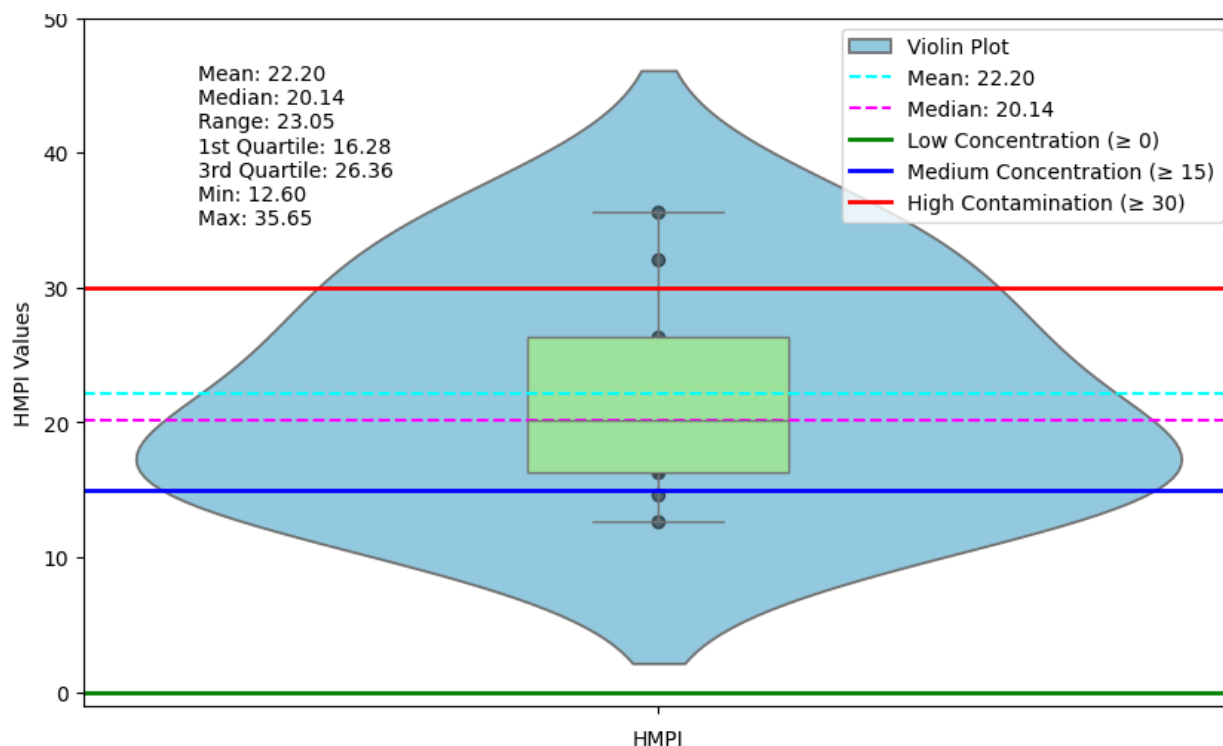


Fig. 40 Heavy Metal Pollution Index of SMI

The provided violin plot Fig. 41 illustrates the distribution of the Potential Ecological Risk (PER) values for SMI. The mean and median values are significantly different, indicating a skewed distribution. The mean PER value is 113.92, while the median is 33.10. This suggests that a few stations with extremely high PER values are pulling the mean upwards. The range of PER values is 580.23, with a minimum of 17.90 and a maximum of 598.13. The interquartile range (IQR), which represents the middle 50% of the data, is 12.95. The violin plot shows a highly skewed distribution, with a long tail towards the higher PER values.

This suggests that there are a few stations with extremely high ecological risk compared to the majority. The presence of outliers is evident from the extended tail of the violin plot. These outliers likely represent stations with unique environmental conditions or anthropogenic influences. The

PER values suggest a moderate to high level of ecological risk in SMI. The presence of high PER values indicates areas with potential ecological concerns. The variation in PER values across different stations suggests spatial heterogeneity in ecological risk. Factors such as proximity to pollution sources, hydrological conditions, and socioeconomic factors might influence the distribution of pollutants and their impact on the environment. High PER values indicate areas with potential ecological degradation, including loss of biodiversity, habitat degradation, and water quality deterioration.

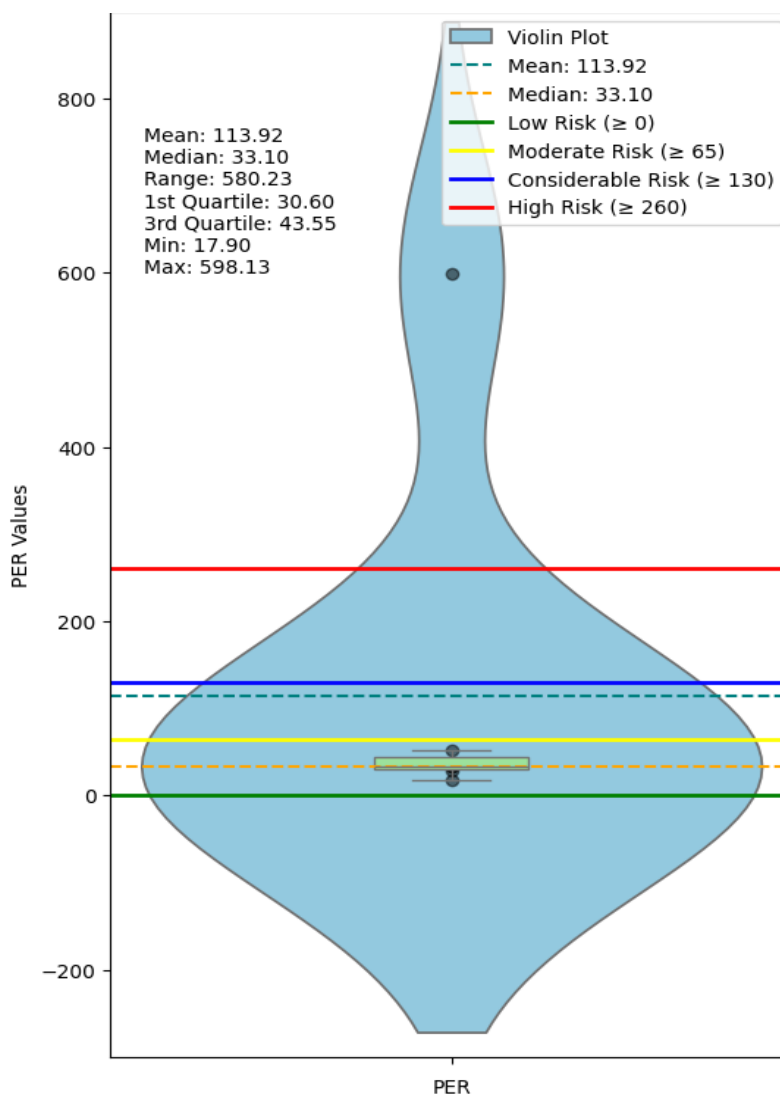


Fig. 41 Potential ecological risk of heavy metal

The provided violin plot in Fig. 42 illustrates the distribution of the Toxicity Risk Index (TRI) values for SMI. The mean and median values are relatively close, indicating a symmetric

distribution. The mean TRI value is approximately 6.96, and the median is around 8.01. The range of TRI values is 6.83, with a minimum of 3.10 and a maximum of 9.92. The interquartile range (IQR), which represents the middle 50% of the data, is 4.08.

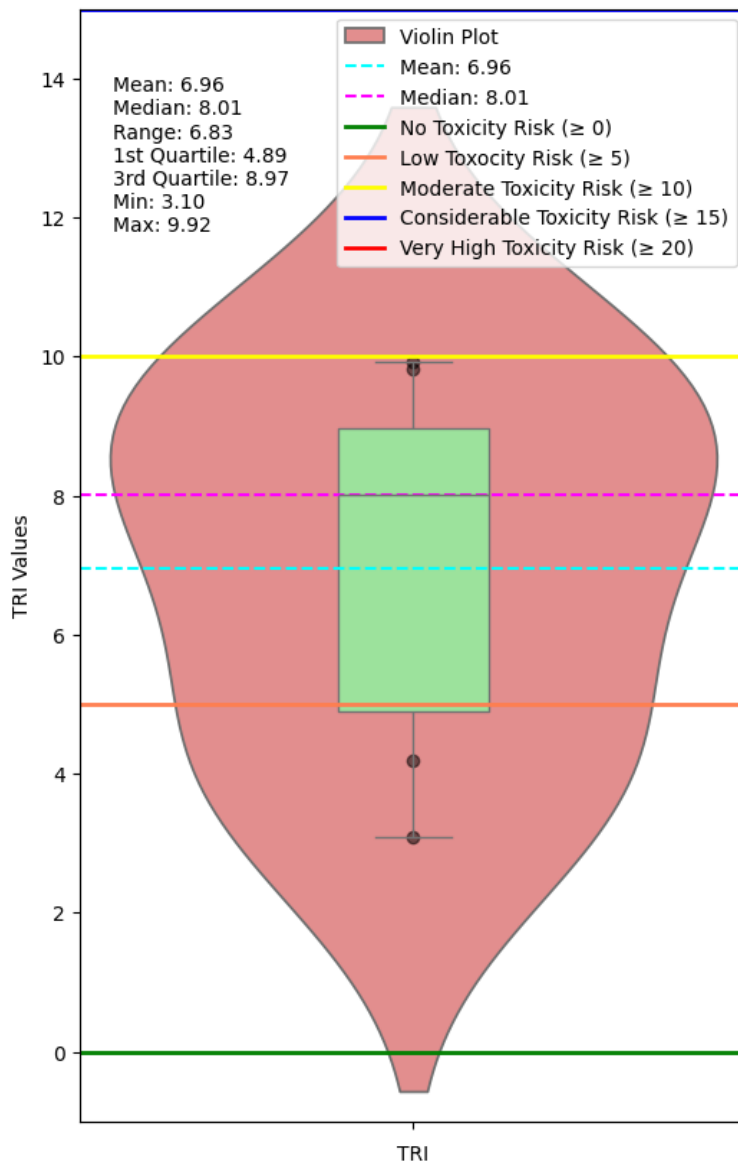


Fig. 42 Toxicity risk of heavy metal pollution

The violin plot shows a somewhat skewed distribution, with a longer tail towards the lower TRI values. This suggests that there might be a few stations with significantly lower levels of toxicity risk compared to the majority. While not explicitly shown in the plot, the presence of outliers could be indicated by the extended tails of the violin plot. Outliers might represent stations with unique environmental conditions or anthropogenic influences. The TRI values suggest a low to moderate

level of toxicity risk in SMI. The majority of stations fall within the "low toxicity risk" or "moderate toxicity risk" categories. However, the presence of higher TRI values indicates areas of concern. The variation in TRI values across different stations suggests spatial heterogeneity in toxicity risk. Factors such as proximity to pollution sources, hydrological conditions, and socioeconomic factors might influence the distribution of pollutants and their toxicity. Elevated levels of toxic substances can have adverse effects on the marine ecosystem, including toxicity to organisms, bioaccumulation, and disruption of food chains.

The provided radar chart in Fig. 43 presents the Geo-Accumulation Index (Igeo) values for different heavy metals at various stations in SMI. Most stations show a moderate to high level of pollution of Hg, with some stations reaching the extremely polluted category. The majority of stations exhibit a moderate to high level of pollution of Cd, with a few stations in the extremely polluted category. The stations generally show a moderate to high level of pollution of Ni. Most stations fall within the moderately polluted category of Lead, with a few in the moderately to strongly polluted range. The stations predominantly show a moderate level of pollution of Cr, with a few in the moderately to strongly polluted category. As in Most stations fall within the moderately polluted category, with a few in the moderately to strongly polluted range. The stations generally show a low to moderate level of pollution of Iron. Zinc in most stations exhibit a moderate level of pollution, with a few in the moderately to strongly polluted range. The stations generally show a low to moderate level of pollution of Mn. Copper in most stations fall within the moderately polluted category, with a few in the moderately to strongly polluted range.

Station 01 shows a high level of pollution for most metals, particularly Hg, Cd, and Ni. Station 02 exhibits a moderate to high level of pollution for most metals. Station 03 manifests a moderate to high level of pollution for most metals, with higher levels for Hg, Cd, and Ni. Station 04 exhibits a moderate level of pollution for most metals. Station 07 shows a moderate level of pollution for most metals, with higher levels for Hg, Cd, and Ni. Station 08 manifests a moderate to high level of pollution for most metals. Station 09 exhibits a moderate level of pollution for most metals, with higher levels for Hg, Cd, and Ni. The radar chart indicates that the study area is generally moderately polluted with heavy metals, with some stations showing higher levels of contamination, particularly for Hg, Cd, and Ni. This suggests that the area may be impacted by

anthropogenic activities, such as industrial emissions, agricultural runoff, or urban wastewater discharge.

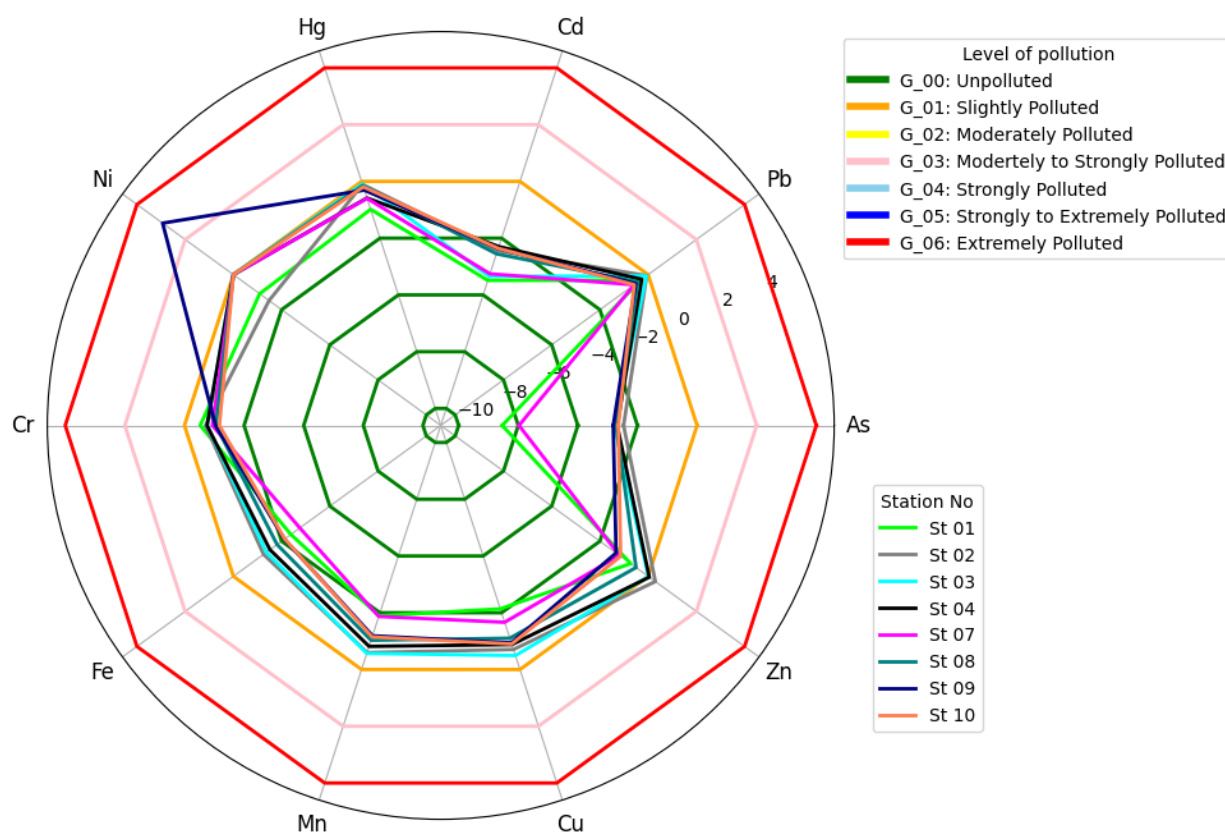


Fig. 43 Rader chart to assess the heavy metal's Geo Accumulation index of SMI

The violin plot in Fig. 44 illustrates the distribution of the ERI across various stations, highlighting differences in risk levels. Stations 01, 03, 04, 07, and 08 primarily show values concentrated in the Low-Risk category. St 01 has a median ERI around 5, with occasional outliers reaching 18, while St 03 and St 04 display similar patterns, with values clustered near 1 and few outliers up to 8. St 07's distribution is skewed lower, with a median around 3. In contrast, St 02, St 08, St 09, and St 10 exhibit slightly higher risk levels. St 02 has a median ERI of 13, with most values falling into the Moderate Risk category and some outliers reaching Considerable Risk. St 08, St 09, and St 10 have medians around 6, 6, and 5, respectively, with the majority of values in the Low-Risk category but some outliers extending to Moderate Risk. The average ERI across all stations is centered around 4, with most values remaining within the Low-Risk category, although a few outliers extend into the Moderate Risk range.

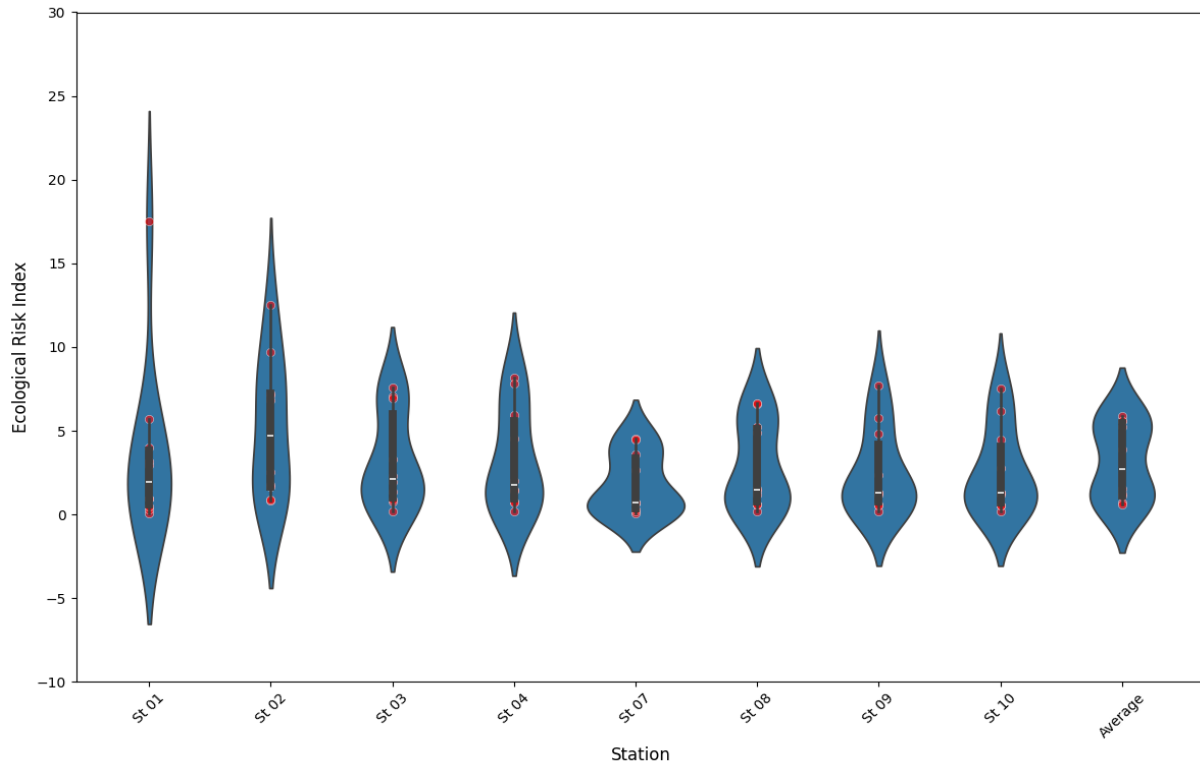


Fig. 44 Station wise ecological risk index

Figure 45 presents box plots depicting the Ecological Risk Index (ERI) values for various elements, including As, Cd, Cr, Cu, Fe, Hg, Mn, Ni, Pb, and Zn. These box plots provide a comprehensive visual summary of the distribution of ERI values, showing key statistical features such as the median, interquartile range, and potential outliers. Among these elements, Fe and As exhibit the highest ERI values, with their box plots extending significantly higher than those of other elements. This indicates that these two elements may pose substantial ecological risks in the study area. For Cd and Cr, the box plots are relatively compact, with most of their ERI values falling within the "Low Risk" category. This suggests that these elements are currently associated with a lower level of ecological concern. In contrast, elements such as Cu, Hg, Mn, Ni, Pb, and Zn show a broader range of ERI values. While these values generally remain within moderate risk levels, their variability implies a need for attention, as they could potentially pose moderate to significant ecological risks under changing conditions.

Notably, the ERI values for most elements predominantly fall within the range of 0 to 10, and none of the elements exceed the threshold for critical ecological risk. However, the growing anthropogenic pressures on the island underscore the importance of closely monitoring these

ecological indicators. Proactive measures are necessary to ensure that the ecological health of the island remains intact, as continued human activities could exacerbate the risks associated with these elements over time.

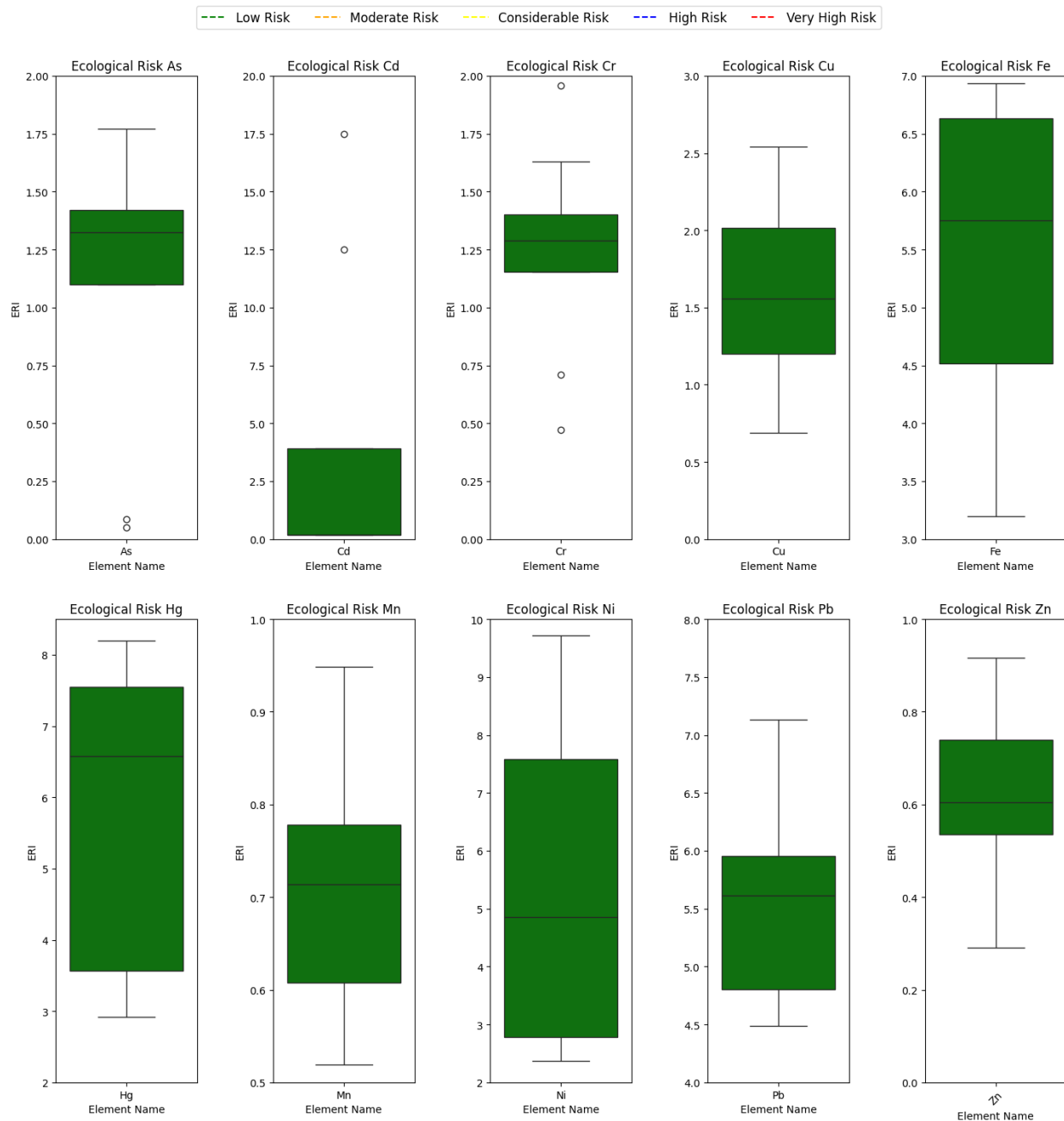


Fig. 45 Element wise Ecological Risk Index

Moreover, SMI having a huge tourism pressure but this island doesn't have any heavy industry, oil or container ship transportations that's why heavy metal concentration is in the state of lower

right now. But the discharge of Naf River may have a contribution of heavy metal introduction in the surrounding environment of the island. The sampling area is also a major concern here. Identifying the sources of Fe and As contamination is essential for designing effective mitigation strategies. Regular monitoring of heavy metal levels and ecological health is crucial to track changes and evaluate remediation efforts. Targeted measures, including pollution control, improved waste management, and habitat restoration, can reduce contamination and protect the environment. Raising public awareness about the risks of heavy metal pollution and the importance of environmental protection can further promote sustainable practices.

5.3 Comparison with Baseline Study

The baseline studies provide valuable insights into the environmental conditions of the region, serving as a foundation for comparison with current research findings. Vegetation health has undergone significant fluctuations over time, with notable reductions in canopy cover observed in 2016, followed by a modest recovery by 2019. Climatic conditions, marked by seasonal variations, have exhibited a trend of increasing temperatures since the 2000s, alongside fluctuations in precipitation patterns.

Table 12 Comparative analysis with the baseline studies

Parameters	Current Study	Baseline Study	Reference
Vegetation	Fluctuating but in a slight increasing rate	Vegetation health fluctuated significantly, with canopy cover reducing in 2016 but improving slightly by 2019.	(Roy, 2020)
Climatic Condition	1. Precipitation - Decreased 2. Temperature - Increased around 1.28 °C 3. Salinity - Slightly increased	1. Significant seasonal variations in precipitation. 2. Increased temperature observed since the 2000s.	Hasan et al. (2022)

Primary Productivity	1. Abundance: 35718 PPs/L $\pm$ 19473 2. Chlorophyll Concentration - slightly decreased	Chlorophyll-a concentration linked to seasonal and tidal influences, indicating declining trends in certain zones.	Mahmud et al. (2017)
Water Pollution	1. CCME water quality fair (76 - 80) 2. NPI - Low Contamination	Marine pollution levels are low but seasonal spikes occur due to tourism and waste management issues.	Karim et al. (2015)
Heavy Metal pollution	Moderate to considerable	Levels of heavy metals such as Pb, Cd, and Hg are detectable but remain within safe limits in most areas.	Bhuyan et al. (2023)
Microplastic Abundance	2155 MPs/L $\pm$ 927.437	2500 MPs/L	Tajwar et al. (2022)

Primary productivity has shown intricate links to seasonal and tidal influences, with chlorophyll-a concentrations displaying declining trends in certain zones. Pollution levels, while generally low, experience seasonal spikes attributed to tourism activities and inadequate waste management practices. Heavy metal pollution has been detectable in previous assessments, with levels of key contaminants such as lead (Pb), cadmium (Cd), and mercury (Hg) reported to remain within acceptable limits for most areas. Furthermore, microplastic abundance in earlier studies highlights the growing concerns over marine debris, which poses a significant ecological challenge.

These findings, as documented in studies by Roy (2020), Hasan et al. (2022), Mahmud et al. (2017), Karim et al. (2015), Bhuyan et al. (2023), and Tajwar et al. (2022), underscore the importance of continuous monitoring to better understand long-term trends and their implications for environmental health and sustainability.

5.4 Implications for Future Management and Policy Recommendations

Establishing the MPA (MPA) around SMI in Bangladesh holds critical potential for guiding future management and policy to conserve marine biodiversity and promote sustainable practices. Effective MPA management on SMI requires sustainable strategies that mitigate environmental degradation and engage local communities. Integrating local knowledge and involving stakeholders is vital for resource management, focusing on community-based approaches to address coastal management vulnerabilities as well as ensure sustainable Ecotourism activity as SMI has a great tourist attraction (Touhiduzzaman & Rahman, 2017). Involving local communities in decision-making can help ensure sustainable livelihoods and ecosystem health. A land use plan that manages unregulated tourism is essential to prevent environmental damage, with zoning regulations to protect sensitive areas and foster sustainable tourism (Jahan & Quamruzzaman, 2010). Future MPA management should prioritize ecological effectiveness, supporting biodiversity and ecosystem services with clear metrics to track conservation progress (Spalding et al., 2016).

To protect SMI and promote sustainable development, a comprehensive approach is essential. Mitigating heavy metal pollution requires regular monitoring, stricter regulations, targeted remediation, and public awareness to reduce contamination and preserve ecological health. Sustainable land-use planning, coastal zone management, habitat conservation, eco-tourism, and community engagement can balance development with environmental protection. Together, these strategies support the island's resilience and long-term sustainability.

Implications for Future Management

- **Biodiversity Conservation:** The MPA aims to safeguard vital ecosystems, like coral reefs and marine wildlife, to maintain ecological balance, sustain fisheries, and protect vulnerable species such as dolphins and turtles.
- **Integrated Management Approach:** Effective MPA management requires collaboration among stakeholders, with the Forest Department ensuring compliance by integrating local knowledge.

- **Regulation of Human Activities:** The proposed MPA will create designated zones with precise laws governing fishing, vessel traffic, and tourism activities. This restriction is required to reduce overfishing, pollution, and habitat loss caused by unregulated tourism and industrial activity.
- **Community Involvement:** The MPA's success relies on involving local communities in conservation efforts, with fishermen willing to support stricter regulations to sustain their traditions. Community-driven initiatives can enhance monitoring and enforcement processes.

Policy Recommendations

- **Strengthening Enforcement Mechanisms:** To guarantee that MPA requirements are followed, strong enforcement methods must be established. To properly monitor MPA activities, combined patrols involving local authorities, the Coast Guard, and community people may be required.
- **Awareness and Education Programs:** Implementing educational outreach activities can help local people and tourists understand the importance of marine conservation. These projects should prioritize sustainable methods that safeguard marine habitats while also boosting local economies.
- **Sustainable Tourism Development:** Policymakers should encourage sustainable tourist methods that reduce environmental effect while also helping local economies. This includes regulating tourist activities, efficiently managing garbage, and ensuring that tourism development does not jeopardize natural integrity.
- **Long-term Monitoring and Research:** The MPA's adaptive management requires continuous monitoring of marine biodiversity and ecological health. Research activities should be encouraged in order to assess the efficacy of conservation strategies and adjust policies in response to scientific results.
- **Collaboration with NGOs and International Bodies:** Partnerships with non-governmental organizations (NGOs) and international conservation agencies can provide extra resources, experience, and financing for adopting successful management techniques in MPAs.

6. Conclusion

6.1 Summary of key findings

The assessment of Saint Martin's Island Marine Protected Area reveals a complex interplay between natural factors and human activities impacting the ecosystem's health. The data indicates that while natural disruptions such as cyclones and decreasing precipitation rates pose challenges, their effects are less alarming compared to the significant impact of human interventions. The increasing trend of NDVI after 2006 suggests a gradual recovery of vegetation, likely due to the Ecologically Critical Area (ECA) designation's efforts to curb deforestation. However, the persistence of microplastic pollution, unchecked resort and hotel developments, and activities like dry fish processing highlight the limitations of the ECA in fully safeguarding the ecosystem. The fair rating of the CCME Water Quality Index (76-80) underscores that while the waters remain suitable for recreational activities, they are not fit for consumption, reflecting underlying pollution issues. The Heavy Metal Pollution Index indicates medium contamination levels (15-30), and the Geo-Accumulation Index points to moderate to strongly polluted sediments (Grade 03, range 2-3). These findings suggest that anthropogenic sources, including discharges from the Naf River and maritime activities, contribute to the pollution load, posing a moderate ecological risk. Seasonal shifts in chlorophyll concentration, with increasing trends in the pre-monsoon and decreasing trends during monsoon and post-monsoon periods, signal alterations in primary productivity. While increases may enhance fish populations, decreases could lead to eutrophication, oxygen depletion, and food web disruptions. This seasonal imbalance indicates ecosystem stress possibly exacerbated by climate change and human activities.

The ECA designation has had some success in stabilizing certain environmental parameters, such as NDVI, but it has not been wholly effective in mitigating human-induced pressures. This underscores the importance of transitioning to a more robust MPA designation, which would facilitate stricter management and enforcement measures. Proper management and continuous monitoring are crucial to address the identified issues, including pollution control, regulation of coastal developments, and mitigation of human activities impacting the marine environment. By fulfilling the first objective, the study has assessed the ecosystem services provided by the SMPA,

highlighting how the current state affects services such as fisheries productivity and recreational opportunities. The analysis shows that while some services remain intact, others are compromised due to pollution and habitat degradation. In addressing the second objective, the study identifies human activities—such as microplastic pollution, deforestation, unsustainable tourism infrastructure, shipping, and riverine discharges—as significant contributors to the ecosystem's declining health. The evaluation of various pollution indices and ecological risk assessments underscores the extent of human impact. For the third objective, recommendations include strengthening the MPA designation to enhance legal and regulatory frameworks, implementing stricter controls on pollution sources, regulating coastal development, and promoting sustainable practices among local industries and communities. Comparative analysis with the ECA status suggests that while the ECA provided a foundation for conservation, its limitations necessitate a more comprehensive MPA approach to effectively improve ecosystem health.

In conclusion, the study demonstrates that achieving a sustainable balance in the SMPA requires concerted efforts to mitigate human impacts through enhanced management strategies. The fulfillment of the objectives highlights the critical need for immediate action to preserve and restore the island's unique marine ecosystem for future generations.

6.2 Future Research Directions and Gap of Knowledge

The evaluation of ecosystem health within the SMPA of Bangladesh is a complex task that necessitates a comprehensive understanding of various ecological and socio-economic factors. However, significant gaps in knowledge have been identified that hinder the effectiveness of this research.

Firstly, there is a critical lack of comprehensive fish population data. Accurate and up-to-date information on fish species diversity, abundance, and distribution is essential for assessing the overall health of marine ecosystems. Without this data, it becomes challenging to understand the dynamics of fish populations and their role in maintaining ecological balance. In addition to fish population data, there is a notable deficiency in other biodiversity data, which is crucial for evaluating the richness and variety of species within the MPA. Understanding the biodiversity of marine habitats is fundamental for assessing ecosystem resilience and function. We are only able

to get phytoplankton data for the analysis. The absence of this data limits the ability to draw meaningful conclusions about the ecological integrity of the area. The health of coral reefs is a key indicator of marine ecosystem health. However, there is insufficient data on coral health within the MPA. A detailed coral health study is necessary to understand the status of coral communities, their resilience to environmental stressors, and their overall contribution to marine biodiversity.

Moreover, social and economic factors play a significant role in the health of marine ecosystems. However, the current research lacks a thorough socio-economic survey that could provide insights into the livelihoods of local communities, their dependence on marine resources, and the impact of human activities on the ecosystem. This information is vital for developing sustainable management strategies that balance conservation efforts with the needs of local populations. Another significant limitation is the lack of field visits, which are essential for firsthand data collection and observation of the ecosystem's current state. Field visits allow for the direct assessment of coral health, habitat conditions, and the presence of key species. The absence of such visits restricts the ability to collect empirical data, which is critical for an accurate evaluation of ecosystem health.

In conclusion, addressing these knowledge gaps—specifically in fish population dynamics, biodiversity metrics, socio-economic impacts, field data collection, and coral health assessment—is essential for a more robust and comprehensive evaluation of the ecosystem health of the SMI MPA. By bridging these gaps, the research can provide valuable insights that inform effective conservation and management strategies for this critical marine environment. By this extensive study we can evaluate the success of designated MPA and effectiveness of future MPA.

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